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Chapter 23

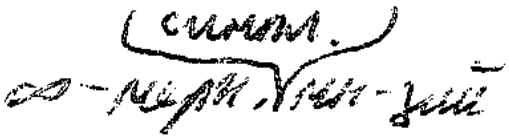
GELFAND-P001074

[source number: 1]

7/II–83

Gelfand Seminar

I. M. G. There is no familiar understanding of the ∞-dimensional symplectic group, ∞-dim. [inserted above: sympl.]



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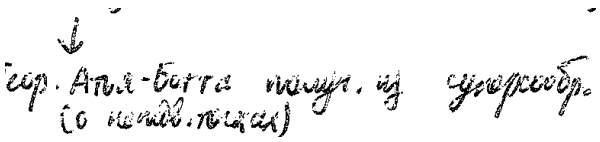
Uncertain reading. abbreviation apparently for “geometry,” ending indistinct

and so on. “Fock space is a toy for mathematicians.”

There are popular lectures by Witten on supermathematics! (preprint; lectures)

↓

The Atiyah–Bott theorem follows from super considerations
(on fixed points)



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Diagram and typed description. Arrow from Witten’s lectures to the Atiyah–Bott theorem.

Kirillov on Neretin’s work

$$T(g)f(z) = f\left(\frac{\alpha z+\gamma}{\beta z+\delta}\right)|\beta z+\delta|^{\lambda}$$

The representation of $SL(2,\mathbb{R})$ extends to a representation of Diff(S^1) (unitarily!).

$$f \longmapsto f(\varphi^{-1}(z))\left|\frac{d\varphi^{-1}(z)}{dz}\right|^{-\lambda/2}$$

[At left with a brace:] this is nonunitary, but



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Diagram and typed description. Annotation and brace before “from this.”

from this, unitary representations are constructed (by considering the action in Fock space —



Diagram and typed description. Separate closing parenthesis at the right of the line.

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— on coherent states.

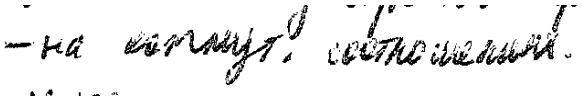


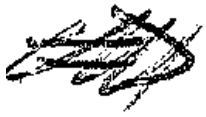
Diagram and typed description. Abbreviation for “coherent” before “states.”

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Idea:

$$\text{Im} \int f(x_1)\overline{g(x_2)}\frac{|x_1 - x_2|^{-2+\lambda}}{\Gamma(-1+\lambda)}\, dx_1\, dx_2 \qquad \text{(symplectic form!)}$$

preserved under fractional linear transformations [crossed-out arrow →?]



Uncertain reading. crossed-out mark resembling a right arrow

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⇒ not preserved under all diffeomorphisms, but of similar form.

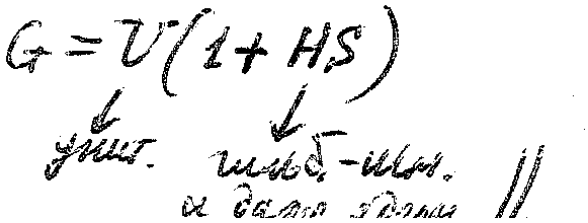


Diagram and typed description.

$G = U(1 + HS)$

unitary Hilbert–Schmidt

[and more general?]

The bottom annotation is provisional. Two oblique strokes appear at right.

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B. Feigin.

Strings and representations of Diff S^1.

V is the space of maps $S^1 \longrightarrow \mathbb{R}^d$. A metric is given on $\mathbb{R}^d$ (Riemannian or Lorentzian). Diff S^1 acts (changes of parameter).

$$f, g \longmapsto \int_{S^1} (f, dg) \tag{1}$$

On V/V_{const} one obtains a nondegenerate form. Put $V' = V/V_{\text{const}}$. [Below V_{const}:] constant maps.

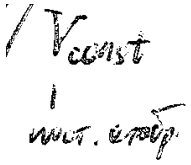


Diagram and typed description. Stroke from V_{const} toward its caption.

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On $V'/\operatorname{Diff} S^1$ there is a Hamiltonian structure (Poisson bracket).

Problem: quantize this situation. [marginal stroke]

Uncertain reading. short oblique mark before a closing parenthesis in the right margin

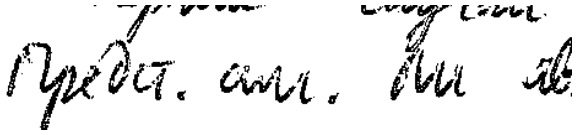


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Kirillov. M is a symplectic manifold. G is a group acting on M and preserving its symplectic structure. $\Rightarrow$ One constructs the reduced manifold M_{red} (symplectic reduction).

idea: $\{\ \}$ on invariant functions is degenerate. One must take a quotient.

B. F. V' is a space with a nondegenerate symplectic form. The quantization method is to choose p, q and realize them in the usual way (Fock space). In finite dimension this is a satisfactory method. Where does it act? The space of invariants. The ∞-dimensional case is problematic. The representation of the symplectic manifold[?] is projective, and invariants do not exist. The space must be changed.



Uncertain reading. abbreviation provisionally read as «мн.» (“manifold”)

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(a Schwinger term appears)

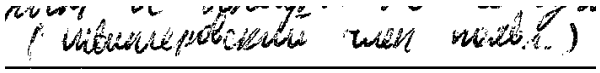


Diagram and typed description. Small note on the Schwinger term below the last line.

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GELFAND-P001075

$\operatorname{sp}(V')$ is the Lie algebra of symplectic transformations.

$\mathcal{L}$ is the algebra of vector fields on S^1.

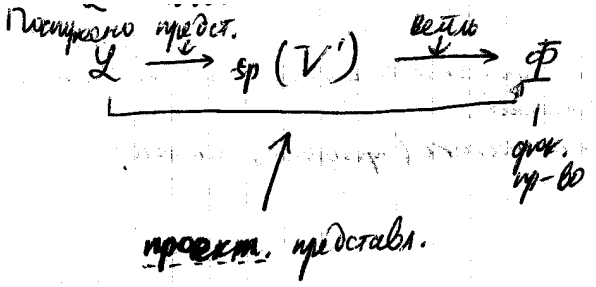
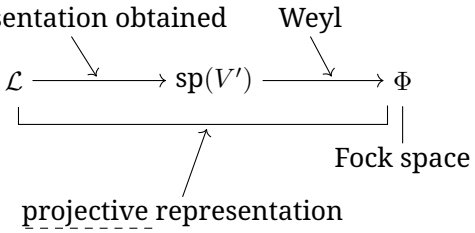


Diagram and typed description.



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The label above the second arrow reads “Weyl.”

Φ is constructed as follows:

V', [crossed out]



Uncertain reading. the crossed-out mark after V' is illegible

Original source details: GELFAND-P001075

$S^1 \longrightarrow \mathbb{R}^d$.

complexify and decompose

$$V' = \underset{\text{(annihilation operators)}}{H^+} \oplus \underset{\text{(creation operators)}}{H^-} \qquad \Phi = \bigoplus S^i H_+.$$

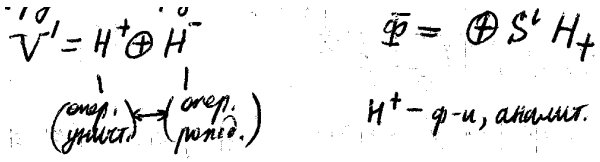
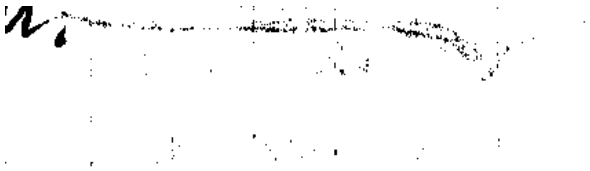


Diagram and typed description. Decomposition of V', operator labels and formula for Φ.

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H^+ consists of functions analytic inside. [faint mark in the margin]



Uncertain reading. a faint broken trace at the right of the line with no legible letters

Original source details: GELFAND-P001075

—— · ——

$$\mathcal{L} = \{L_i, c\}, \quad i \in \mathbb{Z}.$$

$$[L_i, L_j] = (j-i)L_{i+j} + \underbrace{\delta_{i,-j}\frac{i^3 - i}{12}\, c}_{\text{Schwinger term}}.$$

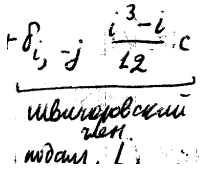


Diagram and typed description. Bracket and caption beneath the central term.

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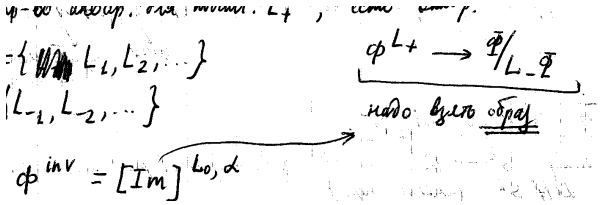
ϕ^{L_+} is the space of invariants for the subalgebra L_+, and there is a map.

$$L_+ = \{\text{[crossed out] } L_1, L_2, \ldots\},$$
$$L_- = \{L_{-1}, L_{-2}, \ldots\}.$$



Uncertain reading. heavily crossed out before L_1, possibly L_0

Original source details: GELFAND-P001075



Original source details: GELFAND-P001075

Diagram and typed description.

$$\underbrace{\phi^{L_+} \longrightarrow \Phi/L_-\Phi}_{\text{take the image}}$$

$$\phi^{\text{inv}} = [\text{Im}]^{L_0,\alpha}$$

Nontrivial at least when $d = 26$.

Problem (I. M. G.). Construct a p-adic analogue of the representation of the diffeomorphism group.

GELFAND-P001076

14/II–83 Gelfand Seminar.

Yu. Neretin.

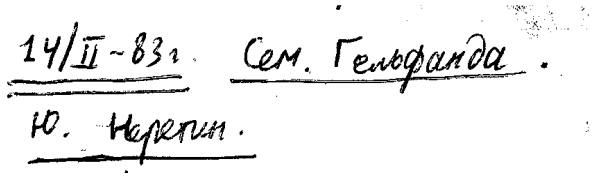


Diagram and typed description. Date and speaker's name.

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$\forall \rho$ almost preserves the inner product.

$$\frac{1}{\Gamma(\rho)} \int_{-\infty}^{\infty} |x_1 - x_2|^{-\rho} f(x_1) \overline{g(x_2)}\, dx_1\, dx_2. \tag{1}$$

([crossed out] the operator will be $O(1+T)$.)



Uncertain reading. illegible crossed-out writing before "operator"

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[Below T, a vertical stroke to the annotation:] H.–S.

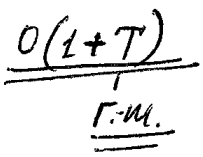


Diagram and typed description. Double underlines and annotation for T.

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Hilbert transform:

$$Hf(s) = \frac{1}{2\pi}\int_0^{2\pi} \operatorname{ctg}\left(\frac{t-s}{2}\right) f(t)\,dt \qquad (f \text{ is a real function}) \tag{2}$$

$$H : \text{const} \longrightarrow 0$$

$\rho = 0, 2 \quad \Rightarrow$ in the space modulo const or with $\int = 0$.

H introduces a complex structure.

In Fourier modes:

$$\| \quad \begin{array}{l} \cos n\varphi \longrightarrow -\sin n\varphi \\ \sin n\varphi \longrightarrow \cos n\varphi \end{array}$$

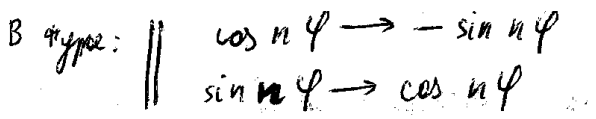


Diagram and typed description. Double stroke to the left of the two lines and overwritten n in the second.

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Inner product

$$\int \ln\left|\sin\frac{t-s}{2}\right| f_1(t) f_2(s)\,dt\,ds$$

$$\int \text{[crossed out]}\ \operatorname{ctg}\left(\frac{t-s}{2}\right) f_1(t) f_2'(s)\,dt\,ds$$



Uncertain reading. heavily crossed-out writing after the integral sign, possibly beginning with ln

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$\Rightarrow H$ defines a complex structure.



Uncertain reading. illegible crossed-out writing between the arrow and H

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[At right:] by decomposition into analytic and antianalytic functions

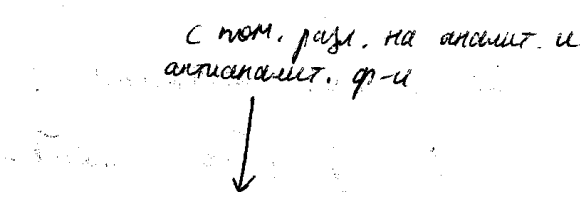


Diagram and typed description. Arrow down from the right-hand note to the remark on Hermitian extension.

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$\Rightarrow$ take the Hermitian extension using this H.



Diagram and typed description. Hatched double stroke at the left of the arrow.

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Im$\langle \cdot\,;\,\cdot \rangle$ is preserved under all diffeomorphisms.

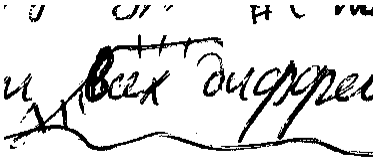


Diagram and typed description. Mark above "all" and wavy underlining.

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I. M. G. Problem: make this a general statement about pseudodifferential operators!! (invariant under [inserted above: all] diffeomorphisms!!)



Diagram and typed description. "All" is inserted above the line.

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GELFAND-P001077

Symplectic form — ?

Yu. N. Projections: $H \pm i$

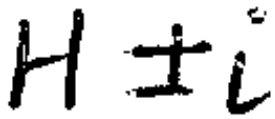


Diagram and typed description. The manuscript reads $H \pm i$.

Original source details: GELFAND-P001077

their images: Lagrangian manifolds.

Symplectic form:

$$\{f_1, f_2\} = \langle f_1, H f_2 \rangle$$

I. M. G. [crossed out]



Uncertain reading. illegible crossed-out writing after the initials

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[Inserted above: "We must also."] consider a generalization of $c[1 + (\text{comp.})]$ ($\Rightarrow$ write the whole series in symbols).

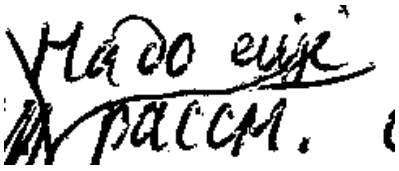


Diagram and typed description. "We must also" is inserted above the line.

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[Crossed-out formula: $\iint \ln \left|\sin \frac{t-s}{2}\right|$.]

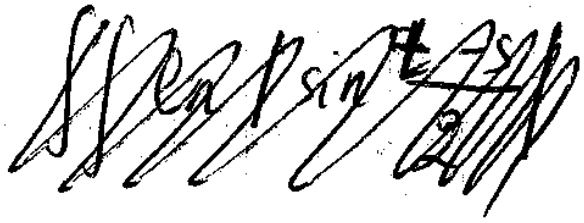
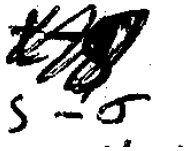


Diagram and typed description. Crossed-out formula containing sin.

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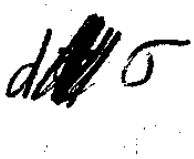
Claim

$$\iint \ln \left|\frac{t-s}{2}\right| f_1(t) f_2(\sigma) \frac{1}{s-\sigma}\, ds\, d\sigma\, dt$$



Uncertain reading. denominator corrected. $s-\sigma$ is written below. The cancelled beginning resembles $t-$

Original source details: GELFAND-P001077



Uncertain reading. a symbol after d is crossed out before σ, possibly t

Original source details: GELFAND-P001077

is preserved under diffeomorphisms

$\Rightarrow$ must be a differential operator (f').

Need: to calculate:

$$\int \ln \left|\frac{t-s}{2}\right| \frac{1}{s-\sigma}\, ds.$$

We should obtain δ'.

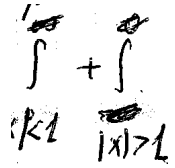


Diagram and typed description. The manuscript reads δ'.

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We must continue analytically!!

$$\int_{-\infty}^{\infty} |x|^{\lambda} f(x)\, dx = \int_{|x|<1} + \int_{|x|>1}$$



Uncertain reading. marks resembling ∞ are crossed out above both integrals on the right. The mark beneath the second may have been $-\infty$

Original source details: GELFAND-P001077

and we continue each analytically.

Idea:

$$\int |t-s|^{\lambda} \ \dots$$

and differentiate with respect to λ!!

— · —

~~V. Ginzburg~~



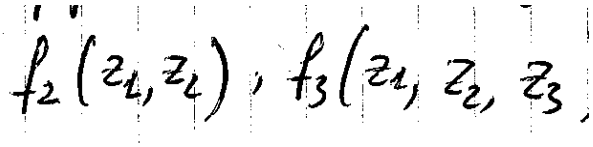
Diagram and typed description. Crossed-out name with a double underline.

Original source details: GELFAND-P001077

Another problem (following Feigin!):

How does a diffeomorphism of S^1 act on a sequence of analytic functions

$$\{f_1(z), f_2(z_1, z_1), f_3(z_1, z_2, z_3), \dots\}$$



Uncertain reading. the second subscript in f_2 looks more like 1. It may be 2

Original source details: GELFAND-P001077

$|z| < 1$.

What effect does smoothness have?

For φ fractional linear, this coincides with the usual one!



Uncertain reading. overwritten mark before "fractional linear," provisionally φ

Original source details: GELFAND-P001077

How can this be realized?

GELFAND-P001078

[source number: 5]

V. Ginzburg.

On the work of Duistermaat, Heckman (Invent. Math., 1982)

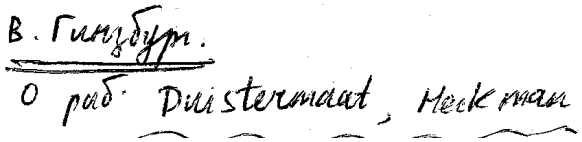


Diagram and typed description. Speaker's name and the authors of the work.

Original source details: GELFAND-P001078

$T = \mathbb{T}^k$ is a torus (k-dimensional). M is a compact symplectic manifold. T acts on M in a Hamiltonian way: it preserves the symplectic structure and every vector field



Uncertain reading. crossed out, possibly «не» or «на»

Original source details: GELFAND-P001078

from the Lie algebra of T determines a field given by a Hamiltonian, and these Hamiltonians commute (with respect to the Poisson bracket).



Diagram and typed description. The Russian text reads “a field given.”

Original source details: GELFAND-P001078

(Commuting fields do not imply commuting functions.)

(This is now called a Poisson Hamiltonian action.)

[At left, beyond the double stroke:] A map arises:

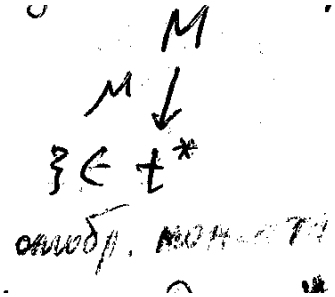


Diagram and typed description.

$$\begin{array}{c} M \\ \mu \big\downarrow \\ \xi \in t^* \end{array}$$

[moment map?]

Original source details: GELFAND-P001078

The caption beneath the diagram is faint. The end of the abbreviation is indistinct.

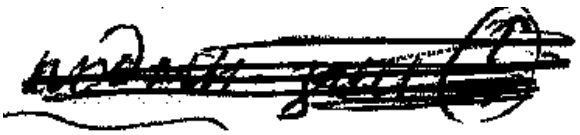
Or: give functions $f_1, \ldots, f_k$: $\{f_i, f_j\} = 0$, The flows of f_i are periodic.



Uncertain reading. mark over the end of the abbreviation “functions,” resembling an asterisk or a correction

Original source details: GELFAND-P001078

Local condition: the fields lie in one Lagrangian



Uncertain reading. crossed-out word beginning “sub…,” probably “subspace”

Original source details: GELFAND-P001078

subspace at every point.
[crossed-out heading]



Uncertain reading. crossed-out heading, possibly “Next” or “Another”

Original source details: GELFAND-P001078

$$\mu(m)(X) = H_X(m) \qquad (X \text{ is any field}) \tag{1}$$



Diagram and typed description. The X in the argument of $\mu(m)$ is retraced.

Original source details: GELFAND-P001078

Or: The cocycle $X_1, X_2 \longrightarrow \omega(X_1, X_2)(m)$ on the Lie algebra must be trivial.

Example: $\{y, -x\}$ give commuting fields $\dfrac{\partial}{\partial x}$, $\dfrac{\partial}{\partial y}$, but their bracket is $\neq 0$.

$$M_\xi = \mu^{-1}(\xi) \quad \text{; on it } \omega \text{ is degenerate;} \tag{2}$$

On M_ξ/T [inserted above: “it”] is already nondegenerate.

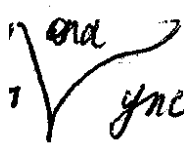


Diagram and typed description. “It” is inserted above the line.

Original source details: GELFAND-P001078

Denote it by ω_ξ. One can calculate volume with respect to it.

Statement of the lemma: ω_ξ is [inserted above: “piecewise”] linear in ξ (mod exact differentials).

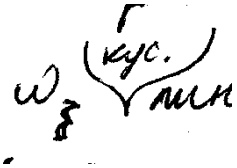


Diagram and typed description. Insertion of the abbreviation for “piecewise” before “linear.”

Original source details: GELFAND-P001078

Corollary. vol(M_ξ) is piecewise polynomial.

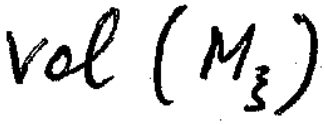
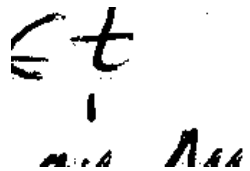


Diagram and typed description. The notation is vol(M_ξ).

Original source details: GELFAND-P001078

(3) Theorem. Consider a vector $X \in t$ and the corresponding Hamiltonian H_X



Original source details: GELFAND-P001078

Diagram and typed description. The abbreviation for “Lie algebra” appears beneath t.

(the lift of X as a function on t^* using μ). Then the integral

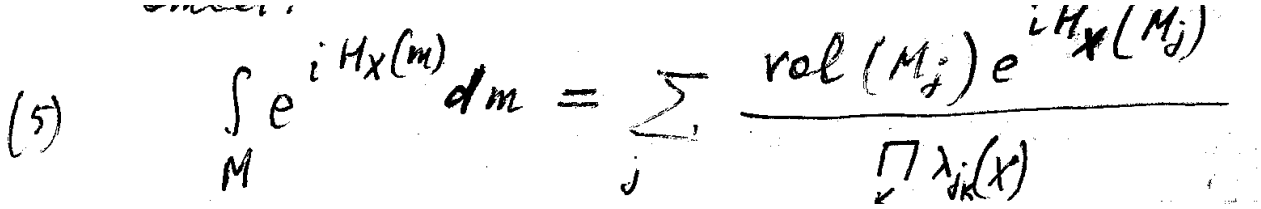
$$\int_M e^{iH_X(m)}\, dm \tag{4}$$

(a function on the Lie algebra)

can be calculated by stationary phase, and this gives an exact answer.

Answer:

$$\int_M e^{iH_X(m)}\, dm = \sum_j \frac{\mathrm{vol}(M_j) e^{iH_X(M_j)}}{\prod_k \lambda_{jk}(X)} \tag{5}$$



Original source details: GELFAND-P001078

Diagram and typed description. Equation (5), with $\mathrm{vol}(M_j)$ and a product in the denominator.

The fixed points of X (= stationary points) form a symplectic submanifold (possibly disconnected). M_j are its connected components. The $\lambda_{jk}(X)$ are linear forms that are computed (Jacobian of the action in transverse directions).



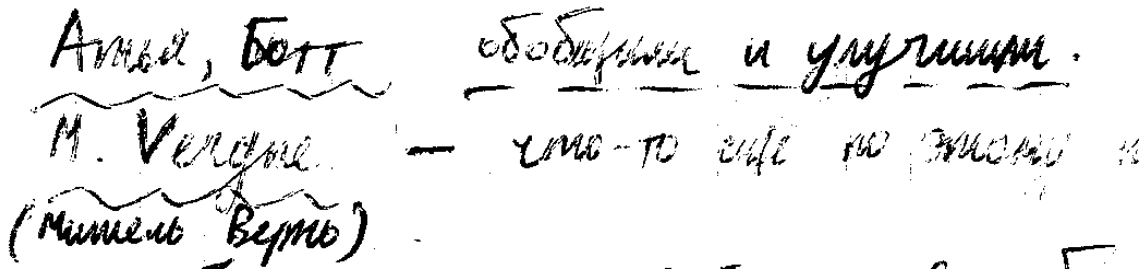
Original source details: GELFAND-P001078

Diagram and typed description. The bottom parenthesis reads “Jacobian of the action in transverse directions.”

GELFAND-P001079

Atiyah, Bott generalized and improved it.

M. Vergne — something else on the subject (Michèle Vergne).



Original source details: GELFAND-P001079

Diagram and typed description. Atiyah, Bott, and Vergne in the manuscript.

An old result of Atiyah and Guillemin–Sternberg: the image of μ — is a convex region!!

(5) $\sim$ Weyl's character formula.

(~ 10 years ago there was a paper by Semenov-Tian-Shansky with similar content.)

(~10 лет назад была раб. Семенова-Тян-Шанского, близкая по содержанию)

Original source details: GELFAND-P001079

Diagram and typed description. Reference to the paper by Semenov-Tian-Shansky.

Idea of the proof: by stationary phase

$$\int_M e^{itH_X(m)}\,dm = \sum \frac{\mathrm{vol}(M_j)e^{itH_X(M_j)}}{\prod_k \lambda_{jk}(X)} + \text{lower-order terms}$$

$\int_M e^{itH_X(m)}\,dm = \sum \frac{\mathrm{vol}(M_j)\,e^{itH_X(M_j)}}{\prod_k \lambda_{jk}(X)}$ + мл. член

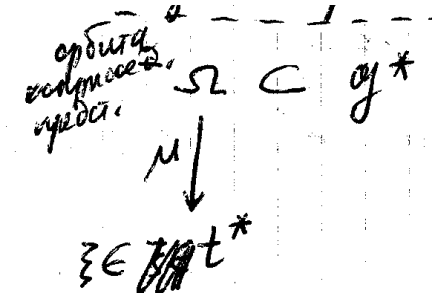
Original source details: GELFAND-P001079

Diagram and typed description. The parameter t in the exponents. The denominator reads $\prod_k \lambda_{jk}(X)$.

Locally quadratic $\Rightarrow$ the remainder is a rapidly decaying function.

Piecewise polynomial $\Rightarrow$ exponentials and polynomials $\Rightarrow$ remainder $= 0$.

Connection with representation theory:

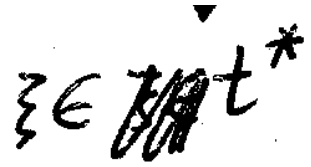


Original source details: GELFAND-P001079

Diagram and typed description.

orbit of the coadjoint representation

$$\begin{array}{c}\Omega \subset g^* \\ \mu \big\downarrow \\ \xi \in t^*\end{array}$$



Original source details: GELFAND-P001079

Uncertain reading. illegible crossed-out notation before t^*

G compact $\Rightarrow \Omega$ compact.

We obtain the Harish-Chandra integral.

Kirillov: this is — the character.

M. Vergne: proof for noncompact orbits of the discrete series.

A. A. Kirillov.

$SU(3)$ $H_0(3) \supset \Omega$

acts on Hermitian matrices (of trace 0), an 8-dimensional manifold

$$h \longrightarrow ghg^{-1}$$



Diagram and typed description. The conjugation formula.

Original source details: GELFAND-P001079

Orbit Ω: 2 invariants $\mathrm{Sp}(A^2)$, $\mathrm{Sp}(A^3)$. [Below «Orbit»: «6-dimensional».]

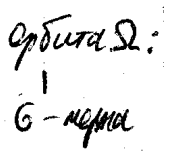


Diagram and typed description. The handwritten note on the orbit dimension.

Original source details: GELFAND-P001079

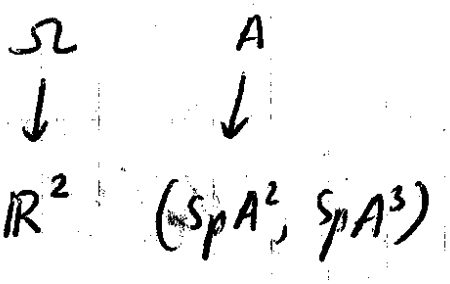


Diagram and typed description.

$$\begin{array}{cc} \Omega & A \\ \downarrow & \swarrow \\ \mathbb{R}^2 & (\mathrm{Sp}\,A^2, \mathrm{Sp}\,A^3) \end{array}$$

Original source details: GELFAND-P001079

The projection of Ω onto $\mathbb{R}^2$ — is an almost regular hexagon.

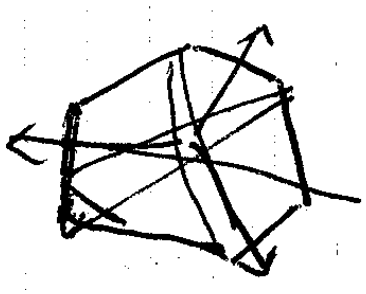
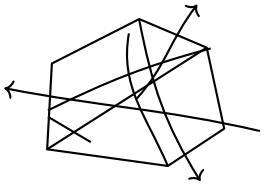


Diagram and typed description.

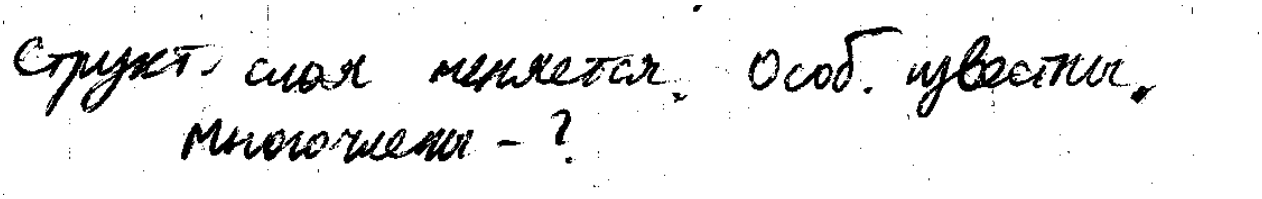


Some connections are uncertain where the strokes overlap.

Original source details: GELFAND-P001079

(All angles are $120°$; every other side has the same length.)

The structure of the fiber changes. The singularities are known,
Polynomials — ?



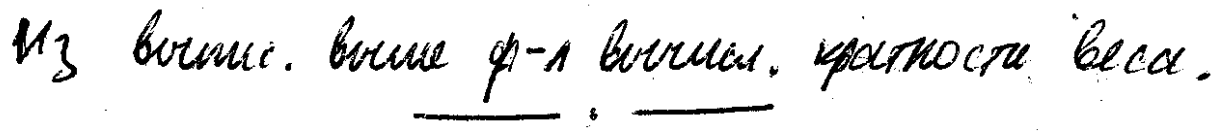
Original source details: GELFAND-P001079

Diagram and typed description. The concluding remark on the structure of the fiber.

GELFAND-P001080

[source number: 6]

From the formulas written out above, calculate the multiplicities of the weight.



Original source details: GELFAND-P001080

Diagram and typed description. The line on weight multiplicities with two underlined fragments.

GELFAND-P001081

21/II-83 Gelfand Seminar.
V. I. Arnold.

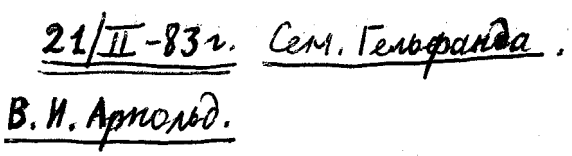
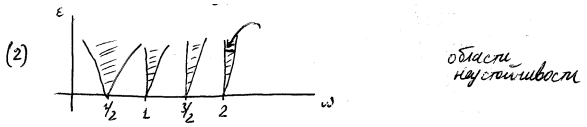


Diagram and typed description. The date, seminar name, and speaker.

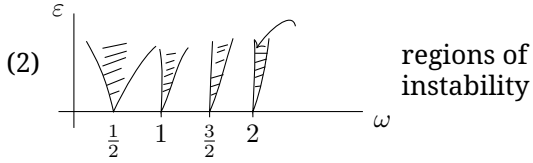
Original source details: GELFAND-P001081

$$(1) \qquad \ddot{x} + (\omega^2 + \varepsilon a(t))x = 0$$
$$a(t + 2\pi) = a(t)$$



Original source details: GELFAND-P001081

Diagram and typed description.



regions of instability

$a(t) = \cos t \quad \Rightarrow$ increasingly high order of contact (of the zone boundaries).



Diagram and typed description.

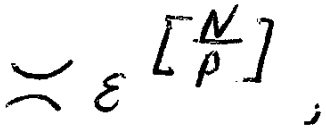


Sketch above the words “zone boundaries.”

Original source details: GELFAND-P001081

(3) Theorem. Width of the Nth zone $\asymp \varepsilon^{[\frac{N}{p}]}$; if a is a trigonometric polynomial

$$\deg a = p.$$



Uncertain reading. The exponent is written $[N/p]$. The brackets are left unexplained.

Original source details: GELFAND-P001081

Mathieu: $p = 1$.



Uncertain reading. The handwritten name appears to read «Маге».

Original source details: GELFAND-P001081

Hill’s equation (a in general position) $\Rightarrow$ first-order contact everywhere

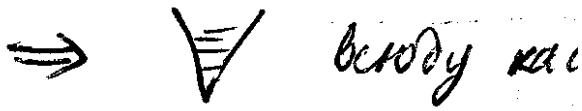


Diagram and typed description.

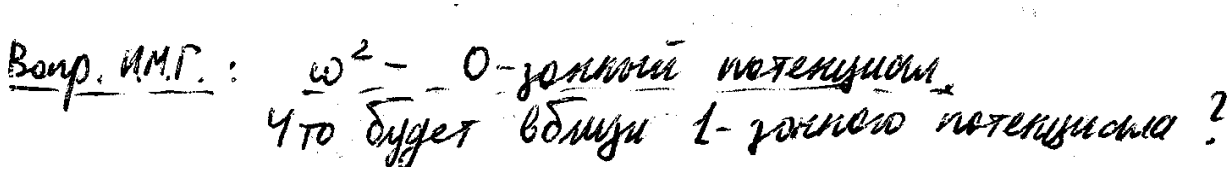


Sketch after the arrow.

Original source details: GELFAND-P001081

(but the width decreases exponentially)

Question by I. M. G.: ω^2 — a 0-zone potential. What happens near a 1-zone potential?

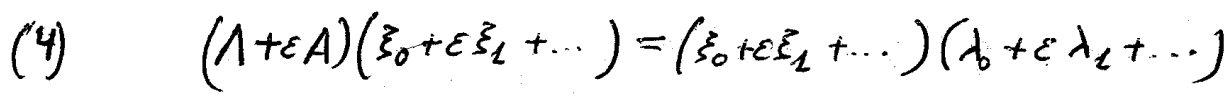


Original source details: GELFAND-P001081

Diagram and typed description. The question about a one-zone potential.

The proof depends on the structure of the Rayleigh–Schrödinger series. We seek the perturbation.

$$(4) \quad (\Lambda + \varepsilon A)(\xi_0 + \varepsilon\xi_1 + \ldots) = (\xi_0 + \varepsilon\xi_1 + \ldots)(\lambda_0 + \varepsilon\lambda_1 + \ldots)$$

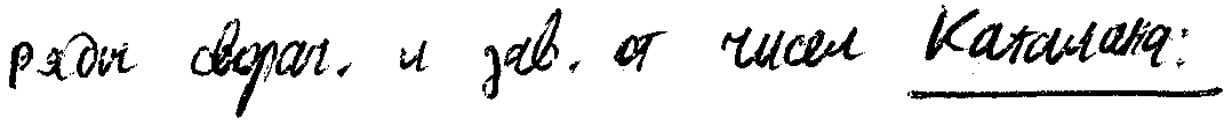


Original source details: GELFAND-P001081

Diagram and typed description. The perturbation series in equation (4).

GELFAND-P001082

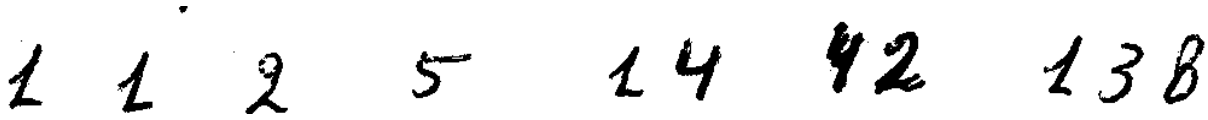
series are [combined?] and depend on the numbers named for Catalan:



Original source details: GELFAND-P001082

Uncertain reading. The abbreviation appears to read «сверт.».

$$1 \quad 1 \quad 2 \quad 5 \quad 14 \quad 42 \quad 138$$



Original source details: GELFAND-P001082

Diagram and typed description. The last number in the manuscript is 138.

Another occurrence, in Griffiths and Harris (they did not notice).

Formula:

$$(5) \qquad a_{n+1} = \sum_{j=1}^{n-1} a_j a_{n-j}$$

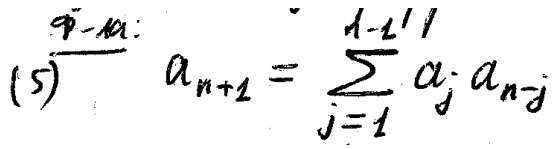


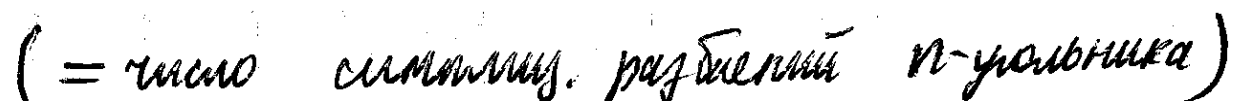
Diagram and typed description. The summation bounds and indices in the manuscript.

Original source details: GELFAND-P001082

Using a generating function we compute

$$a_n = \frac{(2n)!}{n!(n+1)!}$$

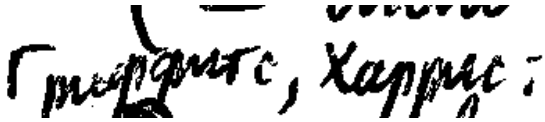
(= the number of simplicial subdivisions of an n-gon)



Original source details: GELFAND-P001082

Diagram and typed description. The polygon index in the manuscript.

[Inserted at the upper left, “Griffiths, Harris”:]
Riemann surfaces of genus g have $3g - 3$ moduli.

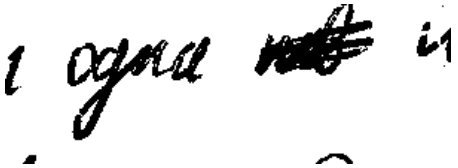


Original source details: GELFAND-P001082

Diagram and typed description. The names inserted above the line.

On $S^2 = \mathbb{C}P^1$ choose branch points, etc.

GH: the number of ways in which one [crossed out] and the same surface was obtained:



Original source details: GELFAND-P001082

Uncertain reading. The crossed-out word may be an abbreviation for “surface.”

$1, 1, 2, 5, \ldots$ — Catalan numbers (but it is not said why)

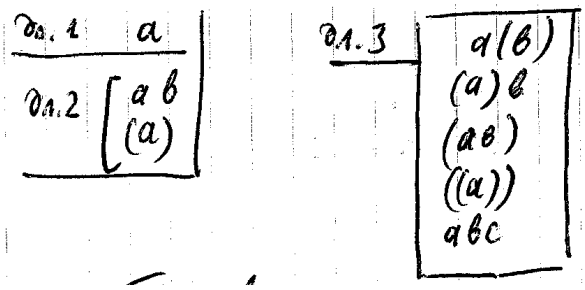
Bracket symbol: an expression properly formed from letters and brackets: example

$$((a))b\,(cd)e$$

$\sum$ of the counts of symbols and opening brackets $= n$ (length)

$\Downarrow$

count = the Catalan number



Original source details: GELFAND-P001082

Diagram and typed description.

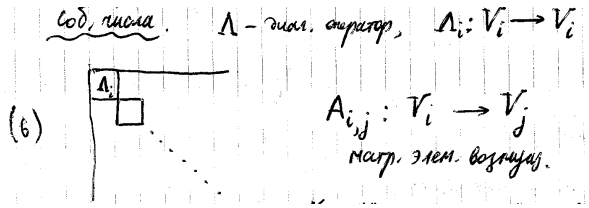
len. 1	a
len. 2	ab
	(a)

len. 3	$a(b)$
	$(a)b$
	(ab)
	$((a))$
	abc

Bracket symbol $\rightsquigarrow$ term of the perturbation series.

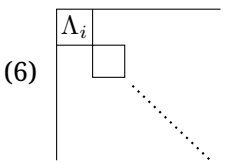
λ_i (eigenvalues) $\quad\rightsquigarrow\quad$ quiver

Eigenvalues. Λ — diagonal operator, $\Lambda_i : V_i \longrightarrow V_i$

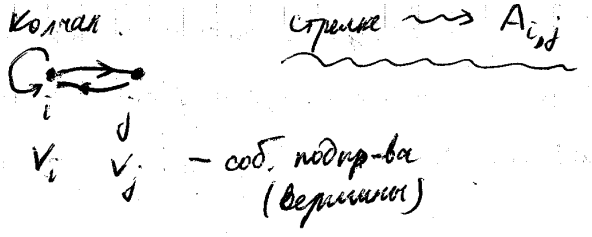


Original source details: GELFAND-P001082

Diagram and typed description.

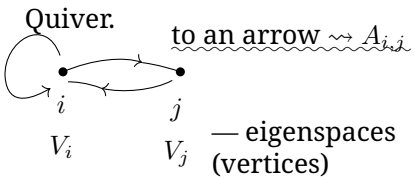


$A_{i,j} : V_i \longrightarrow V_j$
matrix element of the perturbation



Original source details: GELFAND-P001082

Diagram and typed description.



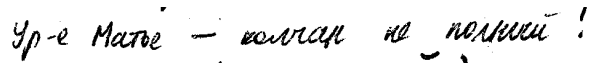
GELFAND-P001083

[source number: 9]

Suppose everything is finite or locally finite

(into $\forall$ vertex there lead finitely many)

Mathieu's equation — the quiver is incomplete!



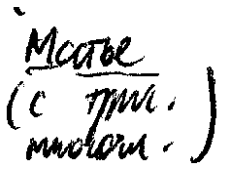
Original source details: GELFAND-P001083

Uncertain reading. The handwritten name appears to read «Маte».

(locally finite)

(the potential — cos and sin and the eigenfunctions too!)

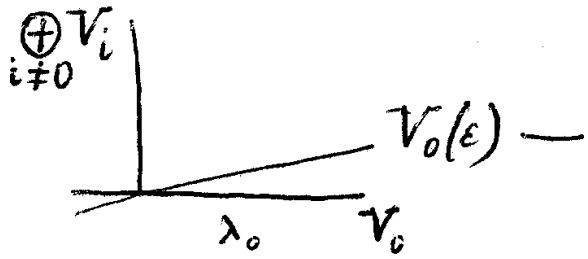
Mathieu (with a trigonometric polynomial) $\Rightarrow$ a matrix of generalized Jacobi type.



Original source details: GELFAND-P001083

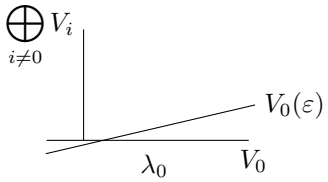
Uncertain reading. The handwritten name appears to read «Маte».

We seek the perturbation of λ_0.

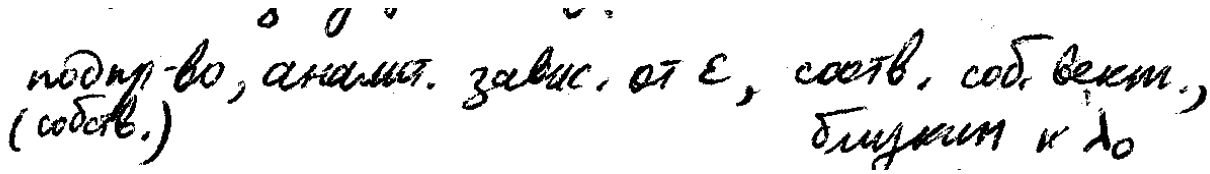


Original source details: GELFAND-P001083

Diagram and typed description.



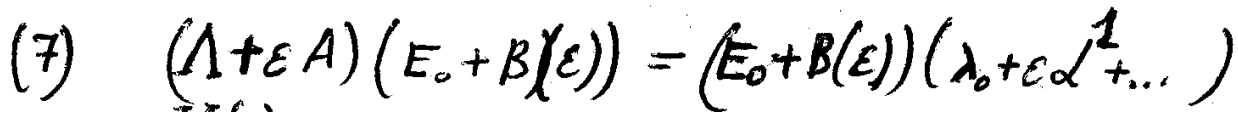
$V_0(\varepsilon)$ — a subspace, [below: "eigen"] depending analytically on ε, corresponding to eigenvectors close to λ_0.



Original source details: GELFAND-P001083

Diagram and typed description. The manuscript abbreviation for "eigenvectors."

$$(7) \quad (\Lambda + \varepsilon A)(E_0 + B(\varepsilon)) = (E_0 + B(\varepsilon))(\lambda_0 + \varepsilon\alpha^1 + \dots)$$



Original source details: GELFAND-P001083

Diagram and typed description. The coefficient α^1 in formula (7).

$V_0(\varepsilon)$ — the graph of the operator

$$B(\varepsilon) : V_0 \longrightarrow \bigoplus_{i \neq 0} V_i \qquad E_0 \text{ — the projector onto } V_0$$

$$\alpha^1 : \text{ operator } V_0 \to V_0, \qquad \alpha^i : V_0 \to V_0$$

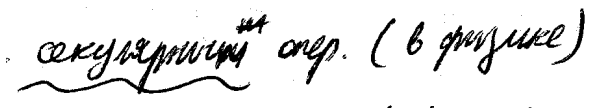
(7) — the condition of invariance of [crossed out] $V_0(\varepsilon)$



Uncertain reading. The crossed-out symbol is illegible.

Original source details: GELFAND-P001083

(8) operator $\lambda_0 + \varepsilon\alpha^1 + \dots$ is called the secular operator (in physics)



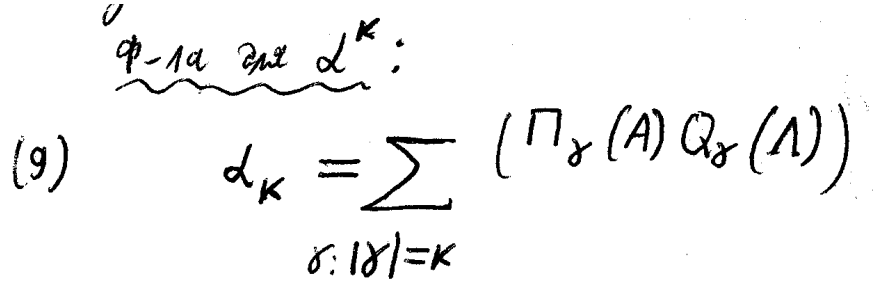
Uncertain reading. A correction is visible above the ending of the Russian word for "secular." The original ending is uncertain.

Original source details: GELFAND-P001083

The restriction of $\Lambda + \varepsilon A$ to $V_0(\varepsilon)$ is similar to the secular operator (8) $\Rightarrow$ they have the same eigenvalues.

Formula for α^k:

$$(9) \qquad \alpha_k = \sum_{\gamma : |\gamma| = k} \left(\Pi_\gamma(A) Q_\gamma(\Lambda)\right)$$

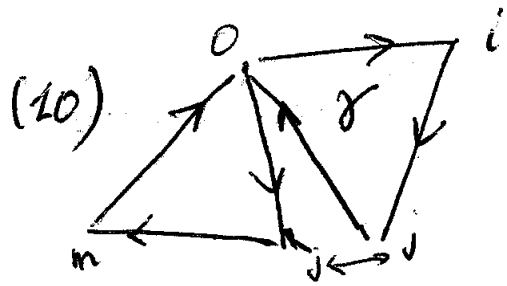


Original source details: GELFAND-P001083

Diagram and typed description. The heading uses a superscript and formula (9) a subscript.

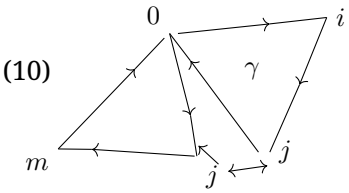
γ — a loop in the quiver from 0 to 0

k — the length of the loop



Original source details: GELFAND-P001083

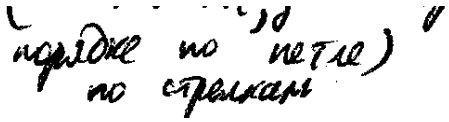
Diagram and typed description.



Both lower labels look like j. The connection of the short diagonal stroke to the adjacent arrows is uncertain.

$$\Pi_\gamma(A) = A_{m0} \dots A_{ij} A_{0i}$$

$\Pi_\gamma(A)$ — the chronological product around the loop ($= \prod A_{i,j}$ in the required order along the loop) [inserted below: «following the arrows».]



Original source details: GELFAND-P001083

Diagram and typed description. The insertion below the note about the order of factors.

$Q_\gamma(\Lambda)$ — a rational function of the eigenvalues

(computed using Catalan numbers!)

Consider a bracket symbol.

Filling it — insert vertices of the quiver in place of the letters.

GELFAND-P001084

A loop is constructed from a filling.

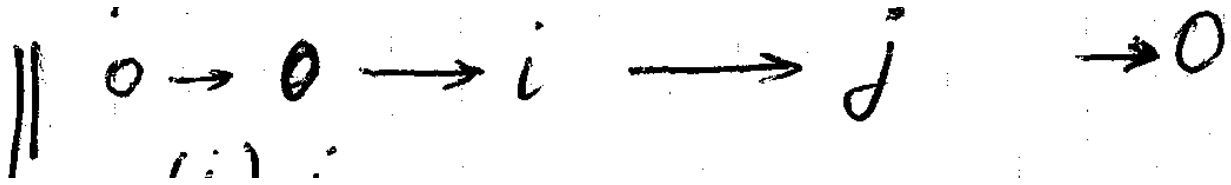


Original source details: GELFAND-P001084

Uncertain reading. The crossed-out mark before “terms” is illegible.

The terms must be indexed by fillings

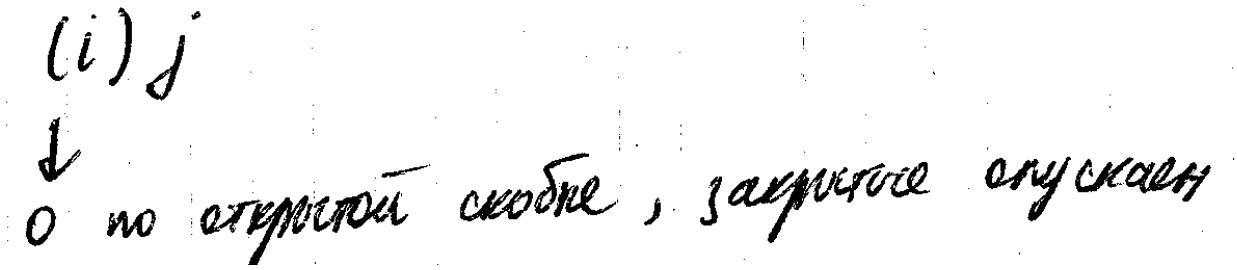
$$\| \quad 0 \longrightarrow 0 \longrightarrow i \longrightarrow j \longrightarrow 0$$



Original source details: GELFAND-P001084

Diagram and typed description. The chain of four arrows with two initial zeros.

$(i)j$
$\downarrow$
0 for an opening bracket, we omit the closing ones



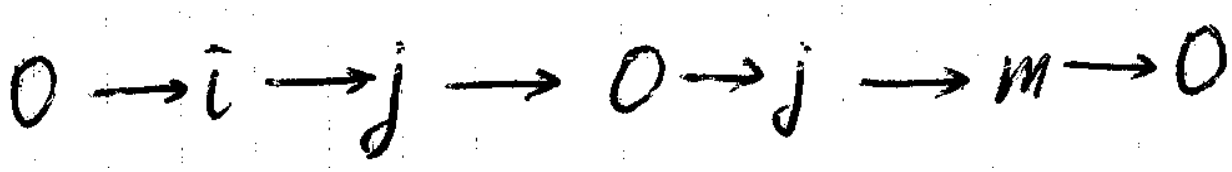
Original source details: GELFAND-P001084

Diagram and typed description. The arrow from the opening bracket to zero and its explanation.

There is one more arrow than the length of the filling.

One loop comes from different fillings.

$$0 \longrightarrow i \longrightarrow j \longrightarrow 0 \longrightarrow j \longrightarrow m \longrightarrow 0$$

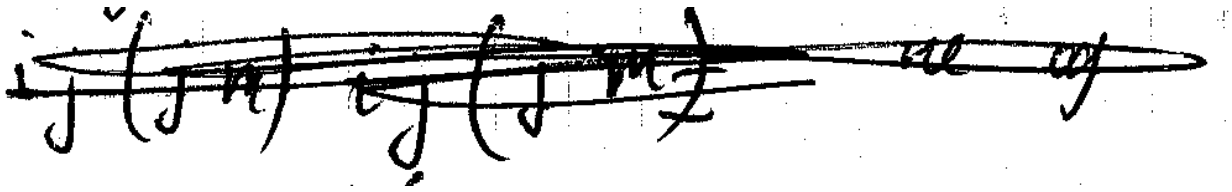


Original source details: GELFAND-P001084

Diagram and typed description. The loop with two occurrences of vertex j.

In fact, the sum runs over fillings rather than loops.

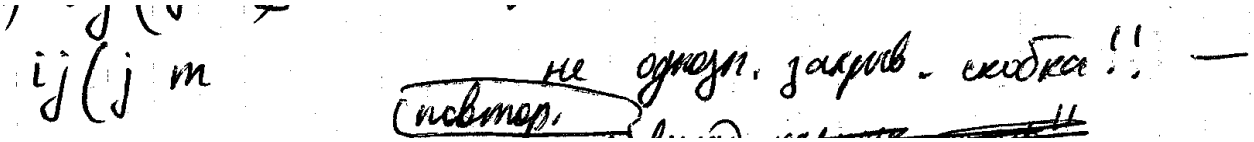
[Line crossed out: $ij(jm)$ $ij(jm)$ not unamb...]



Original source details: GELFAND-P001084

Uncertain reading. Two similar fillings are legible in the crossed-out line. At the right, possibly "not unamb...".

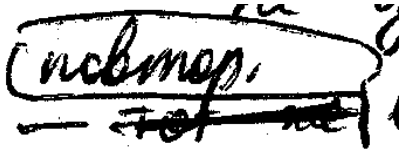
$ij(jm$ ambiguous closing bracket!! —



Original source details: GELFAND-P001084

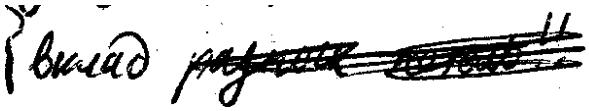
Diagram and typed description. The expression with the closing bracket omitted and its accompanying remark.

repeat [crossed out] contribution [crossed out]



Uncertain reading. Below the circled word, the crossed-out text probably reads “— the same.”

Original source details: GELFAND-P001084



Uncertain reading. After “contribution,” the crossed-out text probably reads “different brackets!!”.

Original source details: GELFAND-P001084

Diagram of a filling: add dashed [~~brackets~~] [inserted below: arrows]



Uncertain reading. The ending of the Russian word for “dashed” is overwritten. The original ending is uncertain.

Original source details: GELFAND-P001084

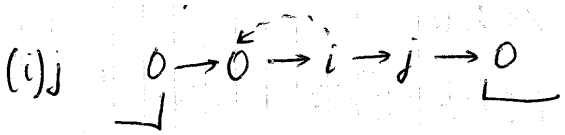


Diagram and typed description.

$$(i)j \qquad 0 \longrightarrow 0 \longrightarrow i \longrightarrow j \longrightarrow 0$$

Original source details: GELFAND-P001084

The outermost zeros do not signify anything

Arrow $\longrightarrow$ to 0, that corresponds to the opening bracket matching [crossed out] the closing bracket (from the letter before the closing bracket)



Uncertain reading. The crossed-out word probably abbreviates “opening.”

Original source details: GELFAND-P001084

$$\alpha_k = \sum_{I,\ |I|=k-1} \frac{\Pi_{\gamma(I)}(A)}{q_I(\Lambda)} \tag{11}$$

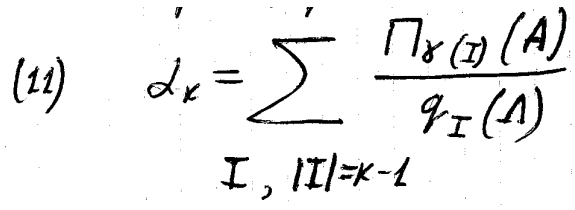
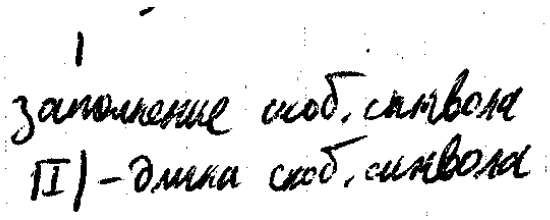


Diagram and typed description. The left side of formula (11) reads α_k.

Original source details: GELFAND-P001084

I
|
filling of a bracket symbol

$|I|$ — length of the bracket symbol

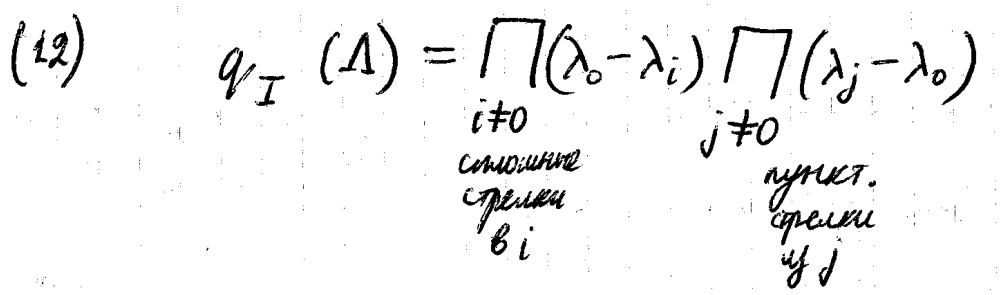


Original source details: GELFAND-P001084

Diagram and typed description. The two annotations for the summation index.

$\Pi_{\gamma(I)}(A)$ — the chronological product around the loop along the solid arrows.

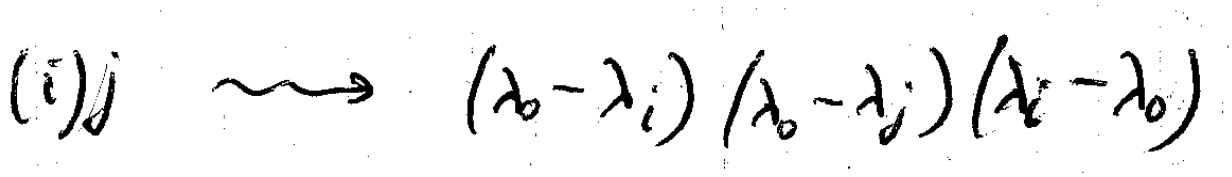
$$q_I(\Lambda) = \prod_{\substack{i\neq 0 \\ \text{solid arrows into } i}} (\lambda_0 - \lambda_i) \prod_{\substack{j\neq 0 \\ \text{dashed arrows from } j}} (\lambda_j - \lambda_0) \tag{12}$$



Original source details: GELFAND-P001084

Diagram and typed description. The factors and annotations for the two products in formula (12).

$$(i)j \rightsquigarrow (\lambda_0 - \lambda_i)(\lambda_0 - \lambda_j)(\lambda_i - \lambda_0)$$



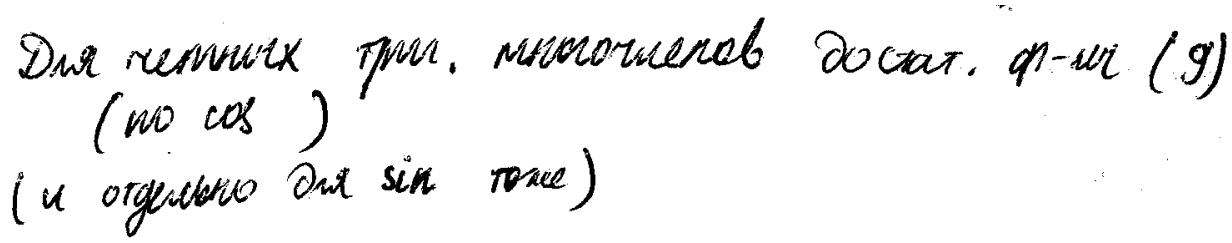
Original source details: GELFAND-P001084

Diagram and typed description. The product for the filling $(i)j$.

GELFAND-P001085

[source number: 10]

For even trigonometric polynomials, formula (9) suffices
[Below the word “even”: (for cos)]
(and separately for sin as well)



Original source details: GELFAND-P001085

Diagram and typed description. The cosine annotation below the word “even.”

$$V = \bigoplus V_i$$

$$\Lambda = \frac{d^2}{dt^2}, \qquad V_k = \{e_k, e_{-k}\}, \qquad e_k = e^{ikt} \tag{13}$$



Original source details: GELFAND-P001085

Diagram and typed description. The notation for the operator and basis in equation (13).

	V_0,	V_1,	V_2	V_3	$\cdots$	V_k
dim	1	2	2	2		2
						$\lambda = -k^2$

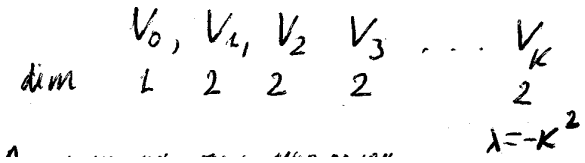


Diagram and typed description. The dimensions and the eigenvalue in the last column.

Original source details: GELFAND-P001085

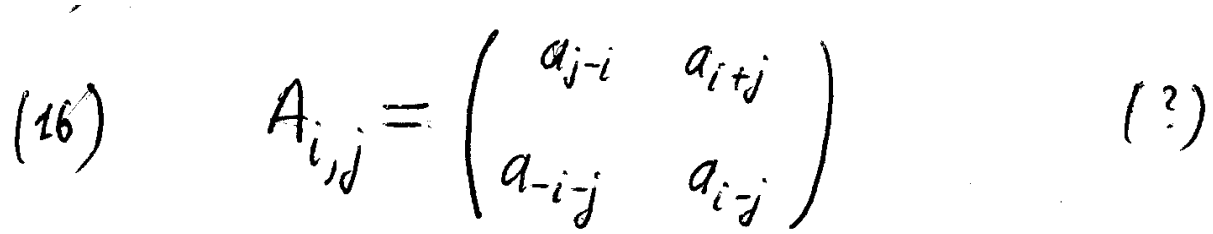
A — multiplication by a trigonometric polynomial

$$a(t) = \sum_{|s| \le p} a_s e_s \tag{14}$$

Let it be even, i.e.

$$a_{-s} = a_s \tag{15}$$

$$A_{i,j} = \begin{pmatrix} a_{j-i} & a_{i+j} \\ a_{-i-j} & a_{i-j} \end{pmatrix} \qquad (?) \tag{16}$$



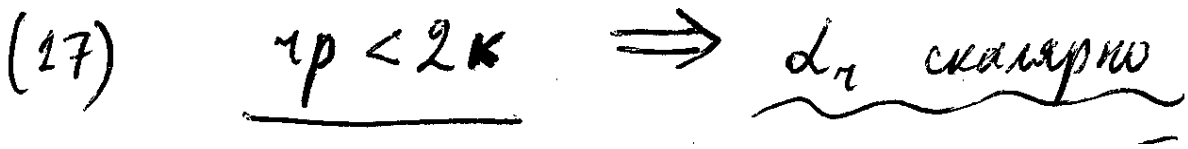
Original source details: GELFAND-P001085

Diagram and typed description. The signs in the subscripts and the question mark beside equation (16).

$(16) \Rightarrow$ if $i + j > p$, the matrices are diagonal and even scalar [written below: (by (15))] in this basis.

If all $A_{i,j}$ in the loop are scalar, then the product is also scalar.

$$\tau p < 2k \quad \Rightarrow \quad \alpha_\tau \text{ is scalar} \tag{17}$$

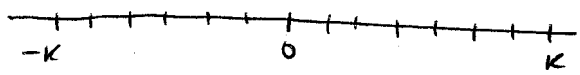


Original source details: GELFAND-P001085

Diagram and typed description. The coefficient α_τ and the underlining in equation (17).

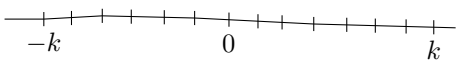
This is a topological lemma (from combinatorial topology).

Proof



Original source details: GELFAND-P001085

Diagram and typed description.



Consider a loop of τ arrows of length $\le p$.

Then for $\forall$ arrow $i + j > p$

(homotopic to 0!)

$i + j \le p \Rightarrow$ a jump across



Original source details: GELFAND-P001085

Uncertain reading. Provisional reading “a jump across.”



Original source details: GELFAND-P001085

Diagram and typed description.



The small mark at the right end looks like i. Arrowheads on the arches are indistinct.

$\Rightarrow$ length $\leq \tau p$ and so on.

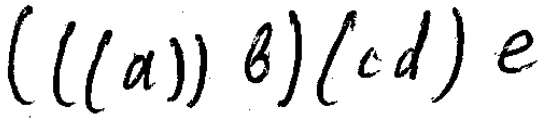
(17) $\Rightarrow$ in this approximation the eigenvalue has yet to split $\Rightarrow$

$\Rightarrow$ the required inequality!

GELFAND-P001086

An involution is also used:

$$(((a))b)(cd)e$$



Original source details: GELFAND-P001086

Diagram and typed description. The original bracket symbol.

use the opening brackets to divide it into fragments

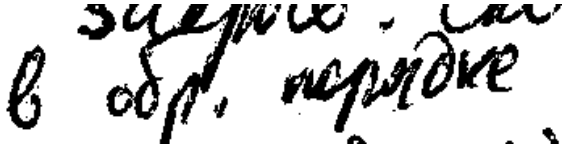
$$a))b) \qquad cd)e$$



Original source details: GELFAND-P001086

Diagram and typed description. The fragments with their opening brackets removed.

Attach the closing brackets to the letter. Rearrange the letters in the fragments [inserted below at the left: in reverse order]



Original source details: GELFAND-P001086

Diagram and typed description. The insertion specifying the reverse order of the letters.

$$b)a)) \qquad ed)c$$



Original source details: GELFAND-P001086

Diagram and typed description. The two fragments with their letters in reverse order.

[crossed out] Put the opening brackets back in their former places

Uncertain reading. The crossed-out abbreviation before “Put” is illegible.



Original source details: GELFAND-P001086

$$(((b)a))(ed)c$$

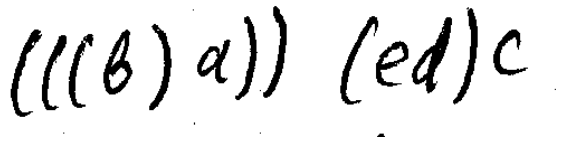
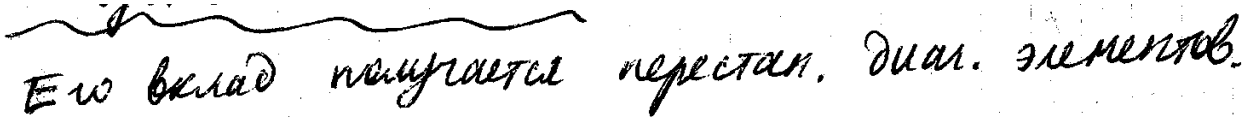


Diagram and typed description. The dual bracket symbol.

Original source details: GELFAND-P001086

We obtain the dual symbol

Its contribution is obtained by swapping the diagonal entries.



Original source details: GELFAND-P001086

Diagram and typed description. The Russian word for “diagonal” is abbreviated at the end of the line.

GELFAND-P001087

28/II-83 Gelfand Seminar.

I. Krichever.

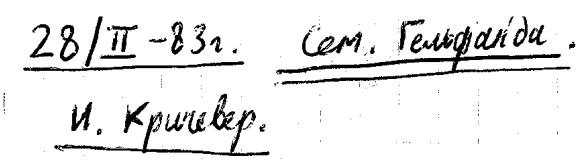


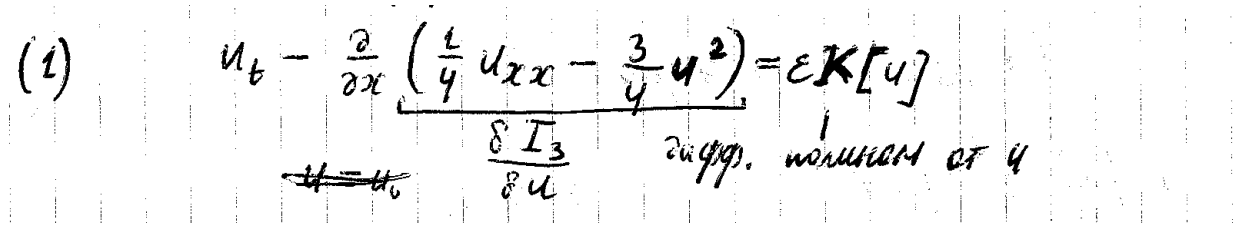
Diagram and typed description. The date, seminar name, and speaker.

Original source details: GELFAND-P001087

The aim is to construct a formal perturbation theory for the perturbation of the KdV equation.

$$(1) \qquad u_t - \frac{\partial}{\partial x}\underbrace{\left(\frac{1}{4}u_{xx} - \frac{3}{4}u^2\right)}_{\frac{\delta I_3}{\delta u}} = \varepsilon K[u]$$

[Below K: differential polynomial in u.]



Original source details: GELFAND-P001087

Diagram and typed description. Equation (1) with a square-ended bracket and the annotation below K.

[Crossed out: ~~$u = u_0$~~.]

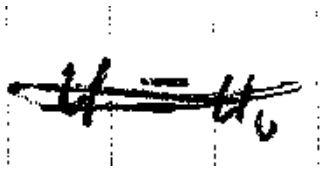


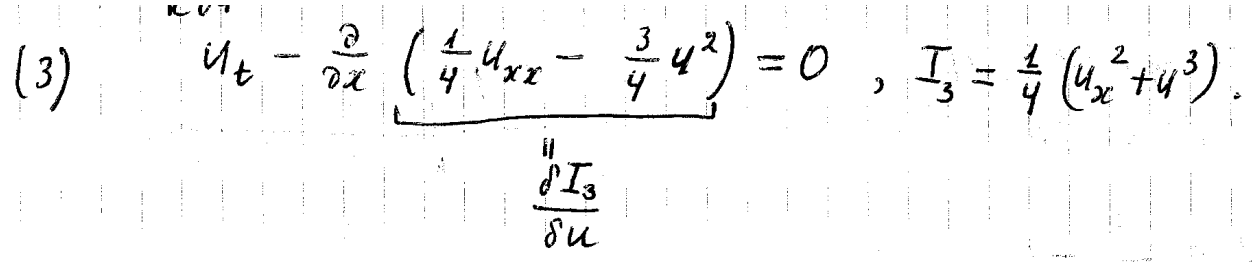
Diagram and typed description. The crossed-out equality before (2).

Original source details: GELFAND-P001087

$$(2) \qquad u = u_0(x,t) + \varepsilon(\ldots)$$

KdV

$$(3) \qquad u_t - \frac{\partial}{\partial x}\underbrace{\left(\frac{1}{4}u_{xx} - \frac{3}{4}u^2\right)}_{\parallel \frac{\delta I_3}{\delta u}} = 0, \qquad I_3 = \frac{1}{4}(u_x^2 + u^3).$$



Original source details: GELFAND-P001087

Uncertain reading. Two short vertical strokes stand between the lower bracket and its label. Their meaning is uncertain.

There is an infinite series of integrals in involution I_k: $I_1 = u$, $I_2 = u^2$, ...



Original source details: GELFAND-P001087

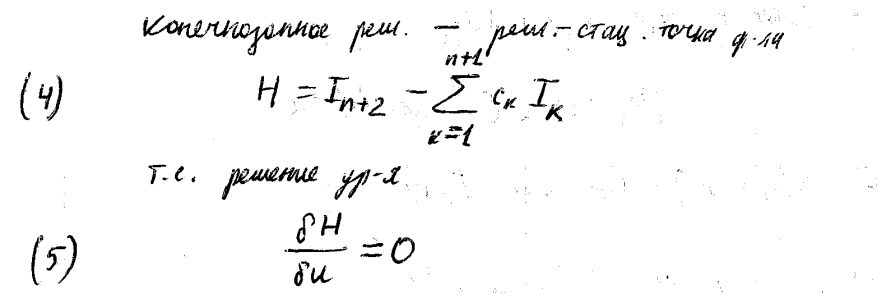
Diagram and typed description. The first integrals, including the overwritten u in I_1.

A finite-gap solution is a solution that is a stationary point of the functional

$$(4) \qquad H = I_{n+2} - \sum_{k=1}^{n+1} c_k I_k$$

i.e. a solution of the equation

$$(5) \qquad \frac{\delta H}{\delta u} = 0$$



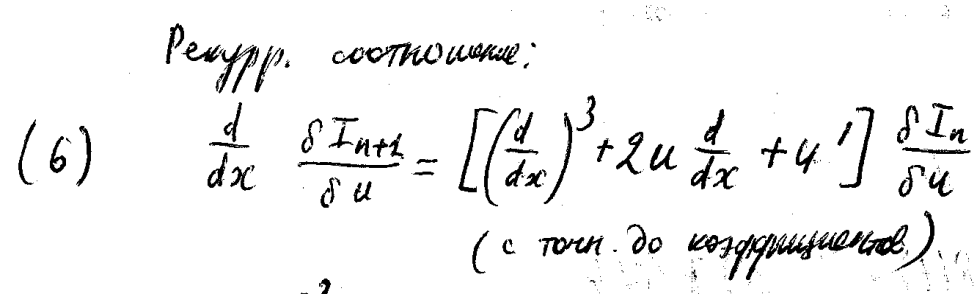
Original source details: GELFAND-P001087

Diagram and typed description. The functional and its stationarity condition in equations (4) and (5).

Recurrence relation:

$$(6) \qquad \frac{d}{dx}\frac{\delta I_{n+1}}{\delta u} = \left[\left(\frac{d}{dx}\right)^3 + 2u\frac{d}{dx} + u'\right]\frac{\delta I_n}{\delta u}$$

(up to coefficients).



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Diagram and typed description. The coefficients, prime, and indices in equation (6).

$$\delta^2 H - ?$$

at a specified stationary point.

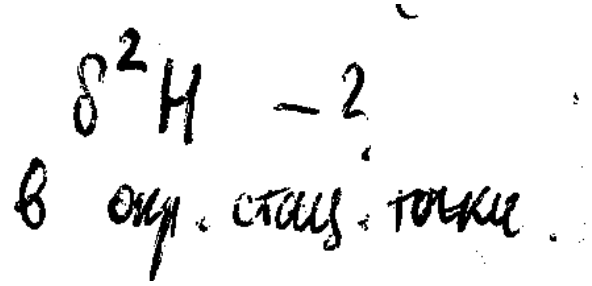


Diagram and typed description. The question about the second variation.

Original source details: GELFAND-P001087

GELFAND-P001088

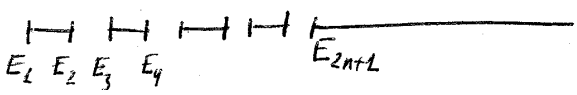
[source number: 15]

$$L = -\frac{d^2}{dx^2} + u_0(x) \qquad \text{spectrum:} \qquad (u_0(x+1) = u_0(x))$$



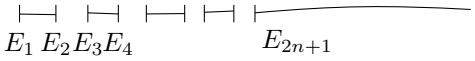
Original source details: GELFAND-P001088

Diagram and typed description. The operator and periodicity condition.



Original source details: GELFAND-P001088

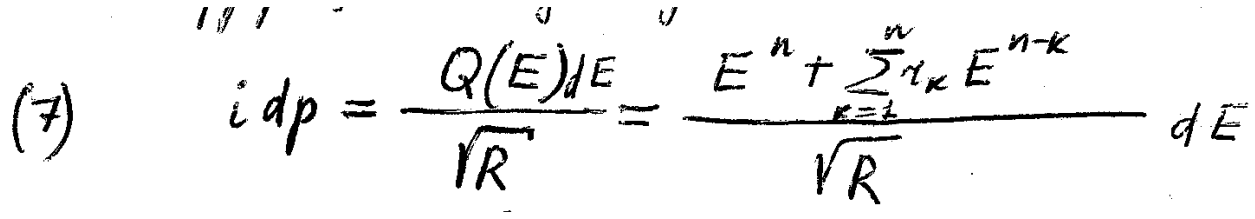
Diagram and typed description.



The remaining endpoints are simple eigenvalues of the periodic and antiperiodic problems.

Quasimomentum differential:

$$(7) \qquad i\,dp = \frac{Q(E)dE}{\sqrt{R}} = \frac{E^n + \sum_{k=1}^{n} r_k E^{n-k}}{\sqrt{R}}\,dE$$

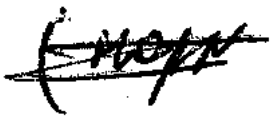


Original source details: GELFAND-P001088

Uncertain reading. The coefficient is provisionally read as r_k. The first letter is ambiguous.

$$R(E) = \prod_{i=1}^{2n+1} (E - E_i)$$

[Crossed out at the right: ~~(norm.~~]



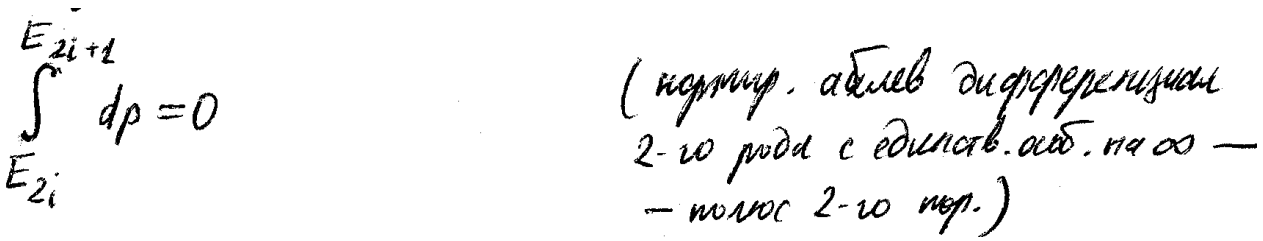
Uncertain reading. The beginning reads "(norm.". The rest of the abbreviation is uncertain.

Original source details: GELFAND-P001088

Conditions uniquely determining dp:

$$\int_{E_{2i}}^{E_{2i+1}} dp = 0$$

(normalized Abelian differential of the second kind with its sole singularity at ∞ — — a second-order pole.)



Original source details: GELFAND-P001088

Diagram and typed description. The vanishing integral and the note on the pole.

Now consider the perturbation $u = u_0 + \delta u$

E_i' — new band edges

$e_s^- < e_s^+$ — new band edges; $\delta e_s = e_s^+ - e_s^-$



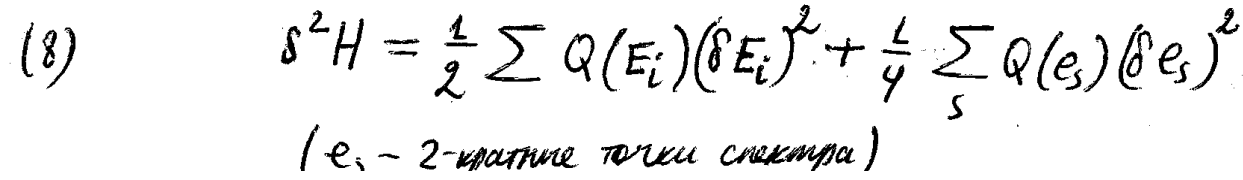
Original source details: GELFAND-P001088

Diagram and typed description. The two new edges and the definition of δe_s.

Then:

$$(8) \qquad \delta^2 H = \frac{1}{2}\sum Q(E_i)(\delta E_i)^2 + \frac{1}{4}\sum_s Q(e_s)(\delta e_s)^2$$

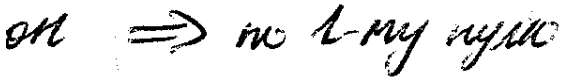
(e_s — double points of the spectrum)



Original source details: GELFAND-P001088

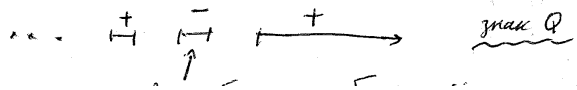
Diagram and typed description. The two sums of the second variation and the note on e_s.

Q changes sign once in $\forall$ of the forbidden zones $\Rightarrow$ one zero in each zone



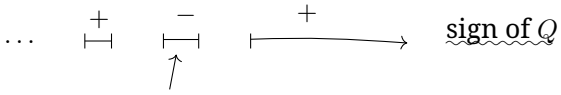
Original source details: GELFAND-P001088

Diagram and typed description. The handwritten phrase “one zero each.”

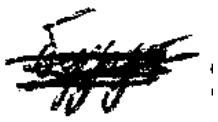


Original source details: GELFAND-P001088

Diagram and typed description.



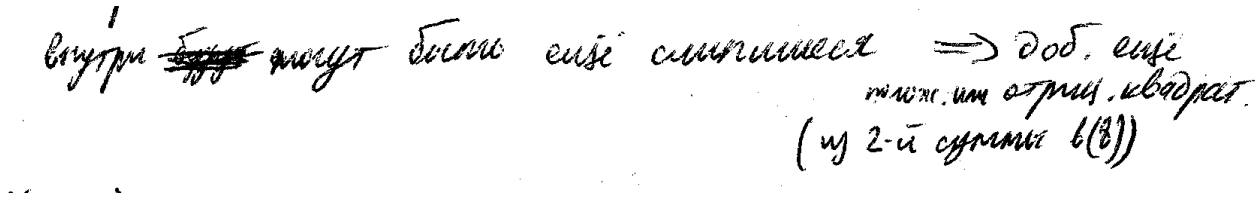
[Below the arrow:] inside [crossed out] there may be other coalesced ones $\Rightarrow$ add another positive or negative square



Original source details: GELFAND-P001088

Uncertain reading. The crossed-out word may read “will be.”

(from the second sum in (8))

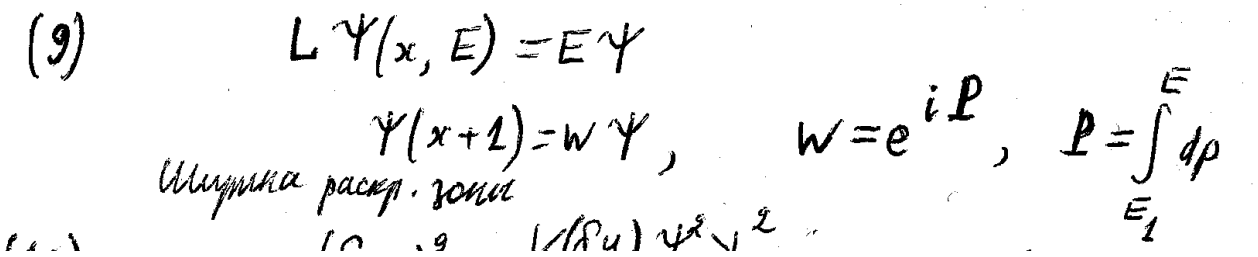


Original source details: GELFAND-P001088

Diagram and typed description. The note below the arrow in the sign diagram.

$$(9) \qquad L\Psi(x,E) = E\Psi$$

$$\Psi(x+1) = w\Psi, \qquad w = e^{iP}, \qquad P = \int_{E_1}^{E} dp$$



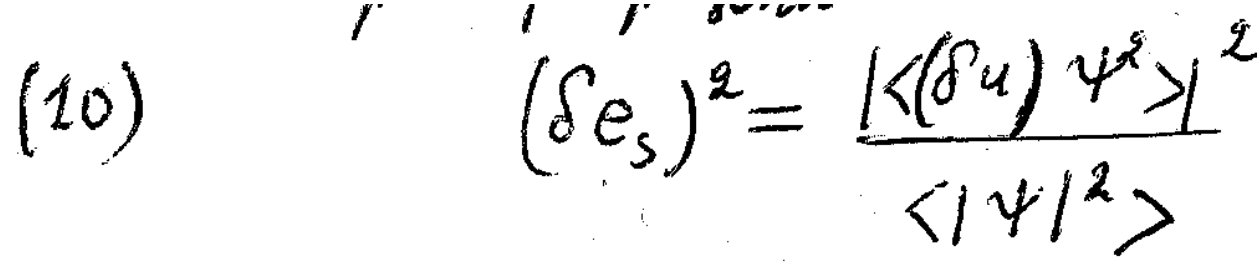
Original source details: GELFAND-P001088

Diagram and typed description. The Bloch equation and multiplier.

Width of the opened gap

$$(10) \qquad (\delta e_s)^2 = \frac{\left|\langle(\delta u)\Psi^2\rangle\right|^2}{\langle|\Psi|^2\rangle}$$

(Ψ — Bloch function, $\Psi(x, e_s)$)

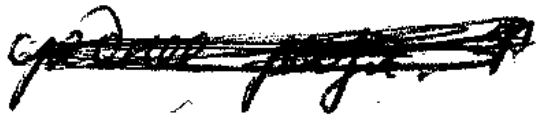


Original source details: GELFAND-P001088

Diagram and typed description. The written denominator is $\langle|\Psi|^2\rangle$ with no outer exponent.

$\langle \quad \rangle$ — averaging over a period

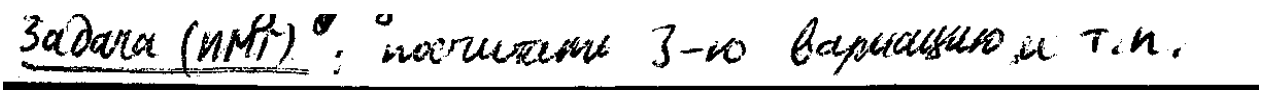
$u_0 = \text{const} \Rightarrow$ we obtain [crossed out: ~~averag.~~ ...] Fourier coefficients for the perturbation.



Undeciphered passage. After the start of "average," one or more words are obscured by the cancellation strokes.

Original source details: GELFAND-P001088

Problem (I. M. G.): compute the third variation, and so on.



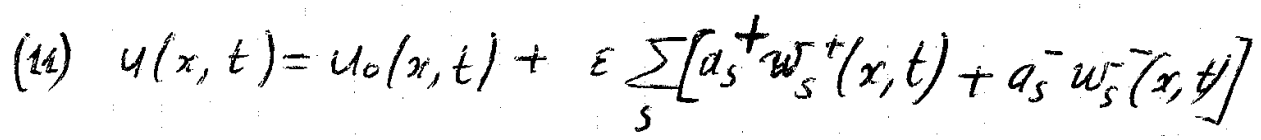
Original source details: GELFAND-P001088

Diagram and typed description. The problem concerning the third variation.

GELFAND-P001089

Return to (1) and let u be a solution. Then to first order in ε

$$u(x,t) = u_0(x,t) + \varepsilon \sum_s \left[a_s^+ w_s^+(x,t) + a_s^- w_s^-(x,t)\right] \tag{11}$$



Original source details: GELFAND-P001089

Diagram and typed description. Signs on the two terms in equation (11).

$w_s^\pm(x,t)$ — solutions of linearized KdV near an n-gap solution. They are found by opening a gap near e_s and taking the derivative with respect to the length of the new gap. [inserted above the line: «$(n+1)$-gap potential».]

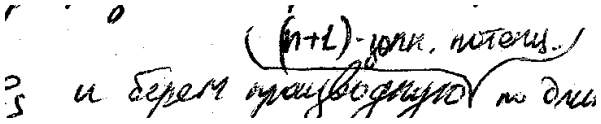
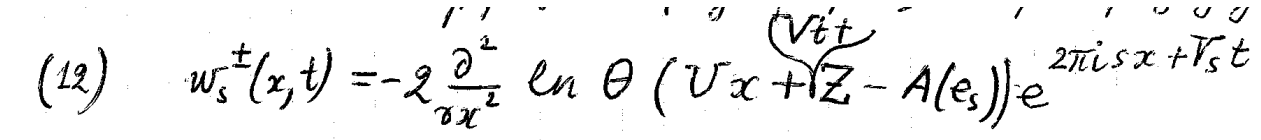


Diagram and typed description. The insertion about the $(n+1)$-gap potential.

Original source details: GELFAND-P001089

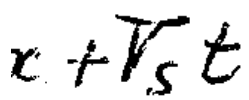
$$w_s^\pm(x,t) = -2\frac{\partial^2}{\partial x^2} \ln \theta\big(Ux + Vt + Z - A(e_s)\big) e^{2\pi i s x + V_s t} \tag{12}$$

[Inserted in the argument of θ: $Vt+$ before Z.]



Original source details: GELFAND-P001089

Diagram and typed description. Equation (12) with the insertion above the line.



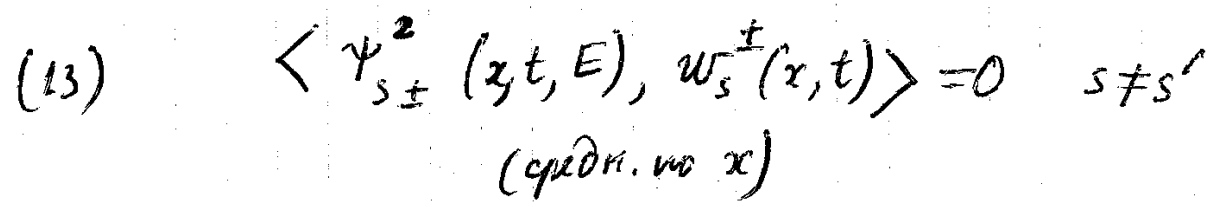
Uncertain reading. The exponent provisionally reads $V_s t$. The first letter is uncertain.

Original source details: GELFAND-P001089

$\Psi^2_{s\pm}(x,t,E)$ — a solution of the adjoint of the linearized KdV.

$$\left\langle \Psi^2_{s\pm}(x,t,E), w^{\pm}_s(x,t) \right\rangle = 0, \qquad s \neq s' \tag{13}$$

(averaged over x)

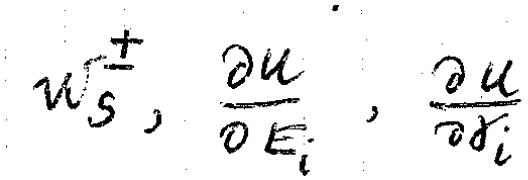


Original source details: GELFAND-P001089

Diagram and typed description. Both factors carry the index s. The condition on the right reads $s \neq s'$.

($\Rightarrow$ there is a basis expansion)

$$w^{\pm}_s, \quad \frac{\partial u}{\partial E_i}, \quad \frac{\partial u}{\partial \gamma_i}$$



Uncertain reading. The last denominator provisionally reads $\partial\gamma_i$. The first letter of the parameter is uncertain.

Original source details: GELFAND-P001089

Wodzicki (Mariusz).

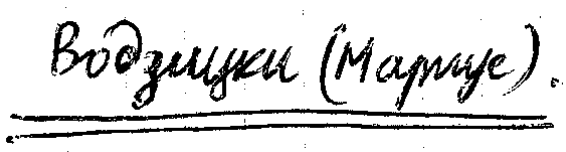


Diagram and typed description. The speaker's name in the manuscript.

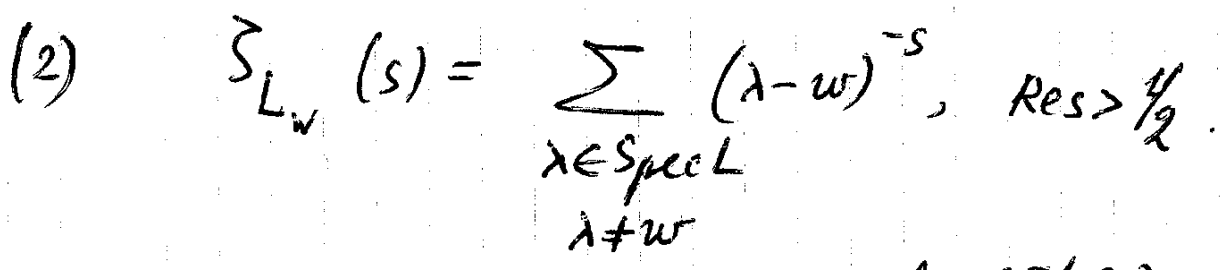
Original source details: GELFAND-P001089

$$L_w = -\frac{d^2}{dx^2} + u(x) - w, \qquad x \in [0,T] \tag{1}$$

(Hill operator; u — T-periodic)

ζ-function:

$$\zeta_{L_w}(s) = \sum_{\substack{\lambda \in \operatorname{Spec} L \\ \lambda \neq w}} (\lambda - w)^{-s}, \qquad \operatorname{Re} s > \tfrac{1}{2}. \tag{2}$$



Original source details: GELFAND-P001089

Diagram and typed description. The zeta function definition and convergence region.

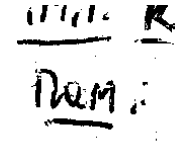
We are interested in the values at $s = 1, 2, 3, \ldots$

Suppose

$$\begin{gathered} u(x) \text{ — finite-gap,} \\ w \in \{\alpha_0, \ldots, \alpha_{2n}\} \text{ — band edges} \end{gathered} \tag{3}$$

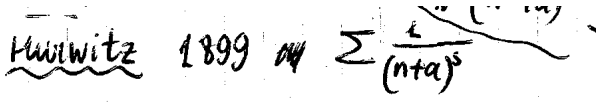
I. M. G. Kostya:

Доп: Hurwitz 1899 [overwritten] $\sum \frac{1}{(n+a)^s}$



Uncertain reading. The underlined abbreviation provisionally reads «Доп:».

Original source details: GELFAND-P001089



Uncertain reading. A short mark after “Hurwitz 1899” is heavily overwritten.

Original source details: GELFAND-P001089

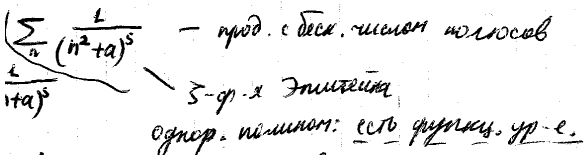


Diagram and typed description.

$\sum_n \frac{1}{(n^2+a)^s}$ — continuation with infinitely many poles

ζ-function of Epstein

Original source details: GELFAND-P001089

homogeneous polynomial: there is a functional equation.

$$\begin{aligned} \frac{1}{(n^2+a)^s} &= \frac{1}{\left(n^2(1+\frac{a}{n^2})\right)^s} = \left(\frac{1}{n^2} - \frac{a}{n^4} + \frac{a^2}{n^6} + \cdots\right)^s \\ &= \frac{1}{n^{2s}}\left(-\frac{as}{n^{2s+2}} + \cdots\right) \end{aligned}$$

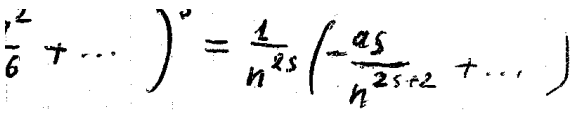
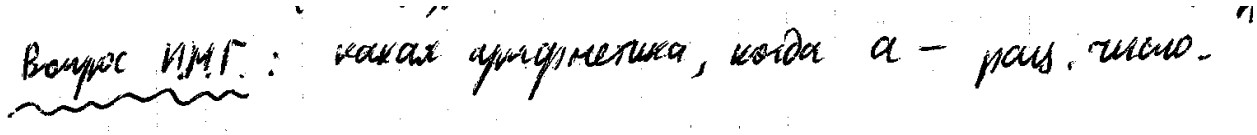


Diagram and typed description. The final equality in the manuscript. No separate 1 is visible before the minus.

Original source details: GELFAND-P001089

I. M. G.’s question: what arithmetic is involved when a — a rational number.



Original source details: GELFAND-P001089

Diagram and typed description. The question about arithmetic for a rational parameter.

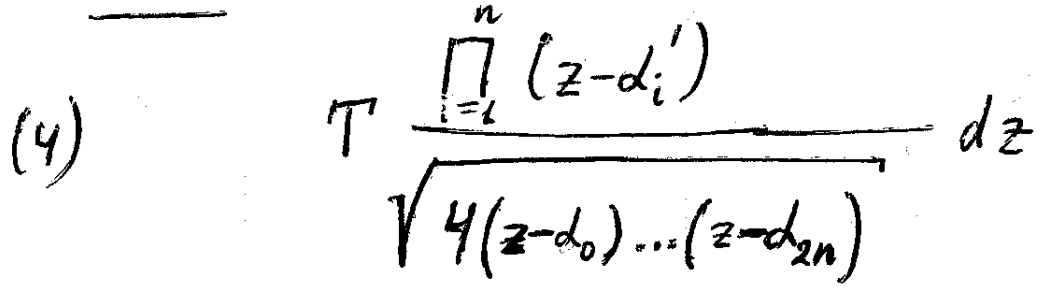
GELFAND-P001090

[source number: 16]

M. W.

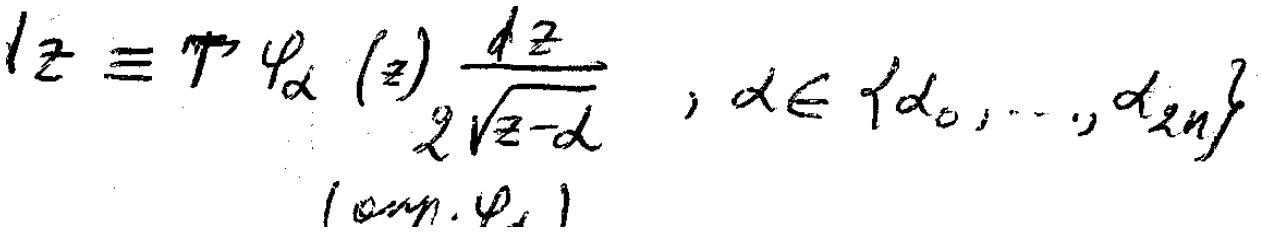
$$T\frac{\prod_{i=1}^{n}(z-\alpha_i')}{\sqrt{4(z-\alpha_0)\cdots(z-\alpha_{2n})}}\,dz \equiv T\varphi_\alpha(z)\frac{dz}{2\sqrt{z-\alpha}}, \qquad \alpha\in\{\alpha_0,\dots,\alpha_{2n}\} \tag{4}$$

(definition of φ_α)



Original source details: GELFAND-P001090

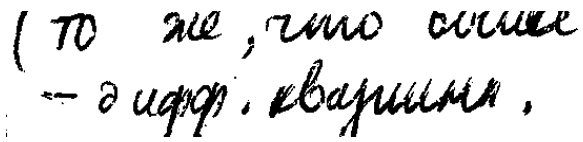
Diagram and typed description. The left-hand side of (4) with its product in the numerator.



Original source details: GELFAND-P001090

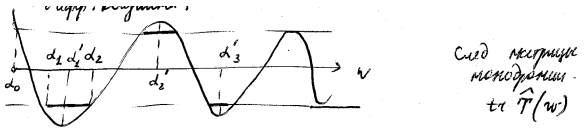
Diagram and typed description. The right-hand side of (4) and the choice of α.

(the same as above in Krichever)
— quasimomentum differential,



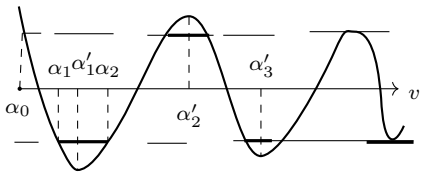
Original source details: GELFAND-P001090

Diagram and typed description. The abbreviated name of the differential.



Original source details: GELFAND-P001090

Diagram and typed description.



Trace of the monodromy matrix. $\operatorname{tr}\widehat{T}(w)$

The axis label beside the arrow resembles v or ν.

α'_j — min and max points of $\operatorname{tr} \widehat{T}(w)$.



Original source details: GELFAND-P001090

Diagram and typed description. The note on the minimum and maximum points.

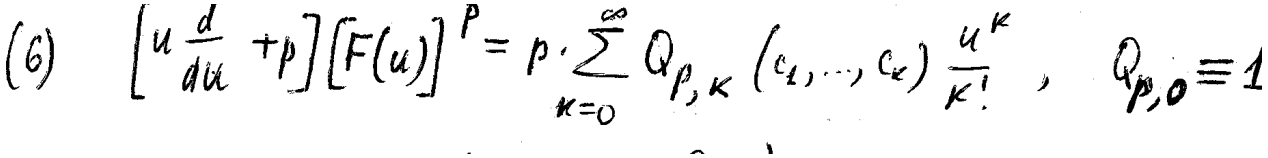
Introduce a universal family of polynomials:

$$F(u) = 1 + \sum_{i=1}^{\infty} c_i \frac{u^i}{i!} \tag{5}$$

and then

$$\left[u\frac{d}{du} + p\right][F(u)]^p = p \cdot \sum_{k=0}^{\infty} Q_{p,k}(c_1, \ldots, c_k)\frac{u^k}{k!}, \qquad Q_{p,0} \equiv 1. \tag{6}$$

(definition of $Q_{p,k}$)

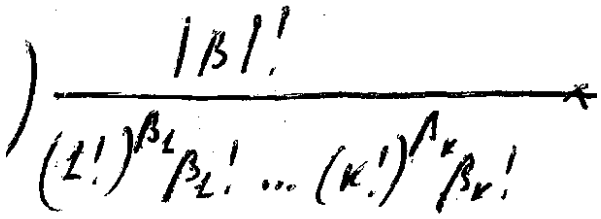


Original source details: GELFAND-P001090

Diagram and typed description. Defining relation (6).

$$\begin{aligned}\widehat{Q}_{p,k} &\equiv \frac{p+2k}{p+k} Q_{p,k}(c_1, \ldots, c_k) \\ &= \frac{p+2k}{p} k! \sum_{\substack{|\beta| \le \min(k,p) \\ \sum_{i=1}^{k} i\beta_i = k}} \binom{p}{|\beta|} \frac{|\beta|!}{(1!)^{\beta_1}\beta_1! \cdots (k!)^{\Lambda_k}\beta_k!} \\ &\quad \times c_1^{\beta_1} \cdots c_k^{\beta_k}\end{aligned} \tag{7}$$

Uncertain reading. The last exponent in the denominator resembles Λ_k.



Original source details: GELFAND-P001090

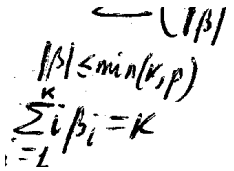


Diagram and typed description. The conditions below the sum.

Original source details: GELFAND-P001090

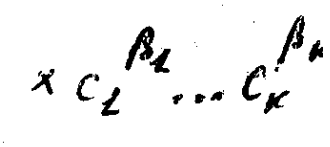


Diagram and typed description. The factors in c_1 and c_k.

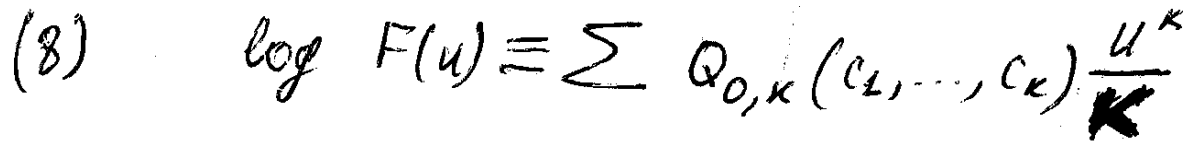
Original source details: GELFAND-P001090

$$\sum [F(u)]^p z^p = \frac{1}{1 - zF(u)} = \Phi(u, z)$$

and we need to take:

$$\left(u\frac{\partial}{\partial u} + z\frac{\partial}{\partial z} \right) \Phi(u, z).$$

$$\log F(u) \equiv \sum Q_{0,k}(c_1, \ldots, c_k) \frac{u^k}{k} \tag{8}$$



Original source details: GELFAND-P001090

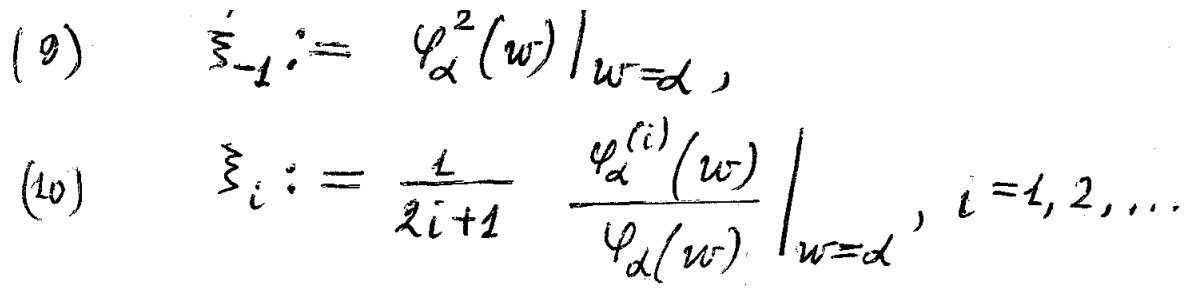
Uncertain reading. The denominator is overwritten and reads k. The earlier strokes are unclear.

(definition of $Q_{0,k}$)

Construct:

$$\xi_{-1} := \varphi_\alpha^2(w)\big|_{w=\alpha}, \tag{9}$$

$$\xi_i := \frac{1}{2i+1} \left. \frac{\varphi_\alpha^{(i)}(w)}{\varphi_\alpha(w)} \right|_{w=\alpha}, \qquad i = 1, 2, \ldots \tag{10}$$

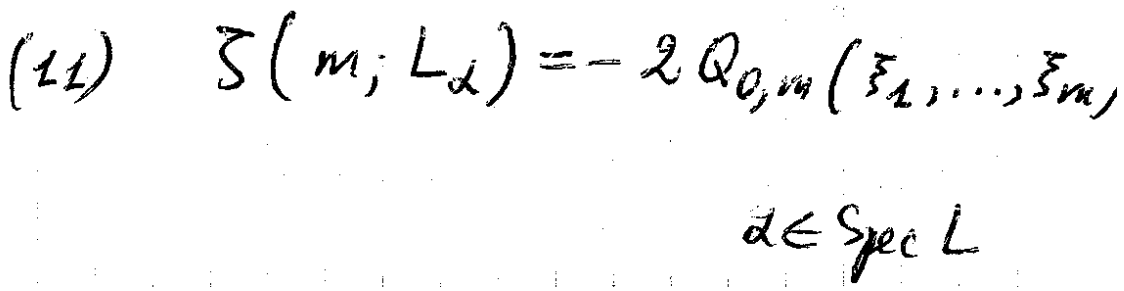


Original source details: GELFAND-P001090

Diagram and typed description. The evaluations and the factor $(2i+1)^{-1}$ in formulas (9) and (10).

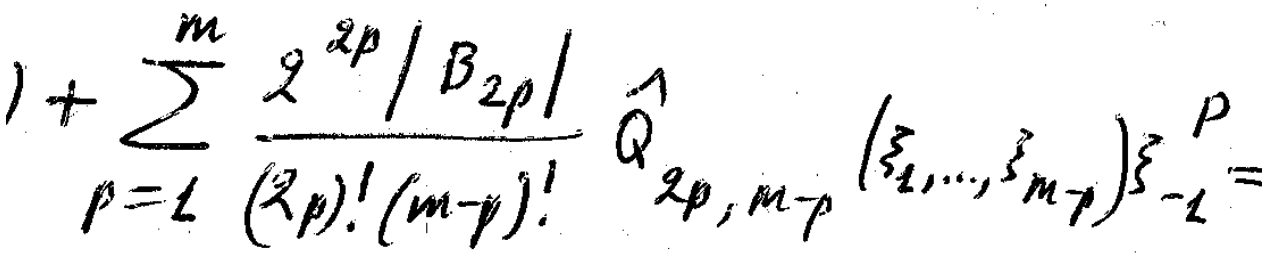
$$\begin{aligned}\zeta(m;L_\alpha) &= -2Q_{0,m}(\xi_1,\dots,\xi_m)\\ &\quad+\sum_{p=1}^{m}\frac{2^{2p}|B_{2p}|}{(2p)!(m-p)!}\widehat{Q}_{2p,m-p}(\xi_1,\dots,\xi_{m-p})\xi_{-1}^{p}\\ &= -2Q_{0,m}(\xi_1,\dots,\xi_m)\\ &\quad+\sum_{p=1}^{m}\frac{\zeta_0(p)}{(m-p)!}\widehat{Q}_{2p,m-p}(\xi_1,\dots,\xi_{m-p})\xi_{-1}^{p}\end{aligned}\tag{11}$$

[Below the first line: $\alpha \in \operatorname{Spec} L$.]



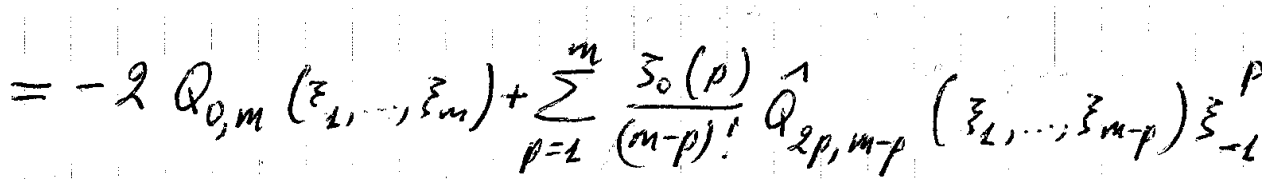
Original source details: GELFAND-P001091

Diagram and typed description. The start of equation (11) and the spectrum annotation.



Original source details: GELFAND-P001091

Diagram and typed description. The Bernoulli number, power of two, and factorials in the first line.



Original source details: GELFAND-P001091

Diagram and typed description. The second line of (11) with $\zeta_0(p)$.

ζ_0 — the ζ-function of $-\frac{d^2}{dx^2}$ on $[0,T]$.

$$\zeta_0(s) = \operatorname{Tr}\left(-\frac{d^2}{dx^2}, [0,T]\right)^{-s}$$

By ζ — the Riemann ζ-function

$$T = 2\pi, \qquad \zeta_0(s) = 2\zeta_{\text{Riemann}}(2s).$$

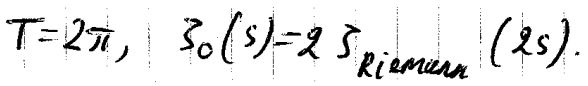


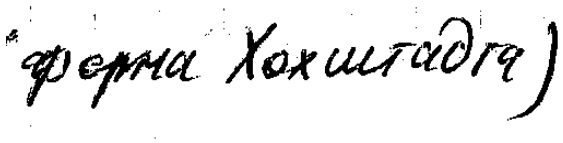
Diagram and typed description. The period is written as $T = 2\pi$.

Original source details: GELFAND-P001091

0-gap $\Rightarrow$ 1 term survives.

— · —

$u(x) \equiv u_0 = \text{const} \Rightarrow$ the formula becomes trivial
(Hochstadt form)



Uncertain reading. The first word is provisionally read as “form.”

Original source details: GELFAND-P001091

$$\frac{T\,dz}{\sqrt{4(z-u_0)}}$$

$$\zeta(m; L_{u_0}) = \frac{2^{2m}|B_{2m}|}{(2m)!}$$

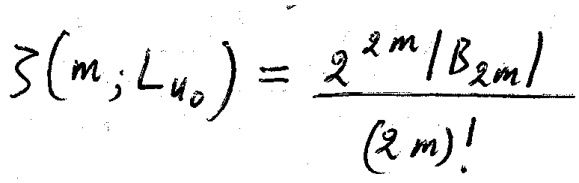


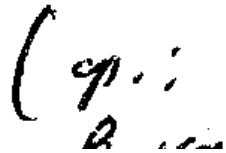
Diagram and typed description. The formula for a constant potential.

Original source details: GELFAND-P001091

1-gap

$$\omega\frac{z-\eta/\omega}{\sqrt{4z^3-g_2z+g_3}}$$

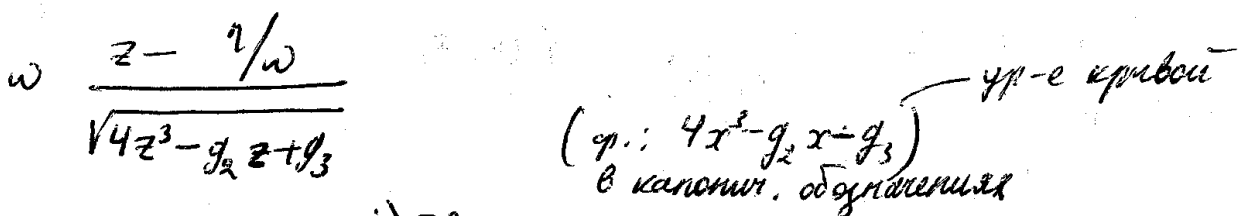
(f.: $4x^3 - g_2x - g_3$) [at the right: «equation of the curve».]



Uncertain reading. The mark before the colon appears to be the abbreviation “f.”.

Original source details: GELFAND-P001091

in canonical notation



Original source details: GELFAND-P001091

Diagram and typed description. The radicand has $+g_3$, and the formula beside it has $-g_3$.

$$\underline{u = 2\wp(x)}$$

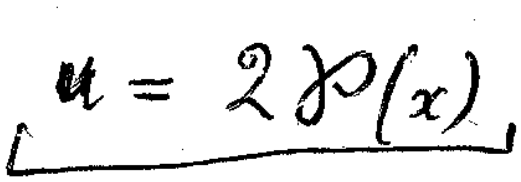


Diagram and typed description. The square-ended bracket below the potential.

Original source details: GELFAND-P001091

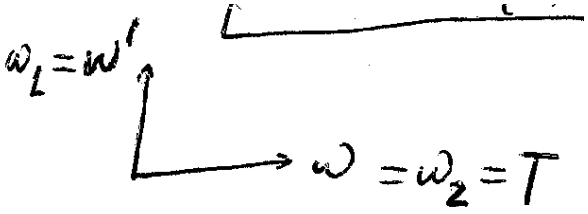
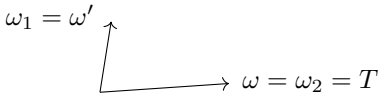


Diagram and typed description.



Original source details: GELFAND-P001091

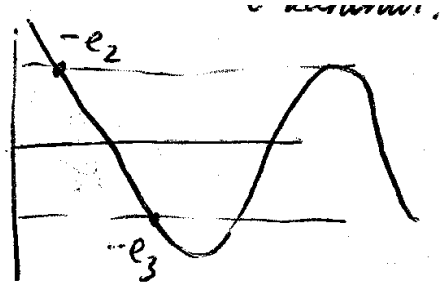
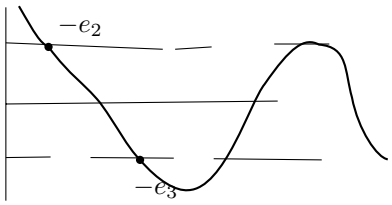


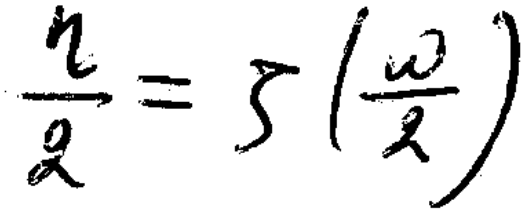
Diagram and typed description.



Original source details: GELFAND-P001091

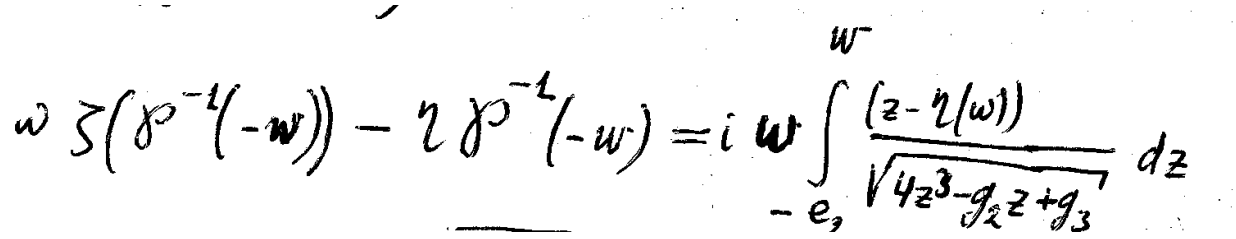
$$\frac{\eta}{2} = \zeta\left(\frac{\omega}{2}\right)$$

Diagram and typed description. The equality with half-period $\omega/2$.



Original source details: GELFAND-P001091

$$\omega\zeta\big(\wp^{-1}(-w)\big) - \eta\wp^{-1}(-w) = i\omega\int_{-e_2}^{w} \frac{\big(z-\eta(\omega)\big)}{\sqrt{4z^3 - g_2 z + g_3}}\,dz$$



Original source details: GELFAND-P001091

Uncertain reading. The numerator is provisionally read as $z - \eta(\omega)$. Two parentheses are visible after ω.

—·—

GELFAND-P001092

[source number: 17]

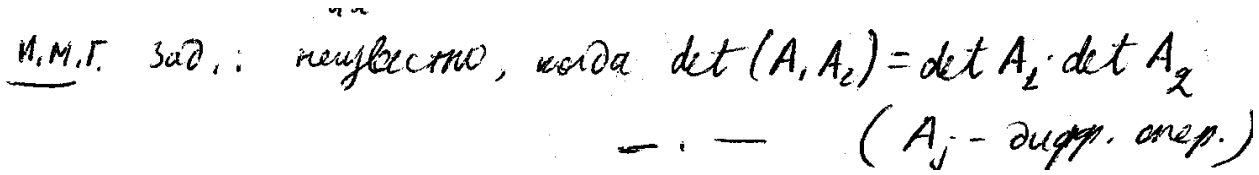
On the Riemann ζ function.

$$L_w \equiv -\frac{d^2}{dx^2} - w$$

I.M.G. problem: it is unknown when

$$\det(A_1 A_2) = \det A_1 \cdot \det A_2$$

—·— (A_j — differential op.)



Original source details: GELFAND-P001092

Uncertain reading. The first subscript on the right is read as 1. The handwriting also allows 2.

$$\zeta(s) = \sum \frac{1}{n^s}, \qquad \zeta(2k) = \frac{|B_{2k}|2^{2k}}{(2k)!}$$

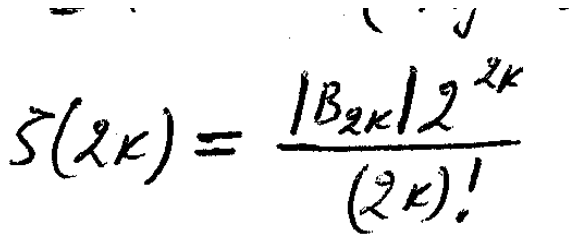


Diagram and typed description. The written value at the even argument $\zeta(2k)$.

Original source details: GELFAND-P001092

Apéry: $\sum \frac{1}{n^3}$ is irrational.

—·—

$$\sum n^2 = 0, \qquad \underbrace{\sum n^3 = -B_2}$$

a puzzle that it
[can be calculated?] and is $\neq 0$.

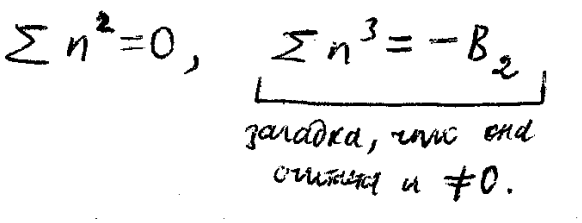


Diagram and typed description. The bracket and note under the sum of cubes.

Original source details: GELFAND-P001092

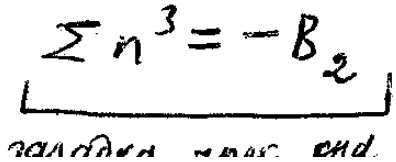
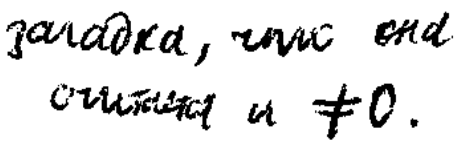


Diagram and typed description. Subscript 2 on B in the sum of cubes.

Original source details: GELFAND-P001092



Uncertain reading. The word before “and” is provisionally read as “can be calculated.”

Original source details: GELFAND-P001092

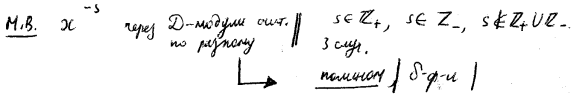


Diagram and typed description.

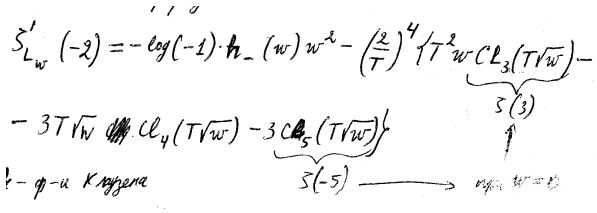
M.V. x^{-s}
calc. via D-modules
in different ways
$\Big\|$
$s \in \mathbb{Z}_+$, $s \in \mathbb{Z}_-$,
$s \notin \mathbb{Z}_+ \cup \mathbb{Z}_-$.
3 cases

→ polynomial | δ-functions |

Original source details: GELFAND-P001092

ζ' corresponds to the logarithm.

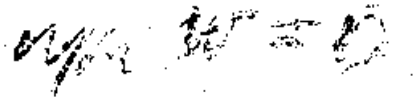
$$\zeta'_{L_w}(-2) = -\log(-1)\cdot h_-(w)\,w^2 - \left(\frac{2}{T}\right)^4 \Big\{ T^2 w \underbrace{\mathrm{Cl}_3(T\sqrt{w})}_{\zeta(3)} - 3T\sqrt{w}\,[\text{cancelled}]\ \mathrm{Cl}_4(T\sqrt{w}) - 3\underbrace{\mathrm{Cl}_5(T\sqrt{w})}_{\zeta(-5)} \Big\}$$



Original source details: GELFAND-P001092

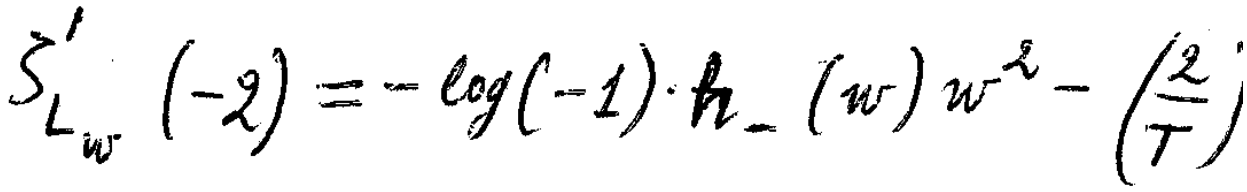
Diagram and typed description.

$$\zeta(-5) \longrightarrow \underset{\uparrow}{\zeta(3)}\ [\text{at } w = 0?]$$



Original source details: GELFAND-P001092

Uncertain reading. The faint annotation provisionally reads “at $w = 0$.”



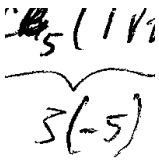
Original source details: GELFAND-P001092

Diagram and typed description. The opening of the formula. The manuscript repeats the minus sign at the line break.



Original source details: GELFAND-P001092

Uncertain reading. Cancelled marks before Cl_4. The start resembles Cl.



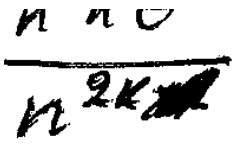
Original source details: GELFAND-P001092

Diagram and typed description. The lower brace is labelled $\zeta(-5)$.

Cl — Clausen functions

$$\mathrm{Cl}_{2k+1}(\theta) = \sum \frac{\cos n\theta}{n^{2k+1}}$$

$$\mathrm{Cl}_{2k}(\theta) = \sum \frac{\sin n\theta}{n^{2k}}$$



Uncertain reading. The cancelled marks after $2k$ in the exponent may read $+1$.

Original source details: GELFAND-P001092

GELFAND-P001093

21/III-83 Gelfand Sem.
S. P. Novikov

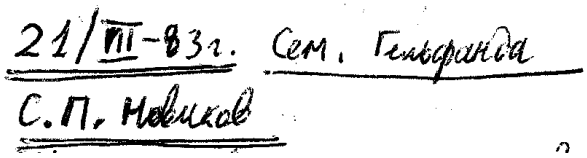


Diagram and typed description. Date, seminar name and speaker.

Original source details: GELFAND-P001093

What does the calculus of variations study?

multivalued functionals are important $\longrightarrow$ Dirac monopole (see Novikov's survey in Russian Mathematical Surveys)

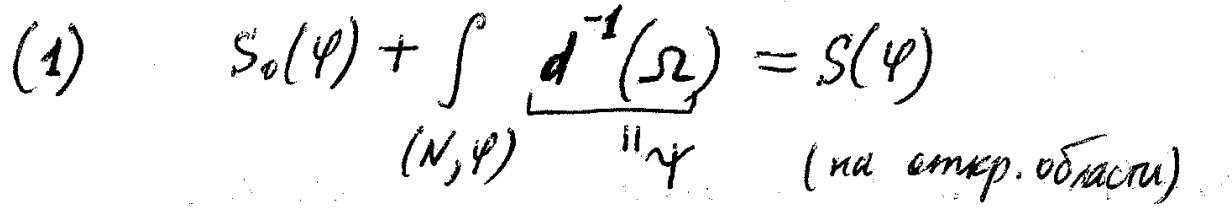
$$\begin{array}{ccc} x \in N^q & \varphi \in M^n & n > q \\ (x^1, \ldots, x^q) & (\varphi^1, \ldots, \varphi^n) & \\ \text{coord.} & & \end{array}$$

$$\varphi(x) : N \longrightarrow M$$

$S_0(\varphi)$ — regular functional

$$(1) \qquad S_0(\varphi) + \int_{(N,\varphi)} \underbrace{d^{-1}(\Omega)}_{\| \gamma} = S(\varphi)$$

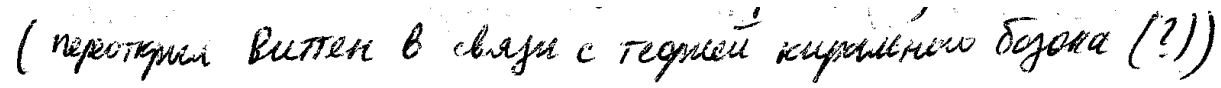
(on an open domain)



Original source details: GELFAND-P001093

Diagram and typed description. Formula (1), the square-ended bracket under $d^{-1}(\Omega)$ and the label γ.

(rediscovered by Witten in connection with the theory of the chiral boson (?))



Original source details: GELFAND-P001093

Diagram and typed description. The note on Witten with its parenthesized question mark.

Ω — $(q+1)$-form, $\quad d\Omega = 0$

Chiral fields. — this situation arises there.

$\varphi \in \mathcal{F}$ — function space.

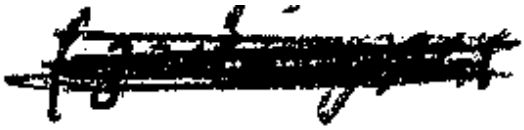
(2) Theorem. The following formula [1?] defines a form (well-defined)



Uncertain reading. The mark after "formula" is overwritten. Possible readings are 1 and A.

Original source details: GELFAND-P001093

δS — a 1-form on the function space (closed)



Undeciphered passage. An illegible entry is cancelled below "function space."

Original source details: GELFAND-P001093

[Volkov?]: $S^1 \to S^2$ (nontriv. case)



Uncertain reading. The word before the colon is provisionally read as "Volkov."

Original source details: GELFAND-P001093

Ω — generalized external field (electromagnetic)

$$M^n = S^2, \qquad \int_{S^2} \Omega \neq 0 \qquad \Rightarrow \qquad \text{Dirac monopole}$$



Uncertain reading. The cancelled notation above S^2 may read M^2.

Original source details: GELFAND-P001093

Multivalued function $\quad$ difference — (const on intersections)

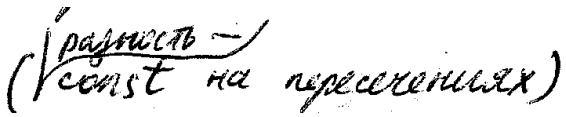


Diagram and typed description. The underlined insertion "difference —" above const.

Original source details: GELFAND-P001093

~~δS closed 1-form~~

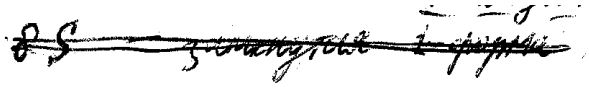


Diagram and typed description. The cancelled line at the bottom of the frame.

Original source details: GELFAND-P001093

GELFAND-P001094

$\exists$ a cover $M = \bigcup_\alpha U_\alpha$.

$$\begin{aligned} &1) \quad \forall \varphi,\ \exists \alpha, \quad \varphi(N) \subset U_\alpha. \\ &2) \quad \Omega = d\psi_\alpha \quad \text{on } U_\alpha. \end{aligned}$$

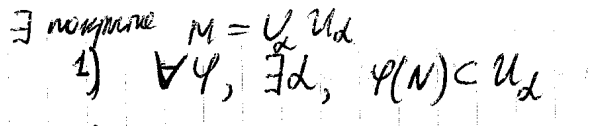


Diagram and typed description. The cover and the first condition on φ.

Original source details: GELFAND-P001094

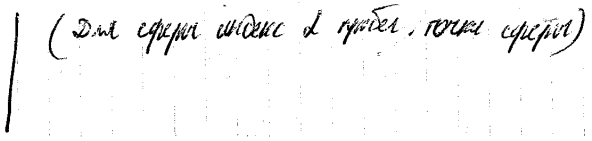


Diagram and typed description.

> (For the sphere, the index α runs over its points)

Original source details: GELFAND-P001094

$$W_\alpha \subset \mathcal{F}, \qquad \varphi(N) \subset U_\alpha.$$

$\{W_\alpha\}$ — a cover of $\mathcal{F}$.

$$S^\alpha = S_0 + \int_N \psi_\alpha.$$

Then $S^\alpha - S^\beta =$ const on $W_\alpha \cap W_\beta$, since

$$S^\alpha - S^\beta = \int_{(N,\varphi)} (\psi_\alpha - \psi_\beta)$$

depends only on the homotopy class, since $d(\psi_\alpha - \psi_\beta) = 0$.



Uncertain reading. A short mark in the left margin may be the numeral 1.

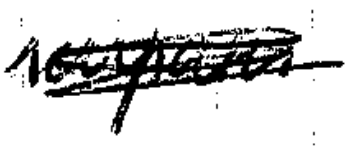
Original source details: GELFAND-P001094

Physical requirement:

$\exp(iS)$ — a single-valued function.

(Feynman amplitude)

Moral: [~~every?~~] the action need not be understood as single-valued.

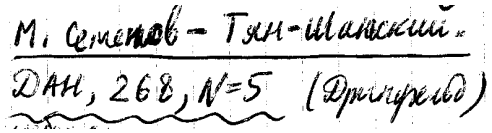


Uncertain reading. The cancelled word may read “every.”

Original source details: GELFAND-P001094

M. Semenov-Tian-Shansky.

DAN, 268, $N = 5$ (Drinfeld)



Original source details: GELFAND-P001094

Diagram and typed description. The speaker's name and the Drinfeld reference, with the issue number as written.

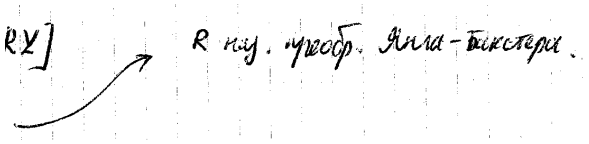
Lie bialgebra.

$\mathfrak{g}$ — Lie alg., $R \in \operatorname{End} \mathfrak{g}$.

$$[X, Y]_R = [RX, Y] + [X, RY]$$

also a Lie algebra

$\mathfrak{g}$, $\mathfrak{g}_R$.

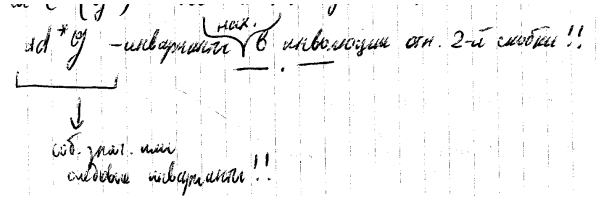


Original source details: GELFAND-P001094

Diagram and typed description.

R is called the
Yang–Baxter transformation.

On $C^\infty(\mathfrak{g}^*)$ — two Poisson brackets.



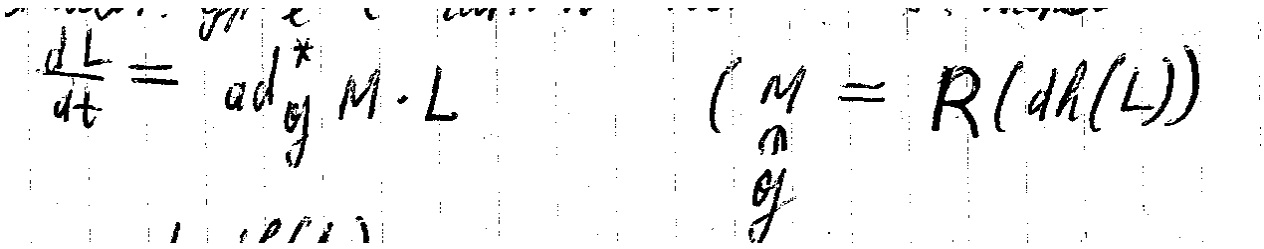
Original source details: GELFAND-P001094

Diagram and typed description.

$\underbrace{\text{ad}^* \mathfrak{g}}$-invariants $\overset{\text{are}}{\vee}$ in involution
w.r.t. the 2nd bracket!!
↓
eigenvalues or
trace invariants!!

Theorem. let h — $\text{ad}^* \mathfrak{g}$-invariant function on $\mathfrak{g}^*$.
then 1) the Hamiltonian eq. with Hamiltonian h has generalized Lax form

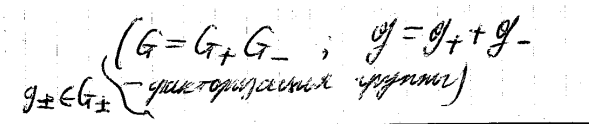
$$\frac{dL}{dt} = \text{ad}^*_{\mathfrak{g}} M \cdot L \qquad \left(\underset{\cup}{M}_{\mathfrak{g}} = R(dh(L))\right.$$



Original source details: GELFAND-P001094

Diagram and typed description. The vertical annotation $\in \mathfrak{g}$ below M.

$$2) \qquad \exp t\, dh(L) = \\ = g_+(t) \cdot g_-^{-1}(t)$$



Original source details: GELFAND-P001094

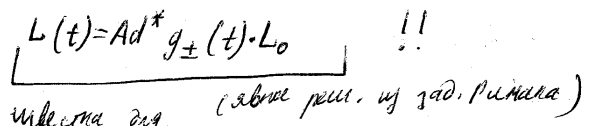
Diagram and typed description.

$g_\pm \in G_\pm$ { $(G = G_+G_-\ ;\ \mathfrak{g} = \mathfrak{g}_+ + \mathfrak{g}_-$ — group factorization)

GELFAND-P001095

[source number: 32]

Then



Original source details: GELFAND-P001095

Diagram and typed description.

$\underbrace{L(t) = \mathrm{Ad}^*\, g_\pm(t) \cdot L_0}$!!

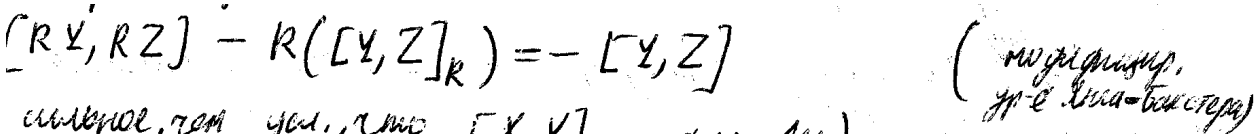
known for (explicit sol. from the Riemann problem)

$R = P_+ - P_-$.

Here: R-matrices satisfying the condition

$$[RY, RZ] - R([Y, Z]_R) = -[Y, Z]$$

(modified Yang–Baxter eq.)



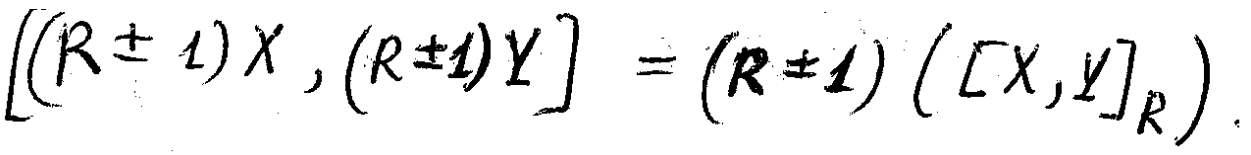
Original source details: GELFAND-P001095

Diagram and typed description. The equation and its right-hand note.

(stronger than the condition that $[X, Y]_R$ — is a Lie alg.).

— · —

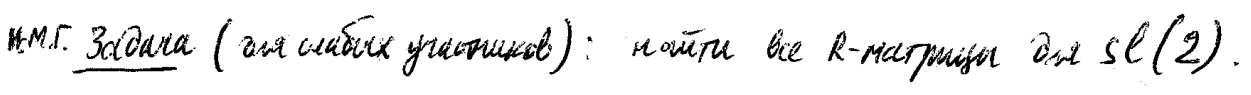
$$[(R \pm 1)X, (R \pm 1)Y] = (R \pm 1)([X, Y]_R).$$



Original source details: GELFAND-P001095

Diagram and typed description. The identity for $R \pm 1$.

I.M.G. Problem (for weaker participants): find all R-matrices for $sl(2)$.



Original source details: GELFAND-P001095

Diagram and typed description. The wording of the problem for $sl(2)$.

— · —

$$\mathfrak{g}_\pm = \operatorname{Im}(R \pm 1)$$

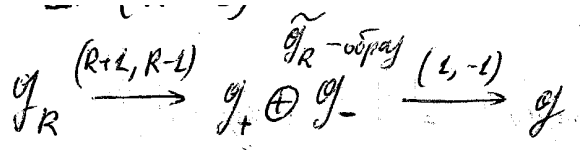


Diagram and typed description.

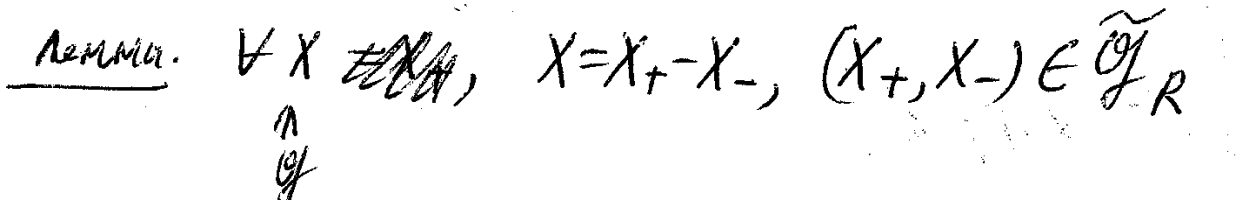
$\widetilde{\mathfrak{g}}_R$ — image

$$\mathfrak{g}_R \xrightarrow{(R+1,\,R-1)} \mathfrak{g}_+ \oplus \mathfrak{g}_- \xrightarrow{(1,\,-1)} \mathfrak{g}$$

Original source details: GELFAND-P001095

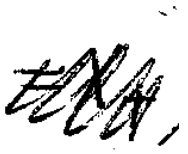
Lemma.

$$\forall\, \underset{\mathfrak{g}}{\cup} X \text{ [cancelled]}, \qquad X = X_+ - X_-, \qquad (X_+, X_-) \in \widetilde{\mathfrak{g}}_R.$$



Original source details: GELFAND-P001095

Diagram and typed description. The vertical annotation $\in \mathfrak{g}$ below X.



Uncertain reading. The cancelled notation after X may read $\in \mathfrak{g}$.

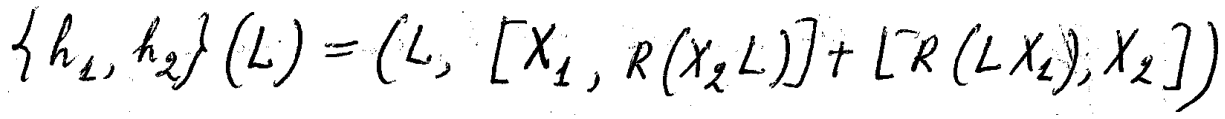
Original source details: GELFAND-P001095

— · —

2nd G.–D. bracket (Gelfand–Dickey)

$\mathfrak{g}$ — Lie alg. with assoc. multiplication and an (invar.) scalar product

$$\{h_1, h_2\}(L) = \big(L,\, [X_1, R(X_2 L)] + [R(L X_1), X_2]\big)$$



Original source details: GELFAND-P001095

Diagram and typed description. The order of the factors $X_2 L$ and $L X_1$ in the two terms.

$$X_i = \operatorname{grad} h_i$$

There are hierarchies of Hamiltonians in Lenard's sense.

— · —

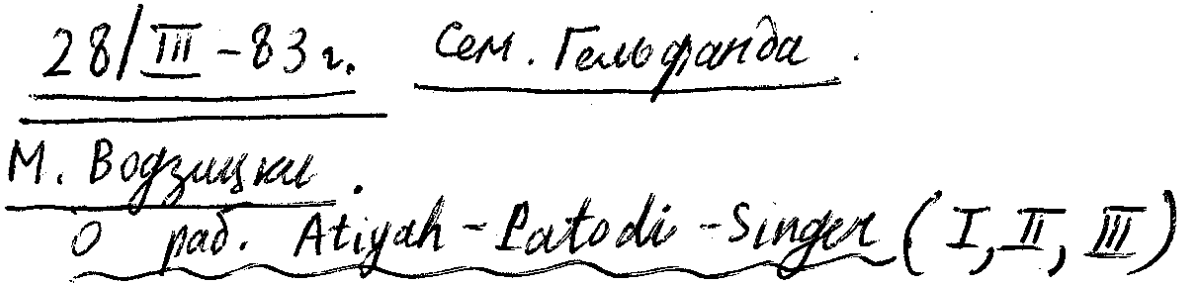
GELFAND-P001096

[source number: 39]

28/III-83 Gelfand Sem.

M. Wodzicki.

On the work of Atiyah–Patodi–Singer (I, II, III)



Original source details: GELFAND-P001096

Diagram and typed description. Date, speaker and reference to Atiyah, Patodi and Singer.

I. X [cancelled], $\partial X = Y$.



Original source details: GELFAND-P001096

Undeciphered passage. The cancelled notation after the first X is illegible.

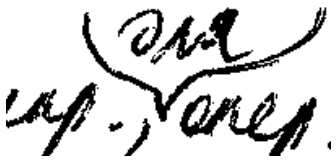
D — 1st-order system on X (elliptic), near the boundary

$$D = \frac{\partial}{\partial u} + A, \qquad u \text{ — normal param.}$$

A — op. on Y.

A is self-adjoint.

Boundary cond. do not always exist (e.g., [inserted: for] the signature op. it is impossible). But nonlocal ones exist.



Original source details: GELFAND-P001096

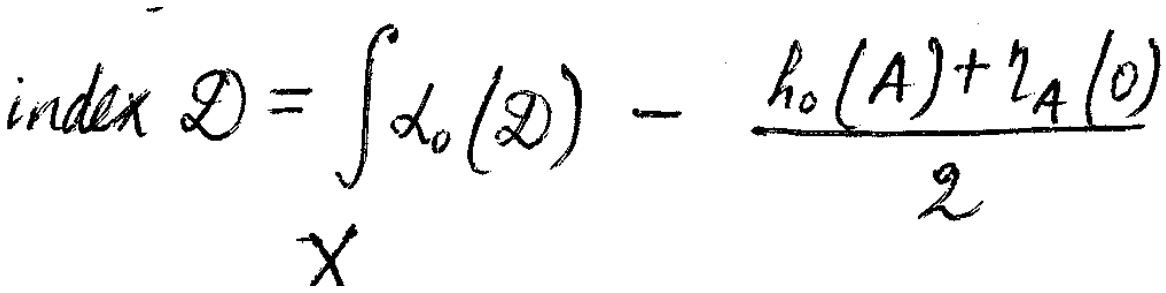
Diagram and typed description. The insertion “for” before “signature op.”

$$(1) \qquad \begin{cases} Du = f \\ P(u|_Y) = 0 \end{cases}$$

(2) P — projection onto eigenvalues ≥ 0 of the operator A.

(3) Theorem. Problem (1) with P as in (2) has finite index and

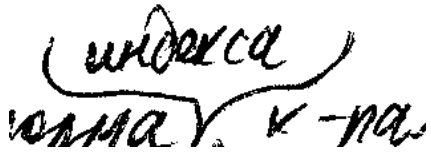
$$\operatorname{index} D = \int_X \alpha_0(D) - \frac{h_0(A) + \eta_A(0)}{2}$$



Original source details: GELFAND-P001096

Diagram and typed description. The index formula with its boundary term.

$\alpha_0(D)$ — differential form [inserted: of the index], written using the Riemannian metric, which has product form near the boundary.



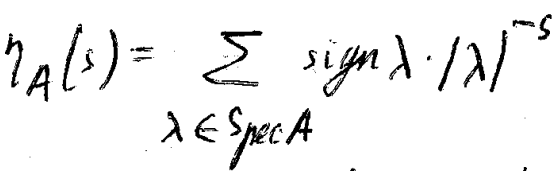
Original source details: GELFAND-P001096

Diagram and typed description. The insertion "of the index" above the line.

$$h_0(A) = \dim \ker A$$

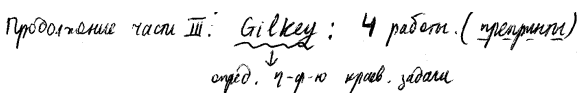
$\eta_A(0)$ — value at 0 of the analytic continuation of the function

$$\eta_A(s) = \sum_{\lambda \in \operatorname{Spec} A} \operatorname{sign} \lambda \cdot |\lambda|^{-s}$$



Original source details: GELFAND-P001096

Diagram and typed description. The summation index is written as $\lambda \in \operatorname{Spec} A$.



Original source details: GELFAND-P001096

Diagram and typed description.

Continuation of part III.

Gilkey: 4 papers. (preprints)

↓

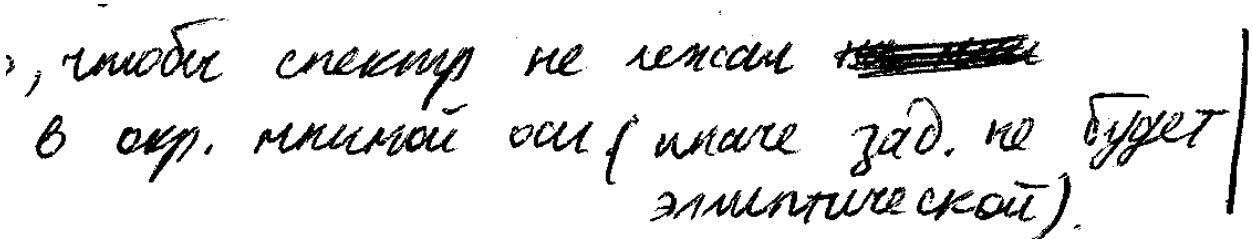
defines the η-function
of the boundary problem

In place of self-adjointness one can require that the spectrum not lie [cancelled] near the imaginary axis (otherwise the problem will not be elliptic).



Original source details: GELFAND-P001096

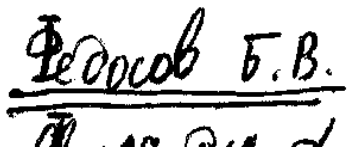
Undeciphered passage. The cancelled entry after “not lie” is illegible.



Original source details: GELFAND-P001096

Diagram and typed description. The vertical rule to the right of the condition on the spectrum.

B. V. Fedosov.



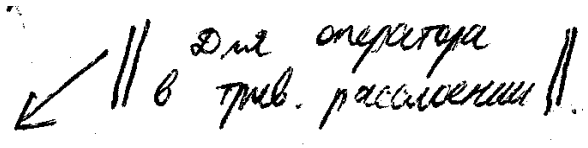
Original source details: GELFAND-P001096

Diagram and typed description. Speaker named before the formula for α_0.

Formula for α_0.

X — manifold (without boundary), SX — bundle of unit tangent vectors,

σ — symbol (function on SX). Using σ, construct a projector P defining the difference element (P — function on T^*X; $p(x,\xi)$)



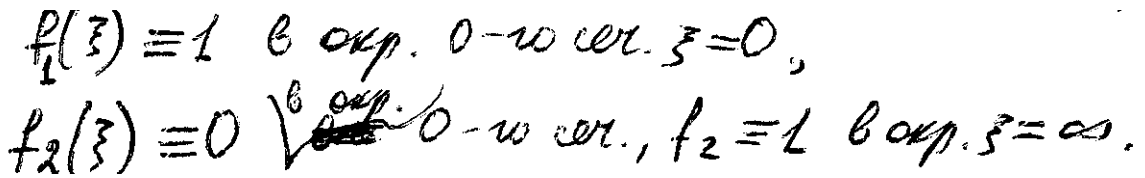
Original source details: GELFAND-P001096

Diagram and typed description.

For the operator
in a trivial bundle

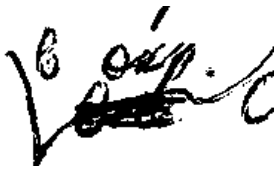
$f_1(\xi) \equiv 1$ near the zero section $\xi = 0$,

$f_2(\xi) \equiv 0$ [~~on?~~] [inserted: near] the zero section, $f_2 \equiv 1$ near $\xi = \infty$.



Original source details: GELFAND-P001096

Diagram and typed description. The conditions on f_1 and f_2 and the insertion “near.”



Original source details: GELFAND-P001096

Uncertain reading. The cancelled word may read “on.”

$f_1 + f_2 \equiv 1.$

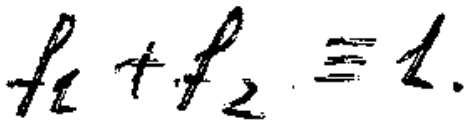


Diagram and typed description. The identity $f_1 + f_2 \equiv 1$ is written without exponents.

Original source details: GELFAND-P001096

$$p = \begin{pmatrix} f_1^2 & f_1 f_2 \sigma \\ f_1 f_2 \sigma^* & f_2^2 \end{pmatrix}$$

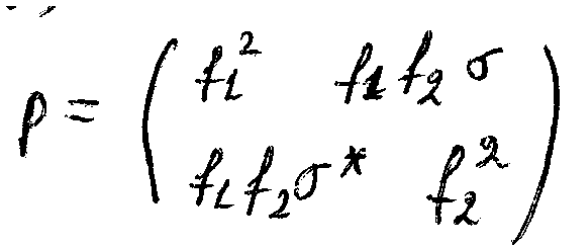


Diagram and typed description. The matrix p with squares on the diagonal.

Original source details: GELFAND-P001096

σ is extended homogeneously (degree 0)

GELFAND-P001097

Again: E_i — bundles over X;

$$A : \Gamma(E_1) \to \Gamma(E_2) \qquad \text{(differential operator)}$$

$p_1(x)$ $p_2(x)$ — projection functions defining the bundles (E_i — their images) (on X)

$\sigma = \sigma(x, \xi)$ — symbol of A (matrix function)

$\sigma p_1 \equiv p_2 \sigma$.

f_1, f_2 — cutoff functions (see above)

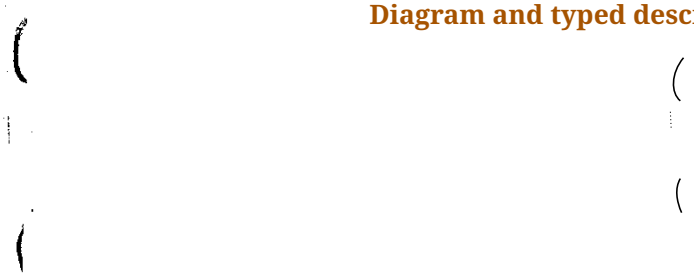


Diagram and typed description.

Original source details: GELFAND-P001097

$$P_1 = \begin{pmatrix} p_1 f_1^2 & \sigma f_1 f_2 \\ \sigma & \end{pmatrix}$$

— written explicitly and defined on all of T^*X.

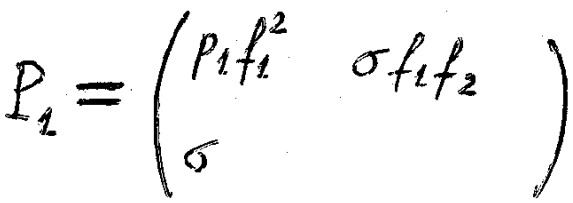


Diagram and typed description. The lower right cell is blank. Only σ is written in the lower left.

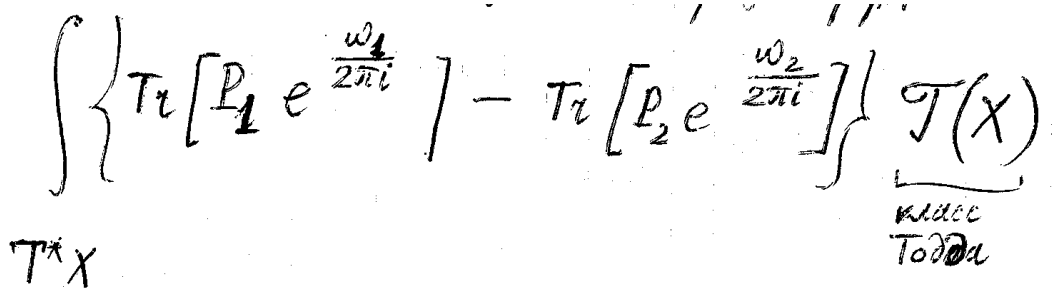
Original source details: GELFAND-P001097

$P_i = P_i(x, \xi)$

$P_1, P_2 \rightsquigarrow$ the curvature form is written from them

$$\omega_i = P_i\, dP_i\, dP_i \qquad \text{— matrix of 2-forms}$$

$$\int_{T^*X} \left\{ \mathrm{Tr}\left[P_1 e^{\frac{\omega_1}{2\pi i}}\right] - \mathrm{Tr}\left[P_2 e^{\frac{\omega_2}{2\pi i}}\right] \right\} \underbrace{\mathcal{T}(X)}_{\text{Todd class}}$$



Original source details: GELFAND-P001097

Diagram and typed description. The integral and the label below $\mathcal{T}(X)$.

$$a = f_1\sigma, \qquad \tau = f_2\sigma^{-1}$$

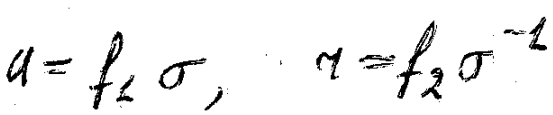
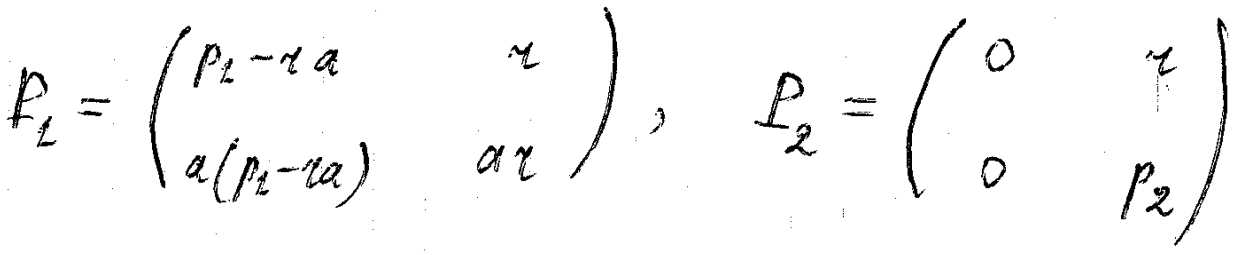


Diagram and typed description. The subscript on f in the definition of a reads 1.

Original source details: GELFAND-P001097

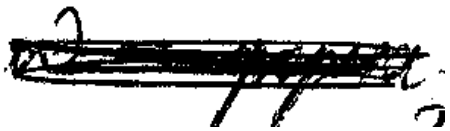
$$P_1 = \begin{pmatrix} p_1 - \tau a & \tau \\ a(p_1 - \tau a) & a\tau \end{pmatrix}, \qquad P_2 = \begin{pmatrix} 0 & \tau \\ 0 & p_2 \end{pmatrix}$$



Original source details: GELFAND-P001097

Diagram and typed description. The full entries of matrices P_1 and P_2.

Todd class:

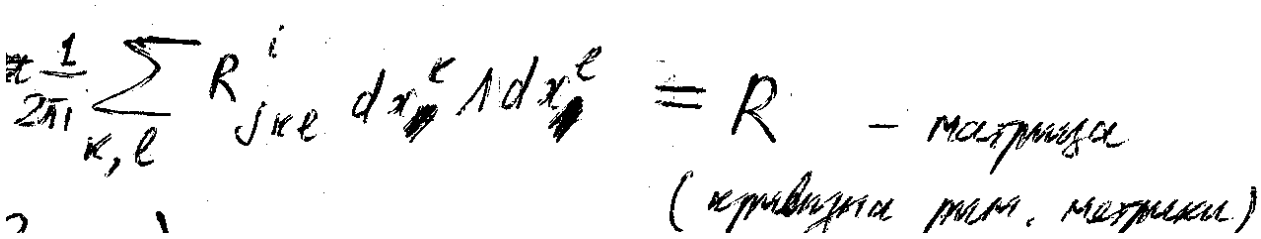


Undeciphered passage. A fragment before the coefficient is cancelled. It may read “2-form.”

Original source details: GELFAND-P001097

$$\frac{1}{2\pi i}\sum_{k,\ell} R^i_{jk\ell}\, dx^k \wedge dx^\ell = R$$

— matrix (curvature of the Riemannian metric)



Original source details: GELFAND-P001097

Diagram and typed description. The curvature matrix with the superscripts written above the differentials.



Original source details: GELFAND-P001097

Uncertain reading. An illegible subscript is cancelled beneath the first x. Its superscript is provisionally read as k.



Original source details: GELFAND-P001097

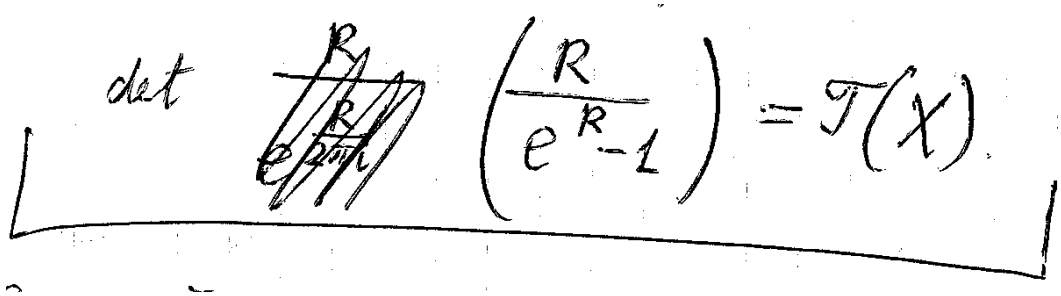
Uncertain reading. An illegible subscript is cancelled beneath the second x.

$$\det \frac{R}{e^{\frac{R}{2\pi i}}}\left(\frac{R}{e^R-1}\right) = \mathcal{T}(X).$$



Original source details: GELFAND-P001097

Diagram and typed description. The cancelled fraction after det.



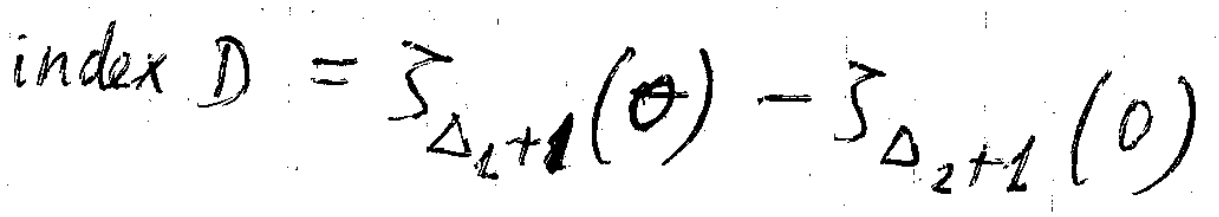
Original source details: GELFAND-P001097

Diagram and typed description. The square-ended bracket under the Todd class formula.

Wodzicki:

$$DD^* = \Delta_2, \qquad D^*D = \Delta_1.$$

$$\operatorname{index} D = \zeta_{\Delta_1+1}(\theta) - \zeta_{\Delta_2+1}(0)$$

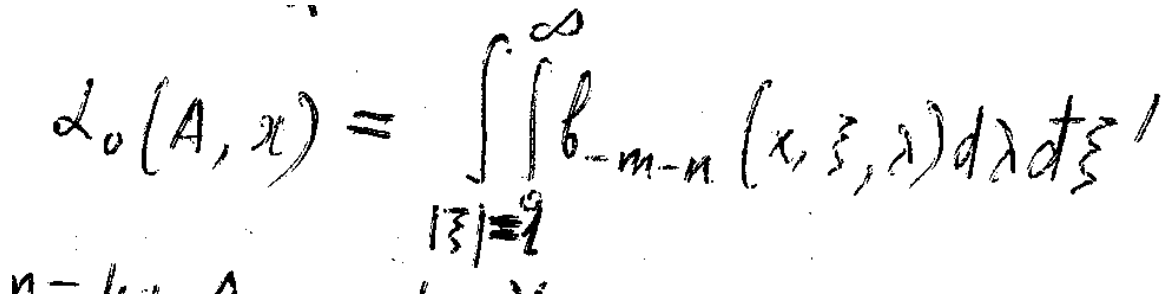


Original source details: GELFAND-P001097

Uncertain reading. The first argument resembles θ or a zero crossed by a horizontal stroke.

$$\zeta_A(0) = \int_X \alpha_0(A, x)$$

$$\alpha_0(A, x) = \int_{|\xi|=1} \int_0^\infty b_{-m-n}(x, \xi, \lambda)\, d\lambda\, d\xi'$$



Original source details: GELFAND-P001097

Diagram and typed description. The limits of integration and the subscript $-m-n$ on b.

$m = \deg A, \qquad n = \dim X$

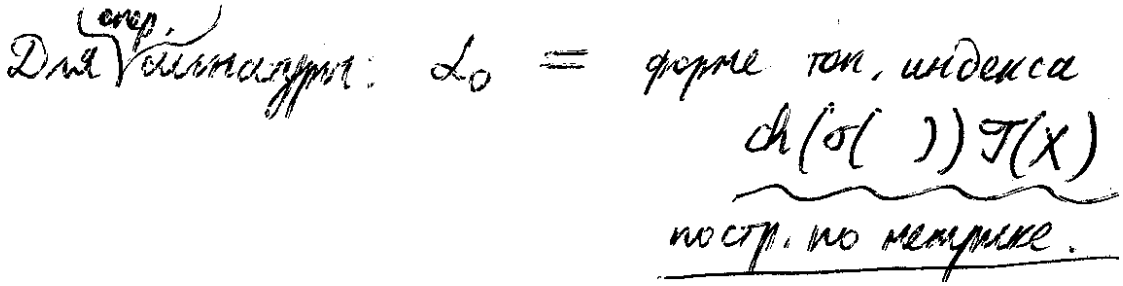
GELFAND-P001098

[source number: 40]

For the signature [inserted: op.]: $\alpha_0 =$ the form for the top. index

$$\underbrace{\operatorname{ch}(\sigma(\,))\mathcal{T}(X)}$$

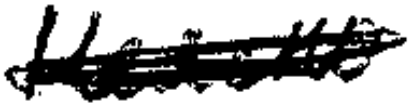
constructed from the metric.



Original source details: GELFAND-P001098

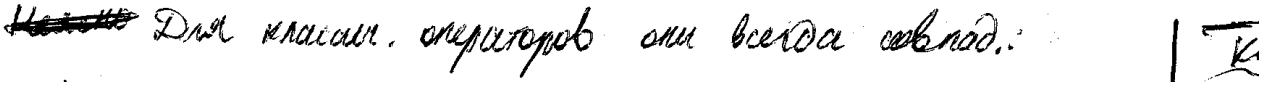
Diagram and typed description. The insertion “op.” and the empty parentheses after σ.

[cancelled] For classical operators they always coincide:



Undeciphered passage. A fragment is cancelled before “For.” Its beginning resembles the Russian «Не».

Original source details: GELFAND-P001098



Original source details: GELFAND-P001098

Diagram and typed description. The line on coincidence for classical operators.

a) signature op.

b) Dirac op.

There is a G-analogue, and there too the local formulas coincide.

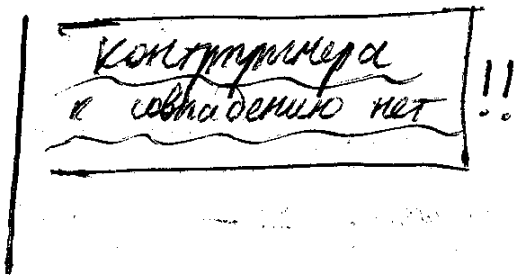


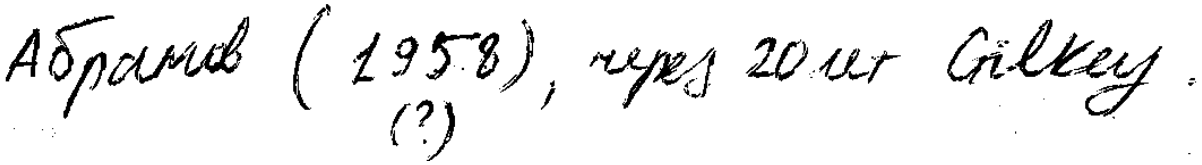
Diagram and typed description.

There is no counterexample to the formulas’ agreement !!

Original source details: GELFAND-P001098

I.M.G. problems: construct formal cohomology for the metric and the symbol

Only for the metric: Abramov (1958), 20 years later Gilkey.
(?)



Original source details: GELFAND-P001098

Diagram and typed description. A question mark is written below the date 1958.

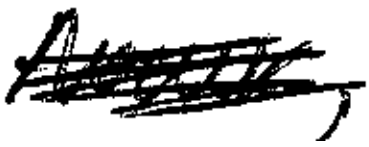
Are there other classes besides the index?

Perhaps this reduces to Gelfand–Fuks cohomology.

Or, the problem: construct local top. invariants.

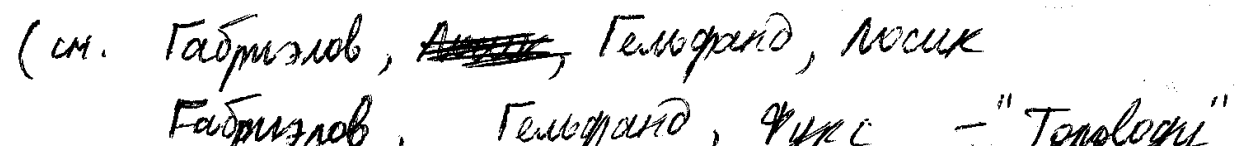
(see Gabrielov, ~~Losik?~~, Gelfand, Losik

Gabrielov, Gelfand, Fuks — «Topology»).



Uncertain reading. The cancelled name is provisionally read as “Losik.”

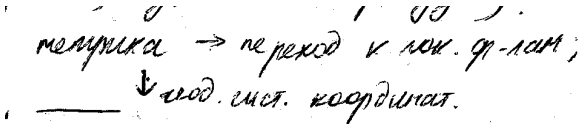
Original source details: GELFAND-P001098



Original source details: GELFAND-P001098

Diagram and typed description. The two groups of names and the reference to the journal Topology.

In the jet space without a metric.



Original source details: GELFAND-P001098

Diagram and typed description.

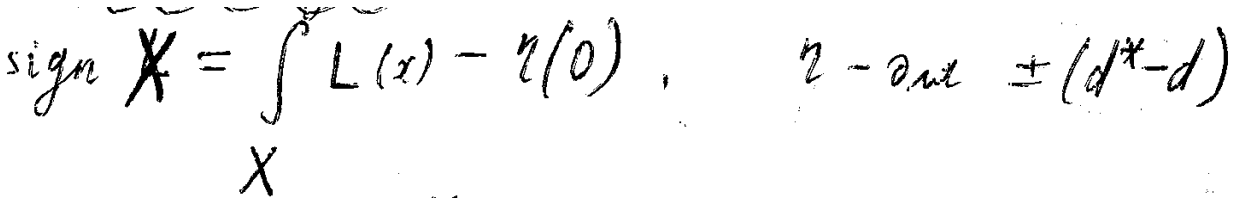
metric ⟶ passage to local formulas.
↓
geod. coordinate system.

— · —

Wodzicki

A. P. S. I. Special case:

$$\operatorname{sign} X = \int_X L(x) - \eta(0), \qquad \eta \text{ — for } \pm(d^* - d)$$

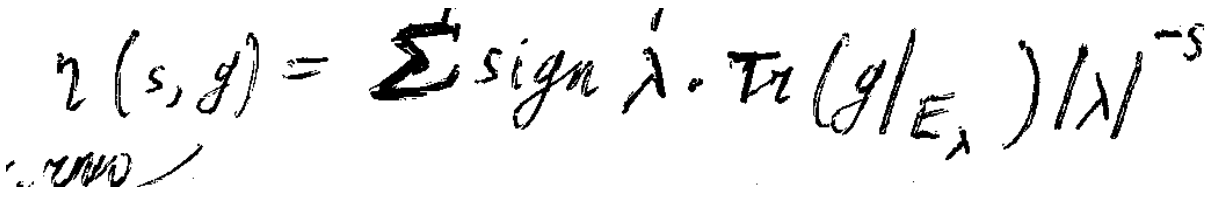


Original source details: GELFAND-P001098

Diagram and typed description. The signature formula and the operator written at its right.

II. — preparatory part and discussion of the functions:

$$\eta(s, g) = \sum \operatorname{sign} \lambda \cdot \operatorname{Tr}(g|_{E_\lambda})|\lambda|^{-s}$$



Original source details: GELFAND-P001098

Diagram and typed description. The trace of the restriction of g to E_λ.

III. [inserted: Prove that] $\eta(0)$ is finite for odd n. A problem like that of a higher index is posed.

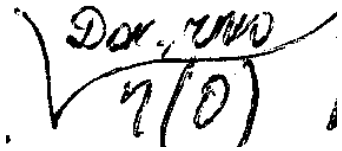


Diagram and typed description. The insertion "Prove that" before $\eta(0)$.

Original source details: GELFAND-P001098

The class is constructed

$$[\sigma_A] \in K^1(T^*X)$$

and from the repr. $\alpha : \pi_1 X \to U(n)$

one constructs

$$[\alpha] \in K^{-1}(T^*X, \mathbb{R}/\mathbb{Z})$$

$$K^0(T^*X, \mathbb{R}/\mathbb{Z}) \to \mathbb{R}/\mathbb{Z}$$

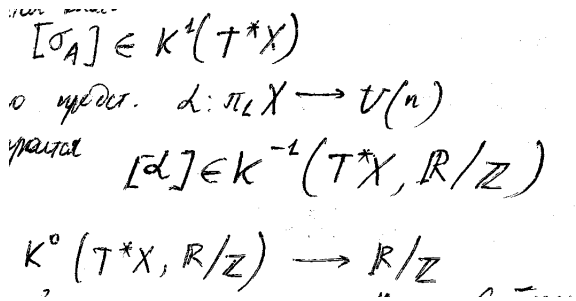


Diagram and typed description. The three K-theory degrees 1, -1 and 0.

Original source details: GELFAND-P001098

and so on.

Some Chern–Simons classes are obtained here.

GELFAND-P001099

Wilson. (integrable systems)

(joint work with G. Segal from Oxford)

[beneath G. Segal: (Graeme)]

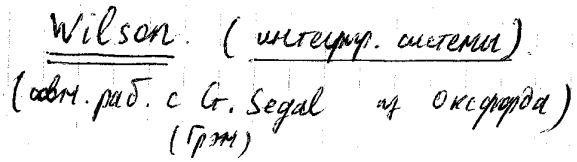


Diagram and typed description. Wilson's name and the note on joint work with G. Segal.

Original source details: GELFAND-P001099

An attempt to understand the work.

Miva; Jimbo, Kashiwara, Date [Sato.]



Original source details: GELFAND-P001099

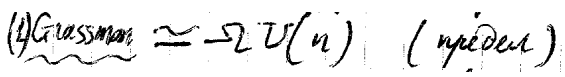
Diagram and typed description. The first name is written "Miva." Segal and Wilson's paper cites the author as Miwa.

Connection with an ∞-dimensional Grassmannian.

$$(1) \quad \underline{\text{Grassman}} \simeq \Omega U(n) \qquad \text{(limit)}$$

space of loops in the unitary group

$$\varinjlim_{n\to\infty} \Omega U(n)$$

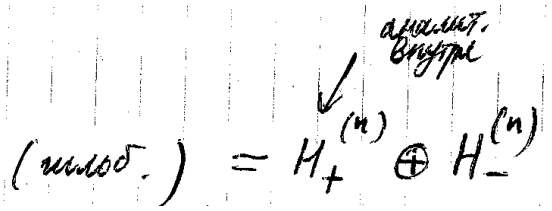


Original source details: GELFAND-P001099

Diagram and typed description. The spelling “Grassman” and the notation $\Omega U(n)$.

Explanation of the isomorphism.

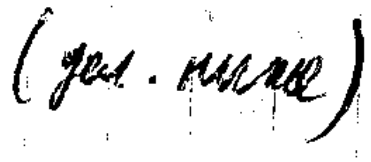
$$H^{(n)} = L^2(S^1;\mathbb{C}^n) \quad \text{(Hilbert)} = \overset{\text{analytic inside}}{\underset{\swarrow}{H_+^{(n)}}} \oplus H_-^{(n)}$$



Original source details: GELFAND-P001099

Diagram and typed description. The arrow from “analytic inside” to $H_+^{(n)}$.

$\gamma : S^1 \to GL(n)$ or $U(n)$ ([group of loops?])



Original source details: GELFAND-P001099

Uncertain reading. The annotation may read “group of loops.”

their group L (continuous)



Original source details: GELFAND-P001099

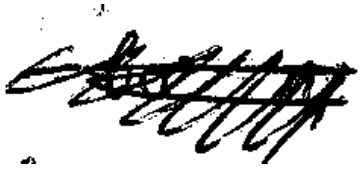
Diagram and typed description. The bracket with raised ends below “continuous.”

L acts on $H^{(n)}$

$$W_\gamma = \gamma(H_+^{(n)}) \qquad \text{— subspace}$$

[at the right] All such W_γ form the Grassmannian $Gr^{(n)}$

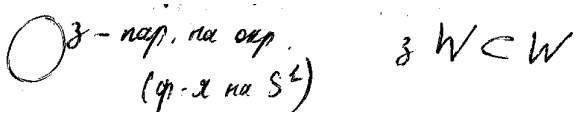
$$W \in Gr^{(n)} = \{W_\gamma \mid \gamma \in L\} \quad \text{[cancelled]}$$



Original source details: GELFAND-P001099

Undeciphered passage. The cancelled fragment after the set notation is illegible.

— homogeneous space with respect to the group of such loops L
Similarly for the unitary group.



Original source details: GELFAND-P001099

Diagram and typed description.



z — param. on the circle (function on S^1)

$zW \subset W$

Diagram and typed description.



Original source details: GELFAND-P001099



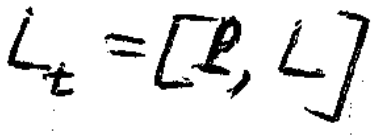
An isolated crossed mark at the right.

W has 2 properties

$$(i) \quad \text{pr} : W \to H_+^{(n)} \quad \text{— Fredholm op.} \quad \text{(Toeplitz)}$$

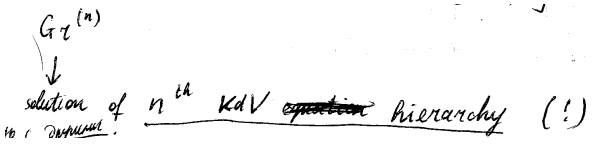
$$(ii) \quad \text{pr} : W \to H_-^{(n)} \qquad \text{compact operator.}$$

Open problem: characterize all loops satisfying $(i), (ii)$.
Connection with the KdV and G.–D. equations.
$Gr^{(n)}$ is related to $L = \partial^n + \ldots$; $\quad L_t = [\ell, L]$.



Original source details: GELFAND-P001099

Uncertain reading. The first entry of the commutator may be ℓ. The handwriting also allows P.



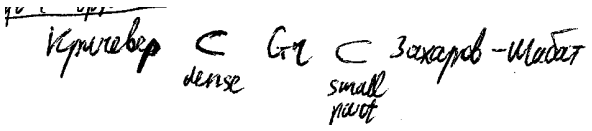
Original source details: GELFAND-P001099

Diagram and typed description.

$Gr^{(n)}$
↓
solution of nth KdV ~~equation~~
hierarchy (!)

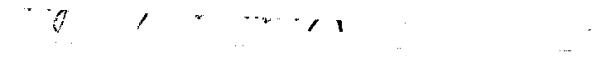
Connection with others.

Krichever $\underset{\text{dense}}{\subset}$ Gr $\underset{\text{small part}}{\subset}$ Zakharov–Shabat



Original source details: GELFAND-P001099

Diagram and typed description. The inclusions labelled “dense” and “small part.”



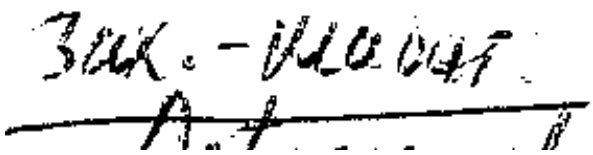
Original source details: GELFAND-P001099

Undeciphered passage. Only scattered strokes remain from the next line at the lower edge.

GELFAND-P001100

[source number: 41]

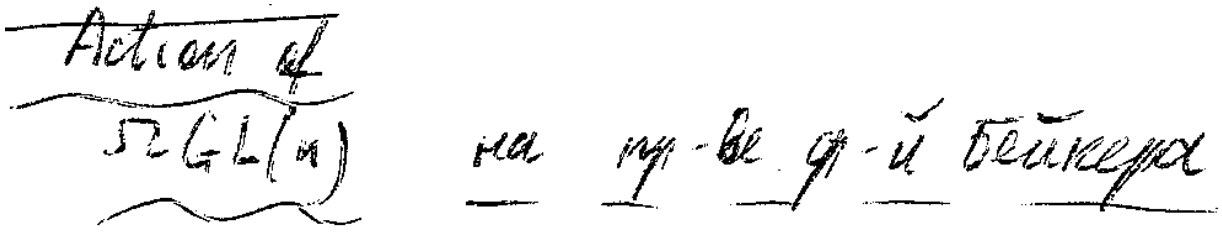
Zak.–Shabat.



Original source details: GELFAND-P001100

Uncertain reading. The first abbreviation may read “Zak.”

Action of
$\Omega GL(n)$ on the space of Baker functions



Original source details: GELFAND-P001100

Diagram and typed description. The heading “Action of” and the group notation.

Gr — orbit of the Z.–Sh. action (dressing of triv. solutions)

List of names:

(1) Gr [~~KdV?~~] and KdV



Original source details: GELFAND-P001100

Uncertain reading. The cancelled entry after Gr may read "KdV."

(2) $MKdV$ — modified KdV

in Drinfeld and Sokolov: replacement of $GL(n)$ by a more general group.

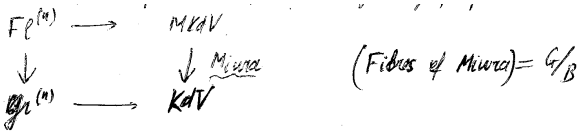


Original source details: GELFAND-P001100

Diagram and typed description. The first divider.

$MKdV$ is associated with the full set of flags rather than Gr.

(quotient of all matrices by triangular matrices that extend analytically downwards)



Original source details: GELFAND-P001100

Diagram and typed description.

$$\begin{array}{ccc} Fl^{(n)} & \longrightarrow & MKdV \\ \downarrow & & \downarrow \text{Miura} \\ Gr^{(n)} & \longrightarrow & KdV \end{array}$$

The first letter at lower left is overwritten. The visible reading is $Gr^{(n)}$.

[at the right] (Fibres of Miura) $= G/B$

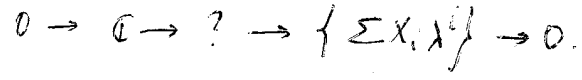
(3) τ-function (a Japanese invention)

Determinant function — on $Gr^{(n)}$.

In Kac–Moody alg., loops appear: $\sum_i X_i \lambda^i$

The extension is written explicitly:

$$0 \to \mathbb{C} \to ? \to \left\{ \sum X_i \lambda^i \right\} \to 0.$$



Original source details: GELFAND-P001100

Diagram and typed description. The exact sequence with a question mark in the middle term.

(4) [~~KdV?~~] Krichever $\subset Gr^{(n)}$

(rational curve)



Uncertain reading. The cancelled entry before "Krichever" may read "KdV."

Original source details: GELFAND-P001100



Diagram and typed description. The second divider.

Original source details: GELFAND-P001100

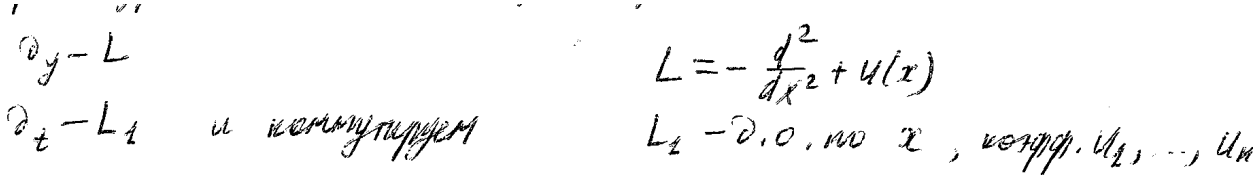
Krichever.

KP describes deformations of arbitrary rings (beyond the special form in KdV).

The hierarchy of KP equations can be understood from the τ-function.

$$\partial_y - L \qquad L = -\frac{d^2}{dx^2} + u(x)$$

$\partial_t - L_1$ and they commute $\qquad L_1$ — differential op. in x, coeffs. $u_1, \ldots, u_n$.



Original source details: GELFAND-P001100

Diagram and typed description. The two commuting operators and the notes at the right.

Property of the τ-function: the commut. system — a system of equations for one scalar function.

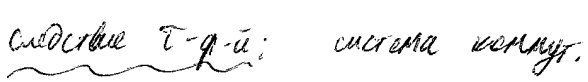


Diagram and typed description. The wavy underlining below the heading.

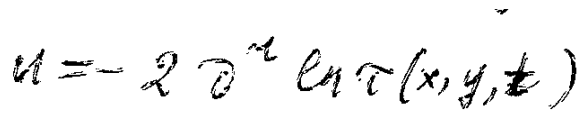
Original source details: GELFAND-P001100



Diagram and typed description. The abbreviation in "the commut. system."

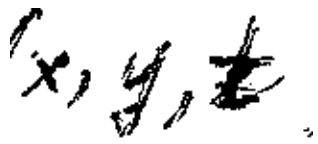
Original source details: GELFAND-P001100

$$u = -2\partial^2 \ln \tau(x, y, t)$$



Uncertain reading. The exponent on ∂ is read as 2. Additional strokes are visible beside the digit.

Original source details: GELFAND-P001100

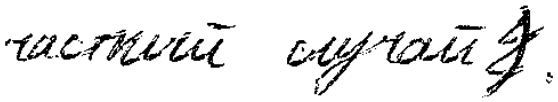


Uncertain reading. The third argument is overwritten as t. The earlier mark is illegible.

Original source details: GELFAND-P001100

τ-function is a function of infinitely many variables.

θ-function arises as a special case of ξ.



Uncertain reading. The final mark resembles ξ. The context allows τ.

Original source details: GELFAND-P001100

GELFAND-P001101

4/IV-83 Gelfand Sem.

Wilson.

(on his work with G. Segal)

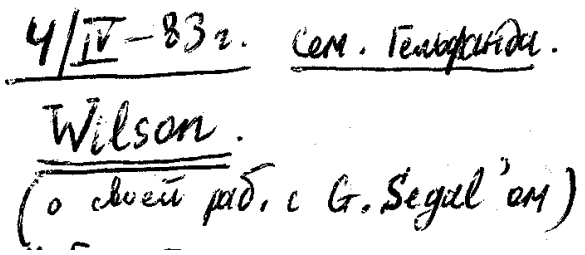


Diagram and typed description. The date, Wilson's name and the note on G. Segal.

Original source details: GELFAND-P001101

I.M.G. τ-function

$$\tau_g(W) \qquad \begin{array}{ll} W \subset H_+ \oplus H_- & \\ W \to H_+ & \text{Fred.} \\ W \to H_- & \text{— comp.} \end{array}$$

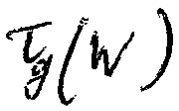


Diagram and typed description. The notation $\tau_g(W)$ before the definition.

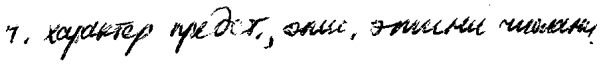
Original source details: GELFAND-P001101

Sato: if it depends on $-n_1, -n_2, \ldots, -n_k$, then one obtains the character of the repr. [described?] by these numbers.



Uncertain reading. The abbreviation after "repr." may mean "described."

Original source details: GELFAND-P001101



Original source details: GELFAND-P001101

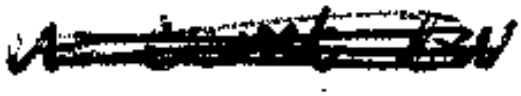
Diagram and typed description. The last part of the remark attributed to Sato.

—·—



Original source details: GELFAND-P001101

Diagram and typed description. Two horizontal strokes and a dot below the sequence of numbers.



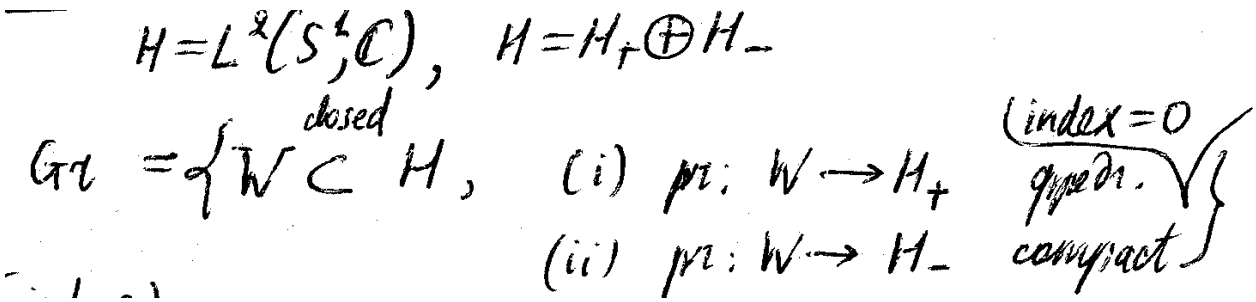
Original source details: GELFAND-P001101

Undeciphered passage. The cancelled entry before the second “Wilson.” is illegible.

Wilson.

Def. $H = L^2(S^1, \mathbb{C}), \quad H = H_+ \oplus H_-$

$$Gr = \left\{ \begin{matrix} \text{closed} \\ W \end{matrix} \subset H, \quad \begin{matrix} (i) & \text{pr} : W \to H_+ & \text{Fred.} \\ (ii) & \text{pr} : W \to H_- & \text{compact} \end{matrix} \right\}$$

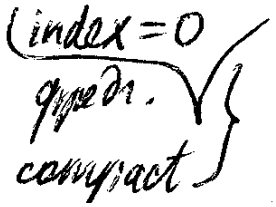


Original source details: GELFAND-P001101

Diagram and typed description. The definition of Gr and the note above condition (i).

Diagram and typed description.

$$\left\{ \begin{matrix} (\text{index} = 0 \\ \text{Fred.} \\ \text{compact} \end{matrix} \right.$$



Original source details: GELFAND-P001101

(without the requirement ind $= 0$)

Gr is disconnected. Its connected components are indexed by the index of op. (i)

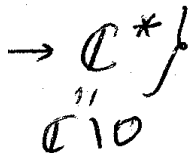
~~Another~~



Original source details: GELFAND-P001101

Diagram and typed description. The cancelled word “Another.”

$$\Gamma = \{\text{analytic functions } S^1 \to \mathbb{C}^*\}$$
$$\mathbb{C}\backslash 0$$



Original source details: GELFAND-P001101

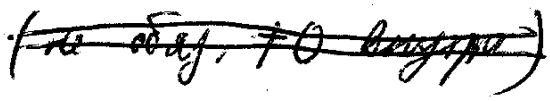
Diagram and typed description. The annotation $\mathbb{C}\backslash 0$ beneath $\mathbb{C}^*$.

$\Gamma_+ \in \Gamma$ [~~KdV?~~] extends analytically inside ~~(not necessarily, $\neq 0$ inside)~~



Original source details: GELFAND-P001101

Uncertain reading. The cancelled entry after Γ may read “KdV.”



Original source details: GELFAND-P001101

Diagram and typed description. The cancelled parenthesis after “inside.”

Γ_+ acts on Gr.

$W \in Gr$

Def.

$$\tau_W : \Gamma_+ \longrightarrow \mathbb{C} \qquad \text{(analytic function!)}$$
$$\tau_W(g) = \det(\text{pr} : g^{-1}W \longrightarrow H_+)$$



Original source details: GELFAND-P001101

Diagram and typed description. The determinant formula with $g^{-1}W$.

One must choose a basis in W.

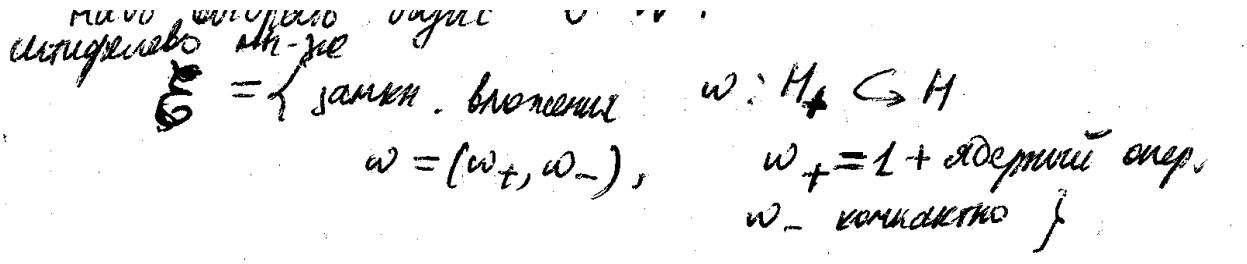
$$\xi = \left\{ \begin{array}{ll} \text{closed embeddings } \omega : H_+ \hookrightarrow H & \\ \omega = (\omega_+, \omega_-), & \omega_+ = 1 + \text{nuclear op.} \\ & \omega_- \quad \text{compact} \end{array} \right\}$$

[above at the left: Stiefel set]



Uncertain reading. The mark before the definition of the Stiefel set may be ξ.

Original source details: GELFAND-P001101



Original source details: GELFAND-P001101

Diagram and typed description. The definition using an embedding and its two components.

GELFAND-P001102

[source number: 48]

One can prove that $\forall$ point of Gr [arises] in this way.



Uncertain reading. The abbreviation is provisionally read as the Russian word for "arises."

Original source details: GELFAND-P001102

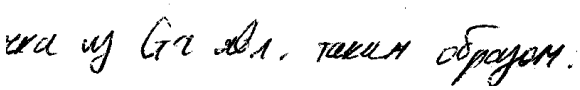


Diagram and typed description. The opening line and its abbreviations.

Original source details: GELFAND-P001102

One can speak of bases: $\omega_i = \omega(z^i)$.

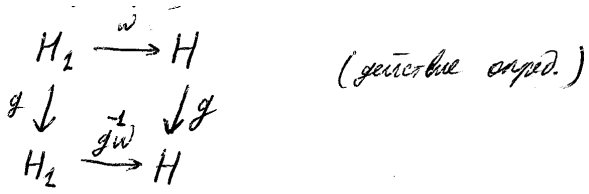


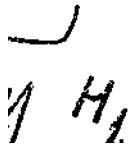
Diagram and typed description.

$$\begin{array}{ccc} H_1 & \xrightarrow{\omega} & H \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g} \\ H_1 & \xrightarrow{g^{-1}\omega} & H \end{array}$$

(action defined)

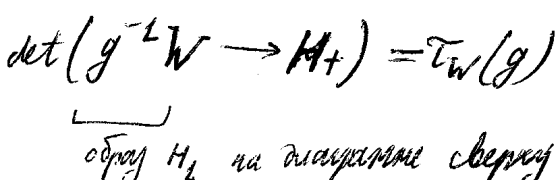
Original source details: GELFAND-P001102

$$\det(\underbrace{g^{-1}W}_{\substack{\text{image of } H_1 \\ \text{in the diagram above}}} \longrightarrow H_+) = \tau_W(g).$$



Original source details: GELFAND-P001102

Uncertain reading. The subscript in the note reads as 1, as in the left column of the diagram.



Original source details: GELFAND-P001102

Diagram and typed description. The determinant and the note below the square-ended bracket.

$g^{-1} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ — notation relative to the decomposition $H_+ \oplus H_-$.



Original source details: GELFAND-P001102

Diagram and typed description. [at the upper right of H_-:]

The τ-function is defined [crossed out: ~~for~~] only on such elements; in fact on the automorphism group of Gr.

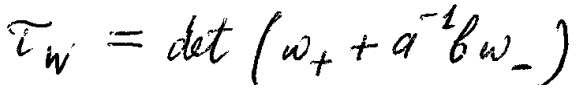


Original source details: GELFAND-P001102

Uncertain reading. The word under the cancellation appears to mean “for.”

Then:

$$\tau_W = \det(\omega_+ + a^{-1}b\omega_-).$$



Original source details: GELFAND-P001102

Diagram and typed description. The formula for τ_W after the block decomposition.

Def. W is transverse $(tr\ H_-)$ if pr : $W \to H_+$ is an isomorphism or

$$W \cap H_- = \{0\}.$$

Such a W is the graph of $\omega_- : H_+ \to H_-$ $\Rightarrow$ canonical choice $\Rightarrow$

$$\tau_W(1) = 1 \qquad \text{(otherwise } 0\text{)}$$

Remark. $\tau_W(g) = 0 \Leftrightarrow g^{-1}W$ is not transverse.

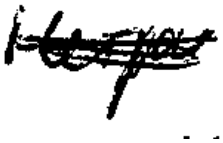
Lemma. $\forall W;\quad \tau_W \not\equiv 0.$



Diagram and typed description. The three crossed horizontal bars in the lemma.

Original source details: GELFAND-P001102

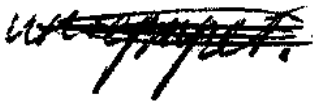
[crossed out: ~~Nontr.~~]



Uncertain reading. The crossed-out opening is provisionally read as an abbreviation for "nontransverse."

Original source details: GELFAND-P001102

Nontransverse subspaces W can be [crossed out:] — an analogue of the θ-divisor on the Jacobian variety of the curve.



Undeciphered passage. The cancelled word after "can be" is difficult to read. The beginning resembles Cyrillic "и" and two strokes descend below the line.

Original source details: GELFAND-P001102

More precisely: $\tau_W(e^{xz}) \not\equiv 0.$



Diagram and typed description. The formula for e^{xz} with the same crossed sign.

Original source details: GELFAND-P001102

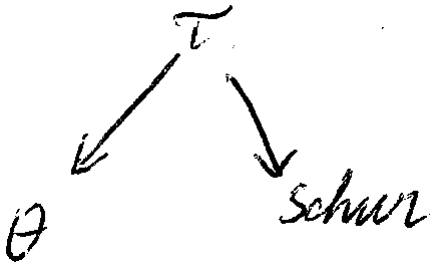
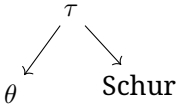


Diagram and typed description.



Original source details: GELFAND-P001102

Schur. Consider special points

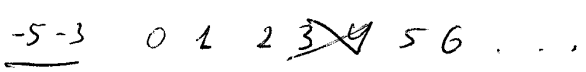
[crossed out: H]



Original source details: GELFAND-P001102

Uncertain reading. The cancelled sign before S resembles H.

$$S = \{s_0 < s_1 < s_2 < \ldots\}, \qquad s_i = i; \quad i \gg 0.$$



Original source details: GELFAND-P001102

Diagram and typed description.

$-5 \quad -3 \quad 0 \quad 1 \quad 2 \quad \cancel{3}\,\cancel{4} \quad 5 \quad 6 \quad \ldots$

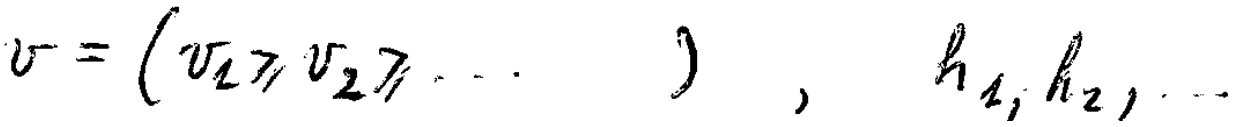
$$H_S = \{z^{s_i}\}_{s_i \in S}.$$

τ-function is a Schur function.

GELFAND-P001103

Suppose there is a partition

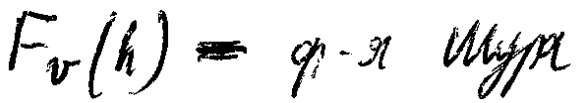
$$\nu = (\nu_1 \geq \nu_2 \geq \ldots), \qquad h_1, h_2, \ldots$$



Original source details: GELFAND-P001103

Diagram and typed description. The partition and the adjacent coefficients.

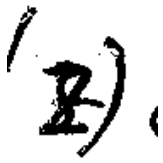
$F_\nu(h)$ [=?] Schur function.



Original source details: GELFAND-P001103

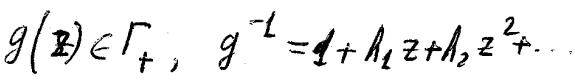
Uncertain reading. The sign is heavily overwritten. It could be an equals sign or a dash.

$$g(z) \in \Gamma_+, \qquad g^{-1} = 1 + h_1 z + h_2 z^2 + \ldots$$



Original source details: GELFAND-P001103

Uncertain reading. The overwritten variable in parentheses is provisionally read as z.



Original source details: GELFAND-P001103

Diagram and typed description. The expansion of g^{-1} with coefficients $h_1, h_2, \ldots$.

$$S \leftrightarrow \nu$$

$$i - s_i \leftrightarrow \nu_i$$

$$\tau_W(g) = F_S(h)$$

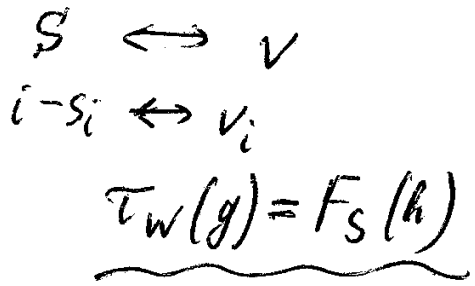


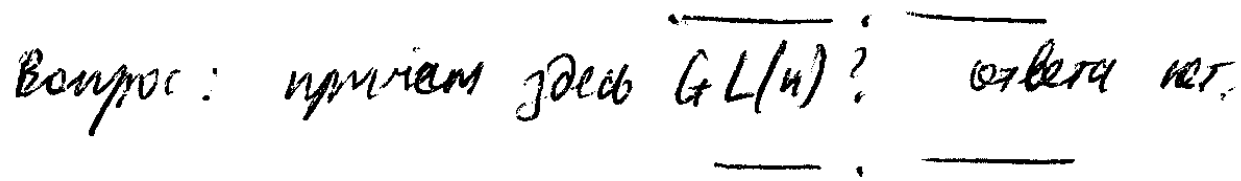
Diagram and typed description. The correspondence $S \leftrightarrow \nu$ and the wavy line beneath the formula for τ_W.

Original source details: GELFAND-P001103

—·—

Question: what does this have to do with $GL(n)$? there is no answer.

—·—

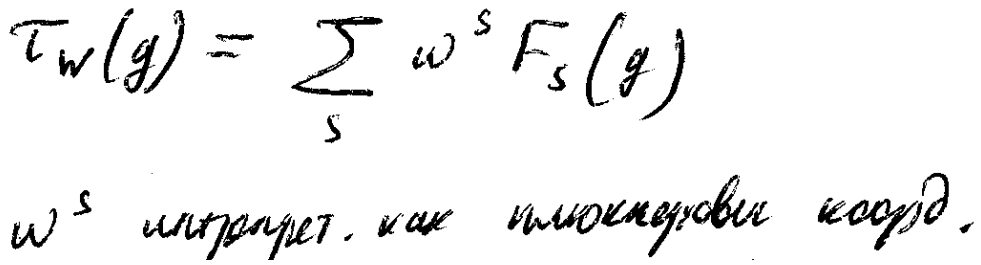


Original source details: GELFAND-P001103

Diagram and typed description. The question and response between two rules.

$$\tau_W(g) = \sum_S w^S F_S(g)$$

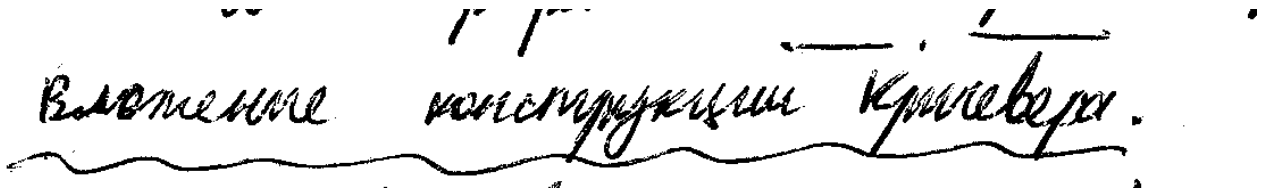
w^S is interpreted as a Plücker coordinate.



Original source details: GELFAND-P001103

Uncertain reading. Both occurrences are read as w^S. The handwritten letter could be ω, and the case of the superscript is uncertain.

Embedding of the Krichever construction.

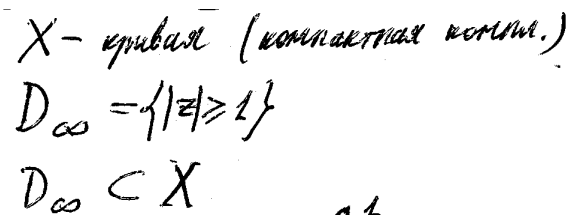


Original source details: GELFAND-P001103

Diagram and typed description. The heading on the Krichever construction.

X — curve (compact complex)

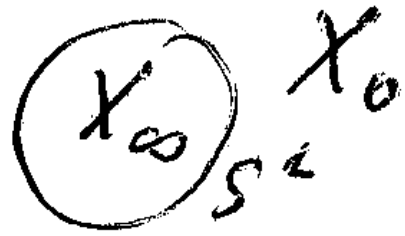
$$D_\infty = \{|z| \geq 1\}, \qquad D_\infty \subset X$$



Original source details: GELFAND-P001103

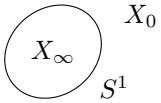
Diagram and typed description. The curve X and the weak inequality in the definition of D_∞.

choose an embedding S^1 — boundary



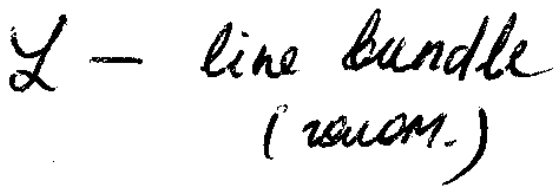
Original source details: GELFAND-P001103

Diagram and typed description.



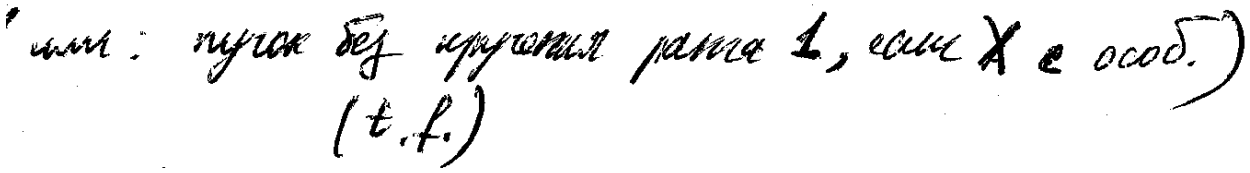
$$X_0 = X \setminus \operatorname{Int} X_\infty.$$

$\mathcal{L}$ — line bundle [below the line: (holom.)] (or: a torsion-free sheaf of rank 1, if X is singular). [Under "torsion-free": (t.f.)]



Original source details: GELFAND-P001103

Diagram and typed description. The original English wording "line bundle" and the abbreviation below it.



Original source details: GELFAND-P001103

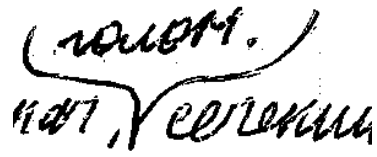
Diagram and typed description. The condition for a singular curve and the annotation "t.f."

$\varphi =$ trivialization of $\mathcal{L}$ over X_∞.

sections over $S^1 \leftrightarrow$ functions.

Construct a point on Gr:

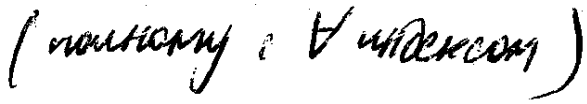
$$W = \{f \in H \mid f \text{ — boundary values of [(holom.)] sections of } \mathcal{L} \text{ over } X_0\}.$$



Original source details: GELFAND-P001103

Diagram and typed description. The insertion "holom." before "sections."

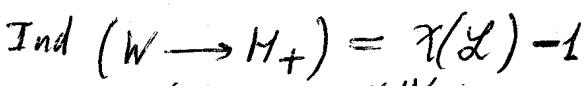
Assertion. $W \in Gr$. (the full one with $\forall$ index)



Original source details: GELFAND-P001103

Uncertain reading. The first word in parentheses is provisionally read as "the full one."

$$\mathrm{Ind}(W \longrightarrow H_+) = \chi(\mathcal{L}) - 1.$$

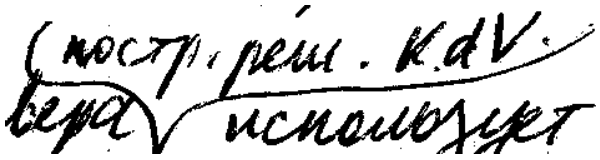


Original source details: GELFAND-P001103

Diagram and typed description. The index of the projection and the term -1.

Remark. Krichever's method [inserted above the line: (constructing KdV solutions)] uses only W, and does not use the curve!

$\Rightarrow$ solutions can be constructed for $\forall W$!!

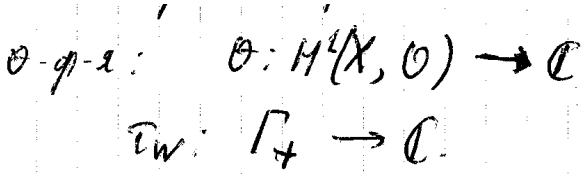


Original source details: GELFAND-P001103

Diagram and typed description. The insertion about constructing KdV solutions.

θ-function:

$$\theta : H^1(X, O) \longrightarrow \mathbb{C}$$
$$\tau_W : \Gamma_+ \longrightarrow \mathbb{C}.$$



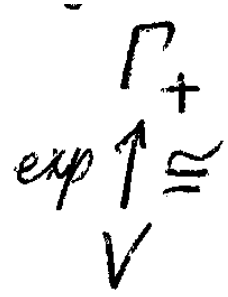
Original source details: GELFAND-P001103

Diagram and typed description. The domains of θ and τ_W.

GELFAND-P001104

[source number: 49]

Connection — ?

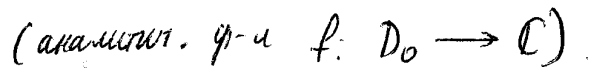


Original source details: GELFAND-P001104

Diagram and typed description.

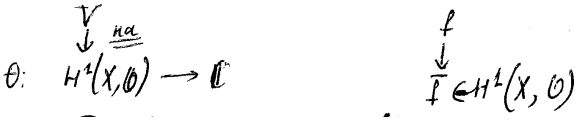
$$\begin{array}{c} \Gamma_+ \\ \exp \uparrow \cong \\ V \end{array}$$

(analytic functions $f\colon D_0 \longrightarrow \mathbb{C}$).



Original source details: GELFAND-P001104

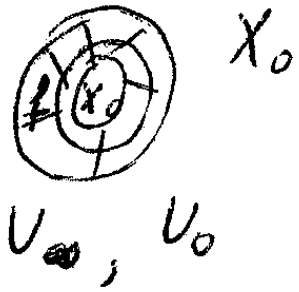
Diagram and typed description. The note with domain D_0.



Original source details: GELFAND-P001104

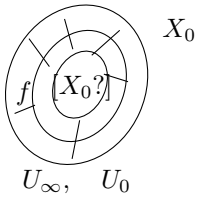
Diagram and typed description.

$$\begin{array}{ccc} V & & f \\ \downarrow \text{onto} & & \downarrow \\ \theta\colon\ H^1(X,O) & \longrightarrow \mathbb{C} & \bar{f} \in H^1(X,O) \end{array}$$



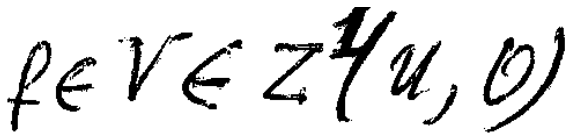
Original source details: GELFAND-P001104

Diagram and typed description.



The inner label is provisionally read as X_0.

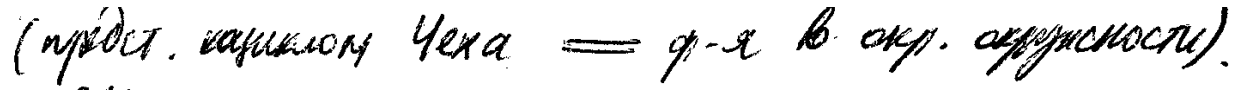
$$f \in V \in Z^1(U,O)$$



Original source details: GELFAND-P001104

Uncertain reading. The manuscript has a sign resembling $\in$ between V and Z^1.

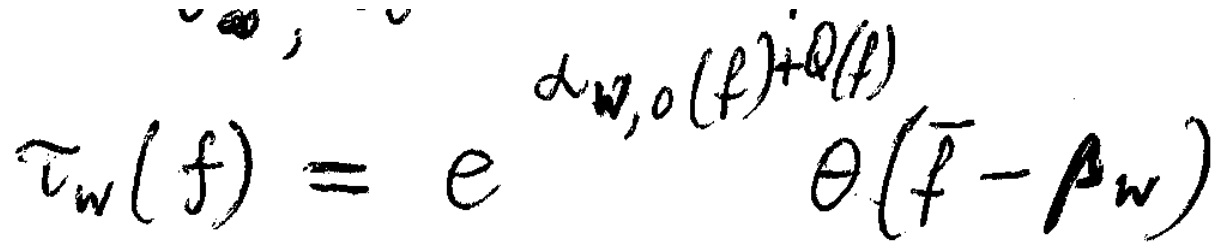
(represented by a Čech cocycle = function in a neighborhood of the circle).



Original source details: GELFAND-P001104

Diagram and typed description. The handwritten note on the Čech cocycle.

$$\widetilde{\tau}_W(f) = e^{\alpha_{W,0}(f)+Q(f)}\theta(\bar{f} - \beta_W)$$



Original source details: GELFAND-P001104

Diagram and typed description. The formula with exponent $\alpha_{W,0}(f) + Q(f)$.

Q depends only on (X, D_∞).

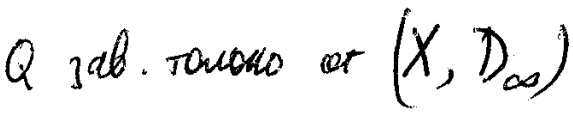
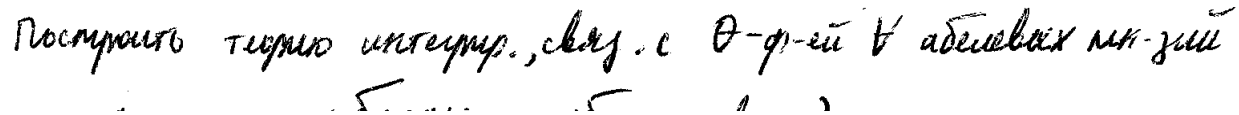


Diagram and typed description. The dependence of Q on (X, D_∞).

Original source details: GELFAND-P001104

Question from I. M. G. Construct a theory of integration connected with the θ-function of $\forall$ abelian varieties



Original source details: GELFAND-P001104

Diagram and typed description. The question about abelian varieties.

(here there arise [crossed out:] only Jacobians of algebraic curves)



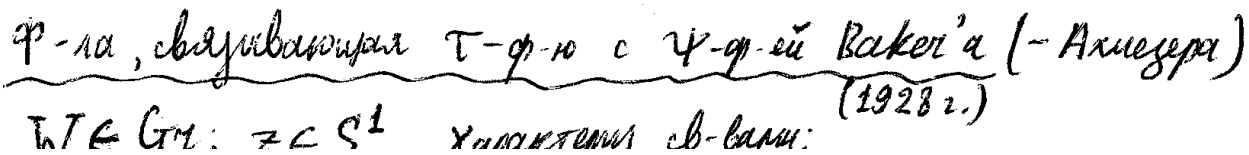
Undeciphered passage. The cancelled mark before “only” is illegible.

Original source details: GELFAND-P001104

—·—

Formula relating the τ-function to Baker’s Ψ-function (— Akhiezer)

[below Baker’s name: (1928)]



Original source details: GELFAND-P001104

Diagram and typed description. The wavy underline below the heading and the addition “— Akhiezer.”

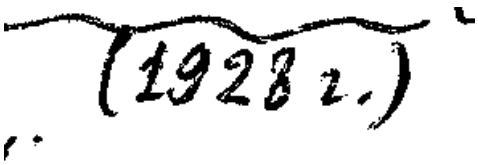


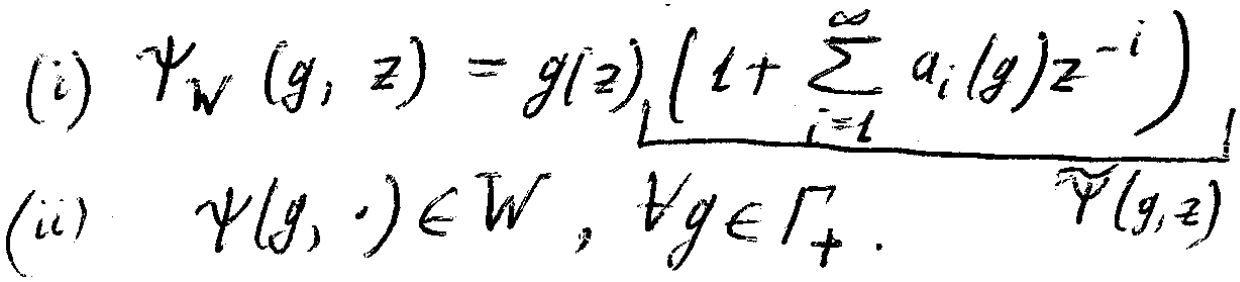
Diagram and typed description. The date below Baker’s name.

Original source details: GELFAND-P001104

$W \in Gr$; $z \in S^1$. Characterized by the properties:

$$\text{(i)} \quad \Psi_W(g,z) = g(z) \underbrace{\left(1 + \sum_{i=1}^{\infty} a_i(g) z^{-i}\right)}_{\widetilde{\Psi}(g,z)}$$

$$\text{(ii)} \quad \Psi(g,\cdot) \in W, \quad \forall g \in \Gamma_+.$$



Original source details: GELFAND-P001104

Diagram and typed description. The two properties and the square-ended bracket above $\widetilde{\Psi}(g,z)$.

Prop. $\forall W$, $\exists!$ such a Ψ_W [crossed out: ~~for any~~].



Uncertain reading. The cancelled ending provisionally reads “for any.”

Original source details: GELFAND-P001104

If $g^{-1}W \cap H_- = \{0\}$, then the value is constructed: [crossed out:] preimage of 1 under the isomorphism



Undeciphered passage. The mark before “preimage” is cancelled and illegible.

Original source details: GELFAND-P001104

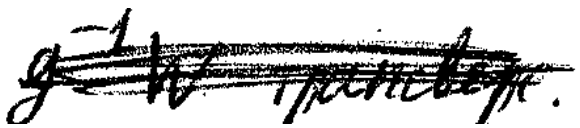
$$g^{-1}W \xrightarrow{\cong} H_+$$



Diagram and typed description. The isomorphism to H_+.

Original source details: GELFAND-P001104

[crossed out: $g^{-1}W$ ~~transv.~~]



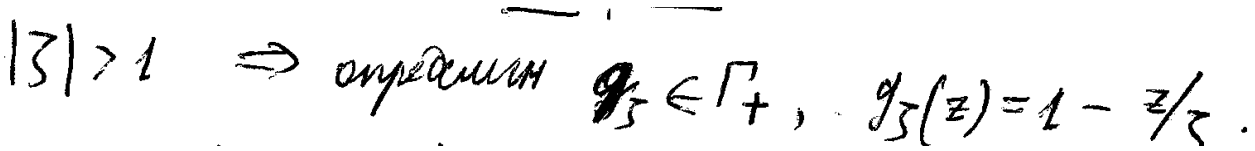
Uncertain reading. The cancelled line provisionally reads “$g^{-1}W$ transv.”

Original source details: GELFAND-P001104

Prop. $\exists!\ \Psi_W(g,z)$ — meromorphic on Γ_+.
Proof as in the case of curves
(there one uses the θ-function, and here the τ-function)

— · —

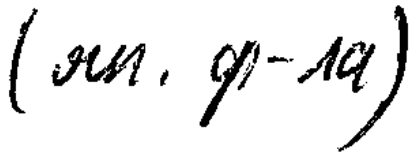
$|\zeta|>1 \quad \Rightarrow$ define $q_\zeta \in \Gamma_+, \quad q_\zeta(z) = 1 - z/\zeta$.



Original source details: GELFAND-P001104

Uncertain reading. The first letter is overwritten. In the formula at right it reads as q.

Prop. (explicit formula). For $|\zeta|>1$



Uncertain reading. The abbreviation is provisionally read as “explicit formula.”

Original source details: GELFAND-P001104

$$\widetilde{\Psi}_W(g,\zeta) = \frac{\tau_W(gq_\zeta)}{\tau_W(g)}$$

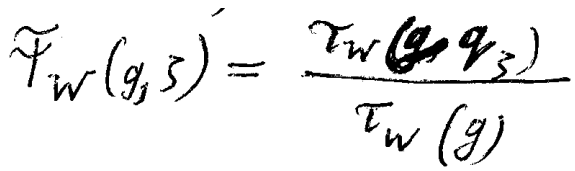


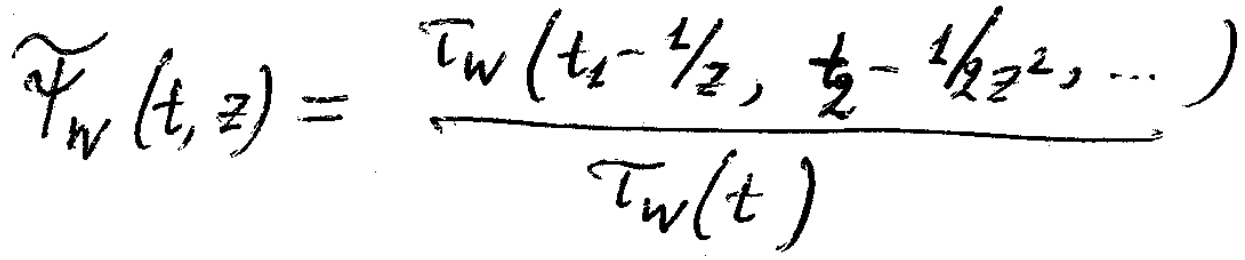
Diagram and typed description. The quotient formula, with an overwritten g in the numerator.

Original source details: GELFAND-P001104

GELFAND-P001105

Or: write it in the form exp and regard everything as functions of t (infinitely many)

$$\widetilde{\Psi}_W(t,z) = \frac{\tau_W(t_1 - 1/z,\ t_2 - 1/2z^2, \ldots)}{\tau_W(t)}$$



Original source details: GELFAND-P001105

Uncertain reading. The second shift is written with a slash as $1/2z^2$, without parentheses.

$$\left(g = e^{\sum_1^\infty t_i z^i}\right).$$

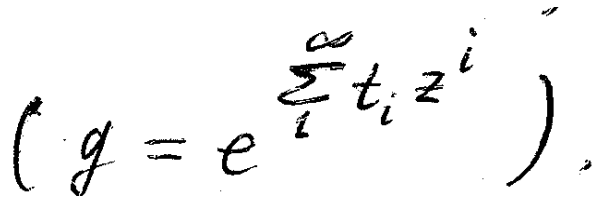


Diagram and typed description. The exponential form of g with lower summation limit 1.

Original source details: GELFAND-P001105

Or:

$$\Psi = \frac{(\text{vertex operator})\tau}{\tau}$$

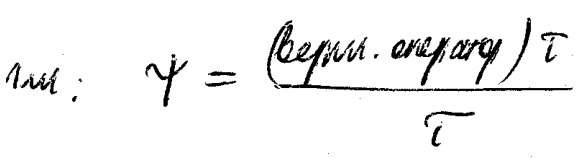
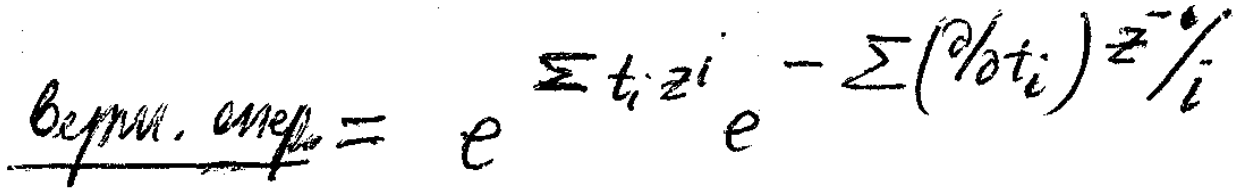


Diagram and typed description. The formula for Ψ in terms of the vertex operator.

Original source details: GELFAND-P001105

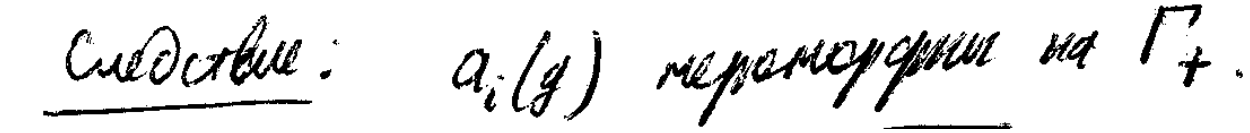
$$\underline{\text{vertex op.}} = e^{\sum t_i z^i} e^{-\sum(\partial/\partial t_i) z^{-i}/i}$$



Original source details: GELFAND-P001105

Diagram and typed description. The two exponentials and the exponent of the second one.

Corollary. $a_i(g)$ are meromorphic on Γ_+.



Original source details: GELFAND-P001105

Diagram and typed description. The corollary on meromorphicity of the coefficients.

—·—

The connection with nonlinear equations is as in Krichever!

—·—

[crossed out: ~~В...~~]



Undeciphered passage. The initial Cyrillic “В” is visible under the cancellation. The remainder is indistinct.

Original source details: GELFAND-P001105

$\forall W$ there is an infinite hierarchy of equations.

—·—

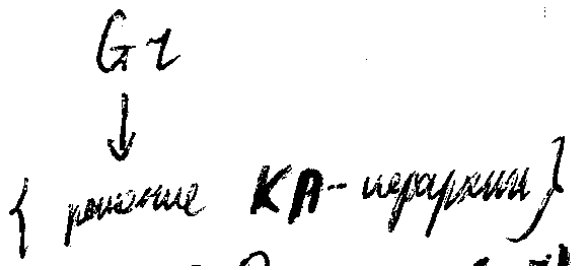


Diagram and typed description.

$$\begin{array}{c} Gr \\ \downarrow \\ \{\text{solution of the KP hierarchy}\} \end{array}$$

Original source details: GELFAND-P001105

$$D = \partial/\partial x; \qquad Q = D + \sum_{1}^{\infty} q_i D^{-i}$$

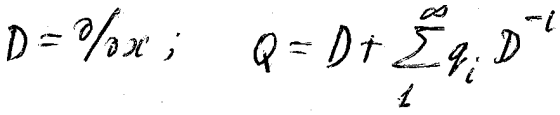


Diagram and typed description. The notation for D and Q.

Original source details: GELFAND-P001105

$$\frac{\partial Q}{\partial t_r} = [(Q^r)_+, Q]$$

[Note below the line from $(Q^r)_+$: differential part of Q^r.]

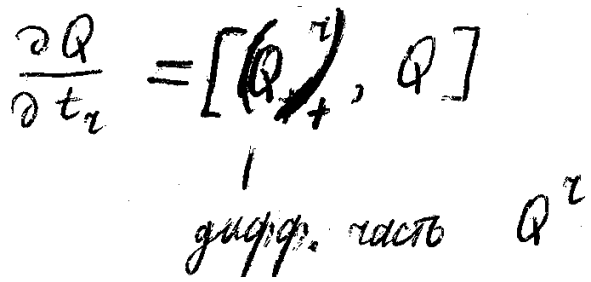


Diagram and typed description. The commutator and the annotation for the positive part.

Original source details: GELFAND-P001105

If $Q^n = L$ is differential, we obtain the Gelfand–Dickey system:

$$\frac{\partial L}{\partial t_r} = [L_+^{r/n}, L]$$

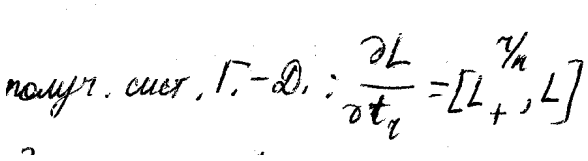
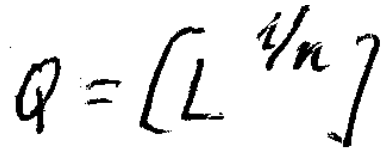


Diagram and typed description. The notation $L_+^{r/n}$ in the commutator.

Original source details: GELFAND-P001105

and conversely, if L is diff. $Q = [L^{1/n}]$, we obtain [crossed out:] the equation written above:



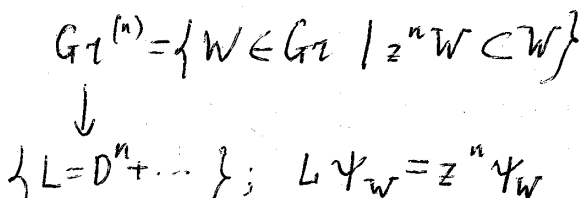
Original source details: GELFAND-P001105

Diagram and typed description. Square brackets in the notation $Q = [L^{1/n}]$.



Original source details: GELFAND-P001105

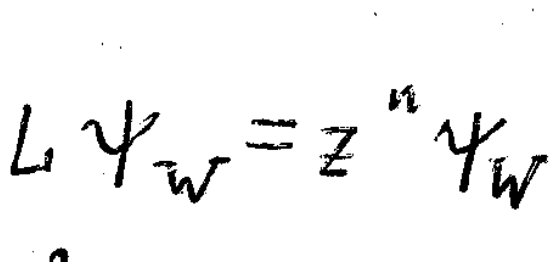
Undeciphered passage. The cancelled mark before “equation” is illegible.



Original source details: GELFAND-P001105

Diagram and typed description.

$$Gr^{(n)} = \{W \in Gr \mid z^n W \subset W\}$$
$$\downarrow$$
$$\{L = D^n + \ldots\}\,; \quad L\Psi_W = z^n \Psi_W$$



Original source details: GELFAND-P001105

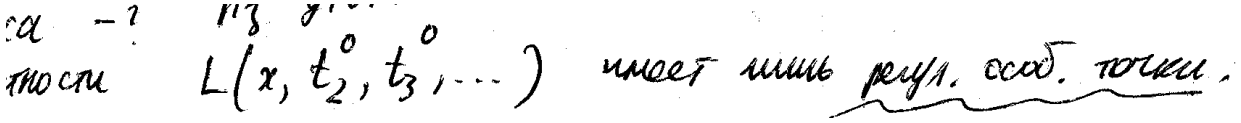
Diagram and typed description. The equation for Ψ_W.

Here:

$$g = e^{xz + t_2 z^2 + \ldots}$$

Poles — ? From the assertion about poles:

in particular $L(x, t_2^0, t_3^0, \ldots)$ has only regular singular points.

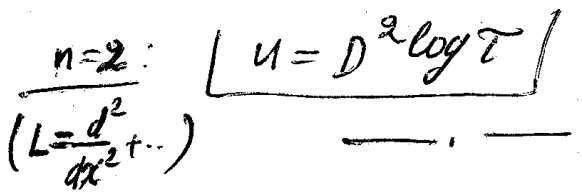


Original source details: GELFAND-P001105

Diagram and typed description. The fixed times t_2^0, t_3^0 and the underlined conclusion.

$$\underline{n = 2}: \quad \boxed{u = D^2 \log \tau}$$

$$\left(L = \frac{d^2}{dx^2} + \ldots\right)$$



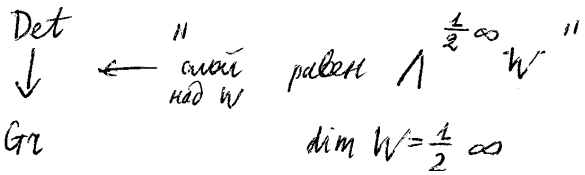
Original source details: GELFAND-P001105

Diagram and typed description. The open-topped bracket below the formula for u and the line about L.

GELFAND-P001106

[source number: 50]

Bundle:

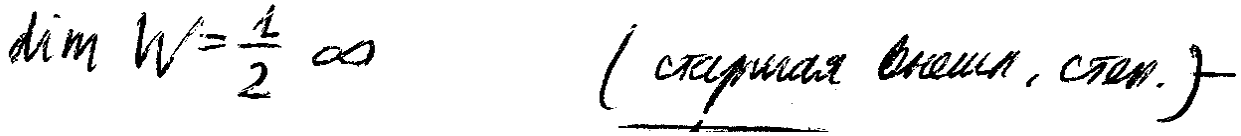


Original source details: GELFAND-P001106

Diagram and typed description.

$$\begin{array}{ccc} Det & & \\ \downarrow & \longleftarrow & \text{“the fibre over } W \text{ equals } \textstyle\bigwedge^{\frac{1}{2}\infty} W\text{”} \\ Gr & & \end{array}$$

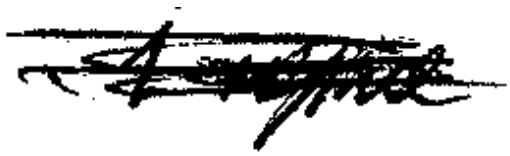
$$\dim W = \frac{1}{2}\infty \qquad \text{(top exterior power)—}$$



Original source details: GELFAND-P001106

Diagram and typed description. The dimension and the note on the top exterior power. A short stroke follows the parenthesis.

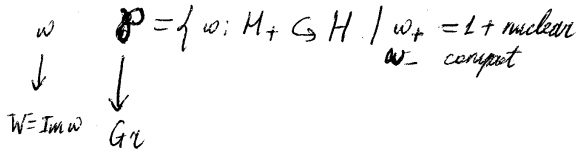
[crossed out below the note at right: ~~one-dimensional~~]



Original source details: GELFAND-P001106

Uncertain reading. The cancelled word is provisionally read as “one-dimensional.”

One-dimensional linear bundle.

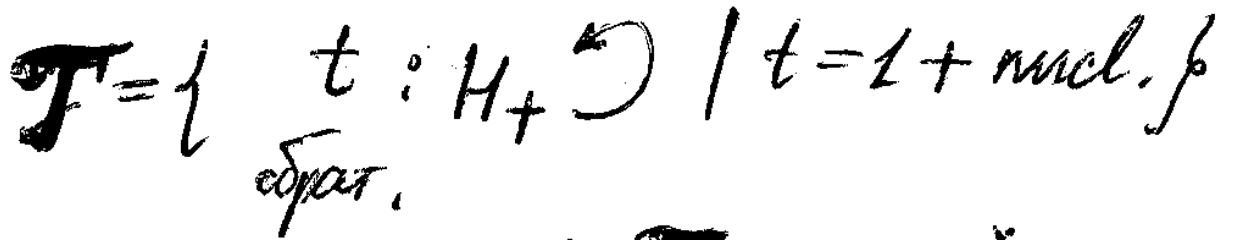


Original source details: GELFAND-P001106

Diagram and typed description.

$$\begin{array}{ccc} \omega & & \mathcal{P} = \{\omega \colon H_+ \hookrightarrow H \mid \\ \downarrow & & \quad \omega_+ = 1 + \text{nuclear} \\ W = \operatorname{Im}\omega & & \quad \omega_- \quad \text{compact} \\ & & \downarrow \\ & & Gr \end{array}$$

$$\mathcal{T} = \left\{ \underset{\text{inv.}}{t} : H_+ \circlearrowleft \;\middle|\; t = 1 + \text{nucl.} \right\}$$



Original source details: GELFAND-P001106

Diagram and typed description. The loop beside H_+ and the abbreviation below t.

$$\det\colon \mathcal{T} \longrightarrow \mathbb{C}^*$$

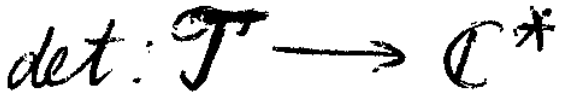


Diagram and typed description. The arrow from $\mathcal{T}$ to $\mathbb{C}^*$.

Original source details: GELFAND-P001106

$$Det = \mathcal{P} \underset{\mathcal{T}}{\times} \mathbb{C}.$$

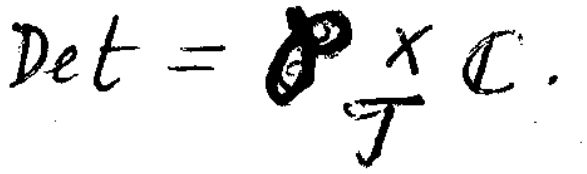
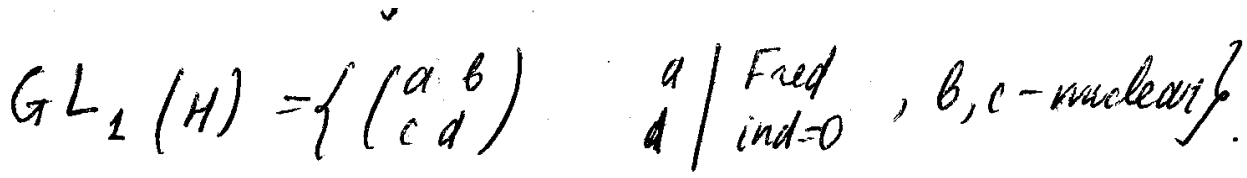


Diagram and typed description. The letter $\mathcal{T}$ is written below the product sign.

Original source details: GELFAND-P001106

$$GL_1(H) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \begin{matrix} a \\ d \end{matrix} \;\middle|\; \begin{matrix} \text{Fred} \\ \text{ind} = 0 \end{matrix}, \quad b, c \text{ — nuclear} \right\}.$$



Original source details: GELFAND-P001106

Diagram and typed description. The blocks a, d and the Fred and ind $= 0$ conditions.

Its action on Gr will be lifted to an action on Det.

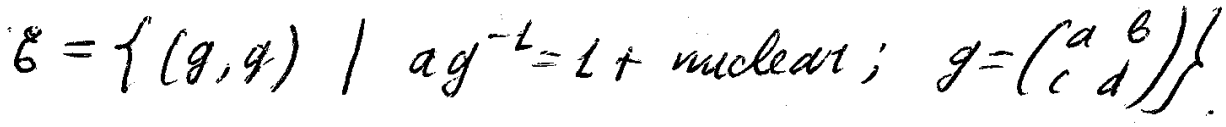


Uncertain reading. The middle of the word "action" has been overwritten.

Original source details: GELFAND-P001106

$$\mathcal{E} \subset GL_1(H) \times GL(H_+)$$

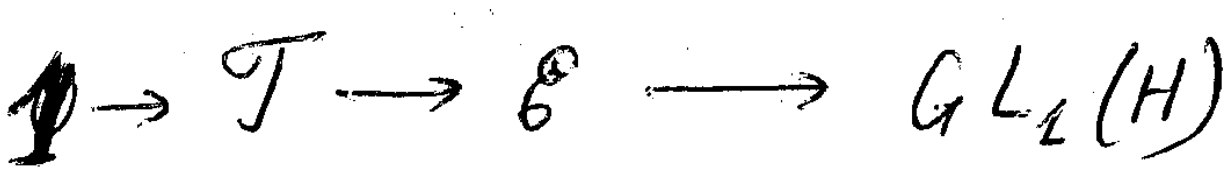
$$\mathcal{E} = \left\{ (g, q) \;\middle|\; aq^{-1} = 1 + \text{nuclear}; \quad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right\}.$$



Original source details: GELFAND-P001106

Diagram and typed description. The condition $aq^{-1} = 1 +$ nuclear for the pair (g, q).

$$1 \longrightarrow \mathcal{T} \longrightarrow \mathcal{E} \longrightarrow GL_1(H)$$



Original source details: GELFAND-P001106

Uncertain reading. The initial digit is overwritten and provisionally read as 1.

Assertion. the extension is nontrivial.

In the finite-dimensional case we obtain a trivial extension.

Action of $\mathcal{E}$ on Det:

$$(g, q)(\omega, \lambda) = (g\omega q^{-1}, \lambda)$$

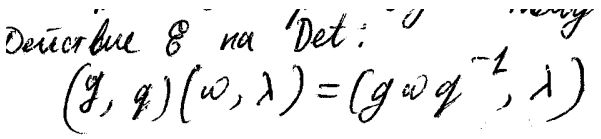


Diagram and typed description. The action on the pair (ω, λ).

Original source details: GELFAND-P001106

from the presentation $Det = \mathcal{P} \underset{\mathcal{T}}{\times} \mathbb{C}$.



Diagram and typed description. The note at the right giving the presentation of Det.

Original source details: GELFAND-P001106

$$\mathcal{T}_0 = \ker \det \mathcal{T} \longrightarrow 1$$



Uncertain reading. The symbols after ker det are $\mathcal{T}$, an arrow and the numeral 1, with no parentheses.

Original source details: GELFAND-P001106

[crossed out:] The elements of $\mathcal{T}_0$ (their image in $\mathcal{E}$) act trivially.



Undeciphered passage. The mark before "The elements" is heavily cancelled and illegible.

Original source details: GELFAND-P001106

$\Rightarrow$ there acts $\mathcal{E}/\mathcal{T}_0$.

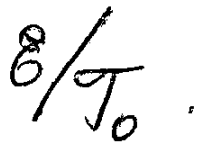
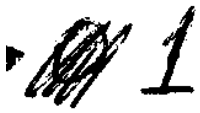


Diagram and typed description. The quotient $\mathcal{E}/\mathcal{T}_0$ on its own line.

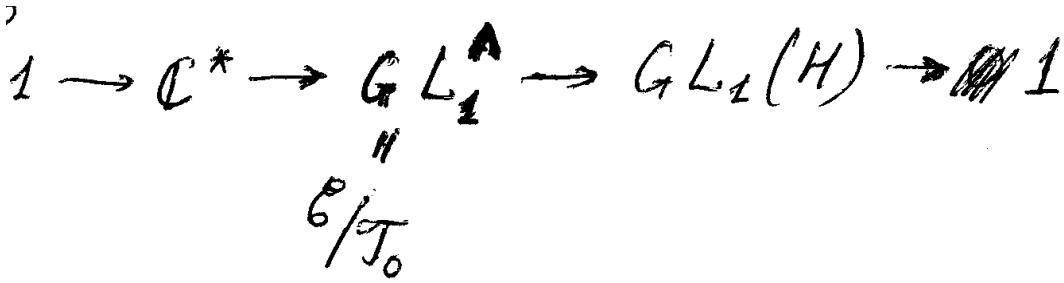
Original source details: GELFAND-P001106

$$1 \longrightarrow \mathbb{C}^* \longrightarrow \widehat{GL_1} \longrightarrow GL_1(H) \longrightarrow 1$$
$$\widehat{GL_1} \;\|\; \mathcal{E}/\mathcal{T}_0$$



Uncertain reading. A sign resembling 0 is cancelled before the final 1.

Original source details: GELFAND-P001106



Original source details: GELFAND-P001106

Diagram and typed description. The exact sequence and the label beneath $\widehat{GL_1}$.

GELFAND-P001107

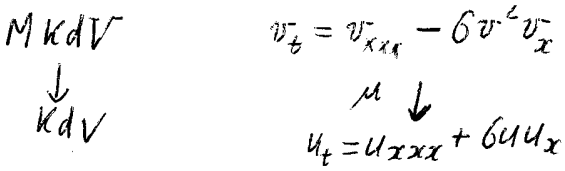


Diagram and typed description.

$$\begin{array}{ccc} \text{MKdV} & & v_t = v_{xxx} - 6v^2 v_x \\ \downarrow & & \mu \downarrow \\ \text{KdV} & & u_t = u_{xxx} + 6uu_x \end{array}$$

Original source details: GELFAND-P001107

$$\underline{u = v_x - v^2}$$

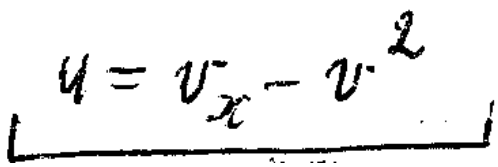
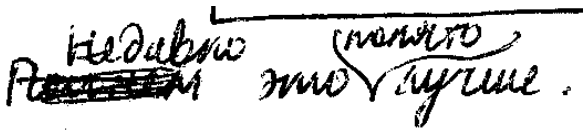


Diagram and typed description. The Miura formula and its square-ended bracket.

Original source details: GELFAND-P001107

[crossed out: ~~Let us consider~~; above it: Recently] this was [inserted: understood] better.



Uncertain reading. The cancelled word is provisionally read as “Let us consider.” “Recently” is written above it.

Original source details: GELFAND-P001107

KdV is connected with

$$\underbrace{D^2 + u = (D - v)(D + v)}_{(u = v_x - v^2)}$$

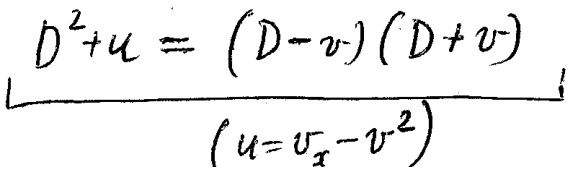
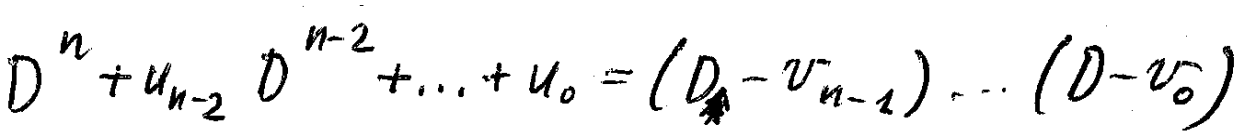


Diagram and typed description. The factorization and the note under the square-ended bracket.

Original source details: GELFAND-P001107

generalization of the right-hand side:

$$D^n + u_{n-2}D^{n-2} + \ldots + u_0 = (D - v_{n-1}) \ldots (D - v_0)$$



Original source details: GELFAND-P001107

Diagram and typed description. The general factorization in the order written.



Undeciphered passage. The cancelled subscript at the first D on the right is illegible.

Original source details: GELFAND-P001107

$$v \longmapsto u$$

generalized Miura transformation.

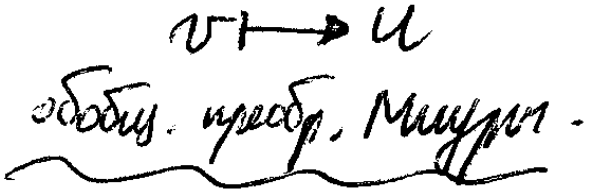


Diagram and typed description. The arrow from v to u and its underlined caption.

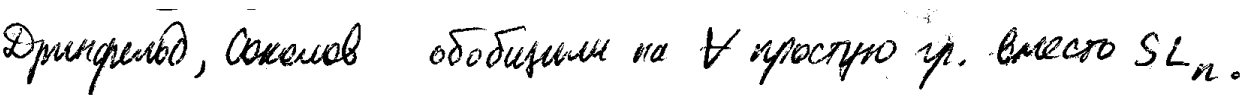
Original source details: GELFAND-P001107

[crossed out:] Drinfeld and Sokolov generalized this to $\forall$ simple group in place of SL_n.



Undeciphered passage. The heavily cancelled mark before Drinfeld's name is illegible.

Original source details: GELFAND-P001107



Original source details: GELFAND-P001107

Diagram and typed description. The statement about Drinfeld, Sokolov and a simple group.

The correspondence is not one-to-one. (diff. eq.)

[below the parentheses: one already has to solve it for $n = 2$.]

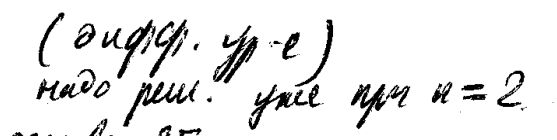
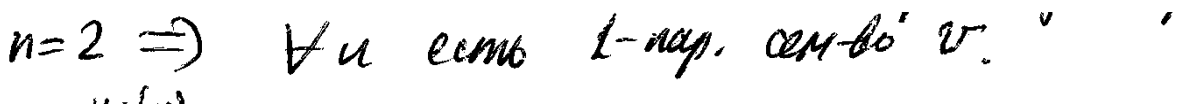


Diagram and typed description. The note below the line about the differential equation.

Original source details: GELFAND-P001107

$$n = 2 \quad \Rightarrow \quad \forall u \text{ there is a one-parameter family of } v.$$

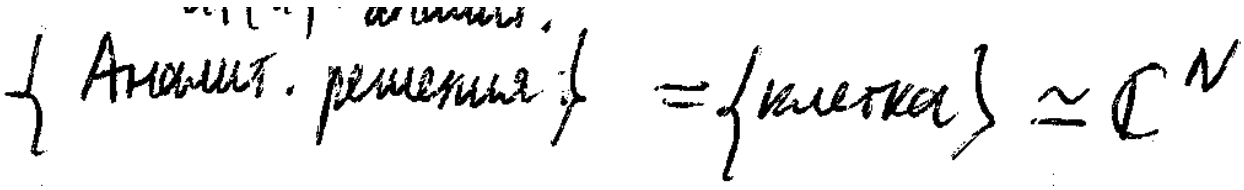


Original source details: GELFAND-P001107

Diagram and typed description. The line on the family for $n = 2$.

$u_i(x)$ — analytic.

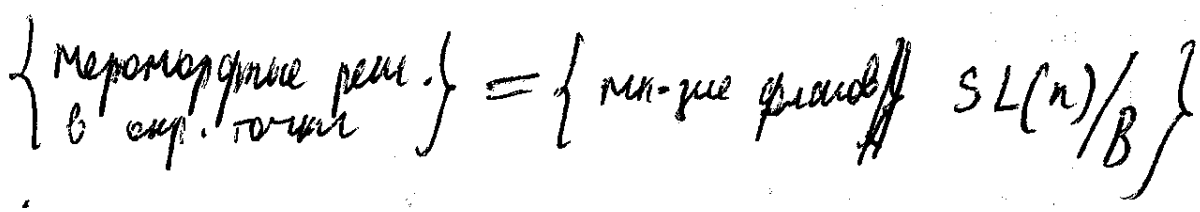
$$\{\text{Analytic solutions}\} = \{\text{cell}\} \simeq \mathbb{C}^N$$



Original source details: GELFAND-P001107

Diagram and typed description. The comparison of analytic solutions with a cell. The sign before $\mathbb{C}^N$ has one wavy and one horizontal stroke.

$$\left\{\begin{matrix}\text{Meromorphic sol.} \\ \text{near a point}\end{matrix}\right\} = \{\text{set of flags } SL(n)/B\}$$



Original source details: GELFAND-P001107

Diagram and typed description. The equality with the set of flags $SL(n)/B$.

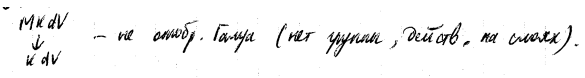


Diagram and typed description. The closing brace after “flags” is cancelled. The retained brace follows $SL(n)/B$.

Original source details: GELFAND-P001107

$\forall n$ there is a system of nonlinear differential equations. Its solutions have the structure of a set of flags.

Main difficulty:

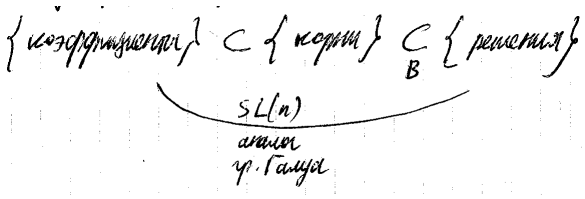


Original source details: GELFAND-P001107

Diagram and typed description.

MKdV
↓
KdV
— not a Galois map
(there is no group acting on the fibres).

Consider the differential field generated by $v_0, \dots, v_{n-1}$.



Original source details: GELFAND-P001107

Diagram and typed description.

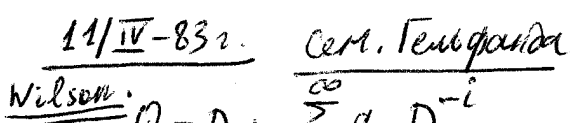
$$\underbrace{\{\text{coefficients}\} \subset \{\text{roots}\} \underset{B}{\subset} \{\text{solutions}\}}_{SL(n)}$$

analogue of the Galois group

GELFAND-P001108

11/IV-83 Gelfand Sem.

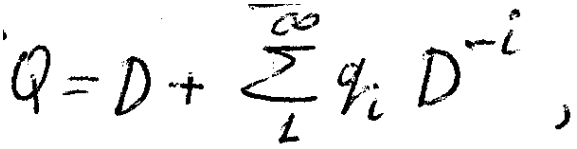
Wilson.



Original source details: GELFAND-P001108

Diagram and typed description. The date 11/IV-83 and the two underlines below Wilson's name.

$$Q = D + \sum_{1}^{\infty} q_i D^{-i}, \qquad L \text{ — diff.} \quad \Rightarrow$$



Original source details: GELFAND-P001108

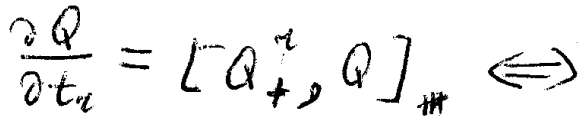
Diagram and typed description. The expansion of Q with lower summation limit 1.



Original source details: GELFAND-P001108

Diagram and typed description. The condition on L and the arrow at the end of the line.

$$\frac{\partial Q}{\partial t_r} = [Q_+^r, Q] \text{ [cancelled]} \quad \Longleftrightarrow \quad L_{t_r} = [L_+^{r/n}, L] \qquad \text{(G.–D.)}$$



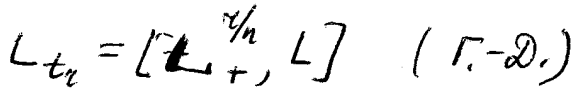
Original source details: GELFAND-P001108

Diagram and typed description. The first commutator and the equivalence sign.

Undeciphered passage. The small cancelled sign after the first square bracket is illegible.



Original source details: GELFAND-P001108



Original source details: GELFAND-P001108

Diagram and typed description. The commutator for L and the abbreviation at the right.

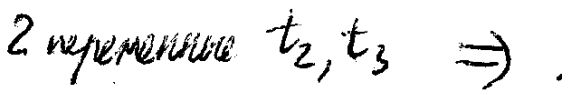


Original source details: GELFAND-P001108

Uncertain reading. The L in the first entry of the second commutator is written over thinner strokes.

Original source details: GELFAND-P001108

$$2 \text{ variables } t_2, t_3 \quad \Rightarrow$$
$$2q_1(x, t_2, t_3) \text{ — solution of K.–P.} \quad \Rightarrow \quad \text{universal KP}$$
$$\underset{y}{} \quad \underset{t}{}$$



Original source details: GELFAND-P001108

Diagram and typed description. The two variables and the rightward arrow.



Original source details: GELFAND-P001108

Diagram and typed description. The factor 2 before q_1 and the labels y, t beneath the arguments.

GELFAND-P001109

V. Ginzburg.

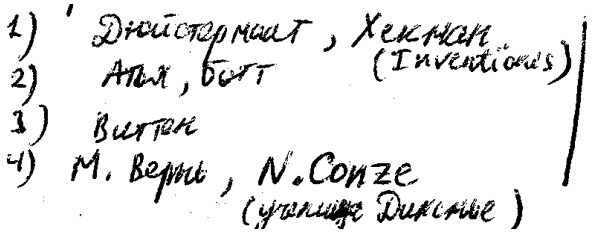


Original source details: GELFAND-P001109

Diagram and typed description. The speaker's name and its underline.

on the works:

1) Duistermaat, Heckman (Inventiones) 2) Atiyah, Bott 3) Witten 4) M. Vergne, N. Conze (Dixmier's pupils)	Berezin, Gelfand (vol. 5, Tr. Moscow Math. Soc.) 1956



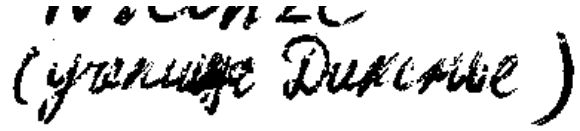
Original source details: GELFAND-P001109

Diagram and typed description. The four entries and the vertical line at the right.



Original source details: GELFAND-P001109

Uncertain reading. The Russian spelling in the first entry is provisionally read as Duistermaat.



Original source details: GELFAND-P001109

Uncertain reading. The parenthesis is provisionally read as "Dixmier's pupils."

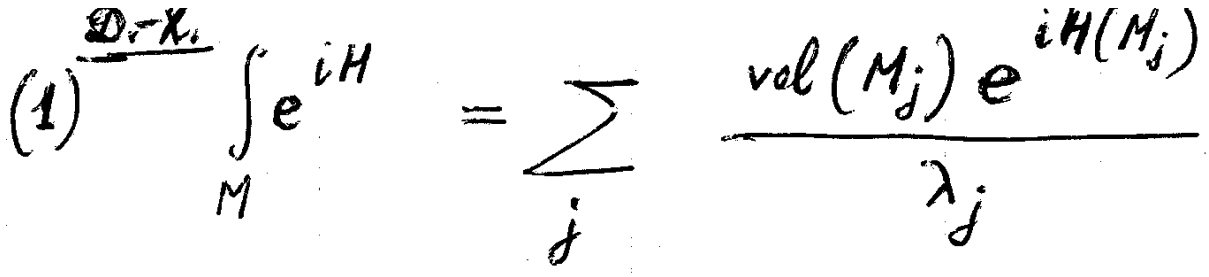


Original source details: GELFAND-P001109

Diagram and typed description. The reference to the right of the list, with the year below the parenthesis.

D.–H.

$$(1) \qquad \int_M e^{iH} = \sum_j \frac{\mathrm{vol}(M_j) e^{iH(M_j)}}{\lambda_j}$$



Original source details: GELFAND-P001109

Diagram and typed description. Formula (1), with exponent iH and denominator λ_j.

M — compact manifold, symplectic.

H — function on M; the condition on it: int. curves of the [inserted: Ham.] field [crossed out:] ξ_H are closed.



Original source details: GELFAND-P001109

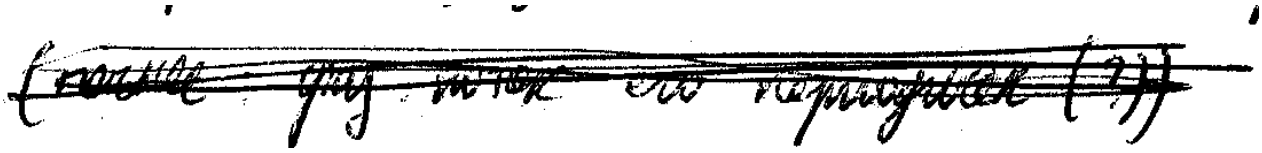
Diagram and typed description. “Ham.” is inserted before “field.”



Original source details: GELFAND-P001109

Undeciphered passage. An illegible notation is cancelled before ξ_H.

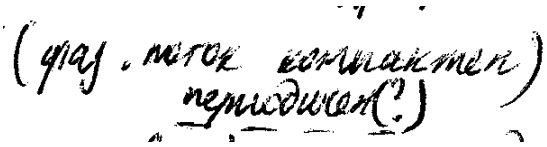
[crossed out: ~~(more precisely, its phase flow is periodic (?))~~.]



Original source details: GELFAND-P001109

Uncertain reading. The cancelled line is provisionally read as “(more precisely, its phase flow is periodic (?)).”

(the phase flow is compact) [added below: periodic (?)]

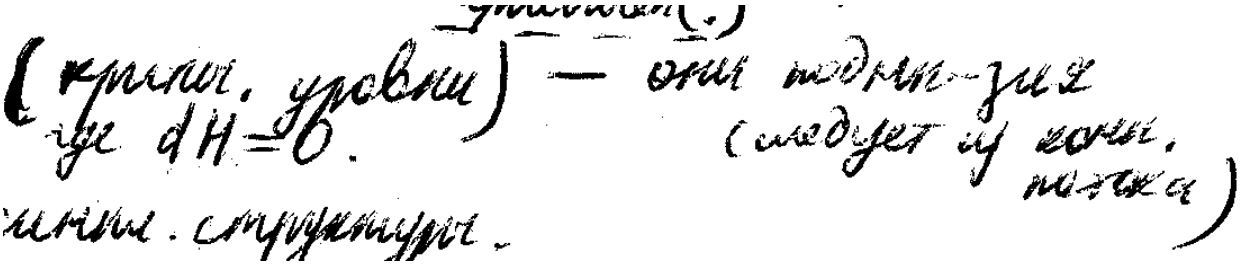


Original source details: GELFAND-P001109

Diagram and typed description. The addition “periodic (?)” below the line about compactness.

M_j — critical sets of the function H (critical levels) — they are submanifolds.

[below “critical levels”: where $dH = 0$. at right: (follows from the compact flow).]



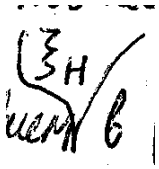
Original source details: GELFAND-P001109

Diagram and typed description. The two notes on the critical levels.

vol M_j — volume with respect to the induced sympl. structure.

H is constant on $M_j \Rightarrow H(M_j)$ — its value.

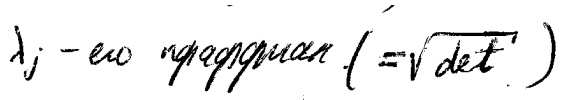
λ_j is determined by the action of [inserted: ξ_H] on (T_xM/T_xM_j) (T_xM_j preserved)



Original source details: GELFAND-P001109

Diagram and typed description. The insertion ξ_H in the line on the action on the quotient space.

The action of ξ_H is skew-symmetric: λ_j — its Pfaffian $(= \sqrt{\det})$.

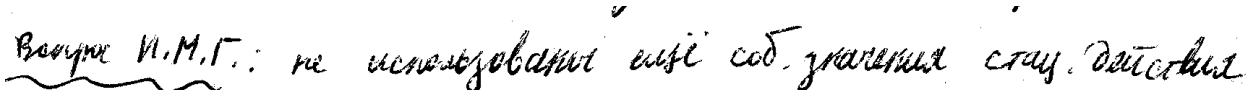


Original source details: GELFAND-P001109

Diagram and typed description. The Pfaffian and square root.

The action is equiv. to a linear one, and for a linear one everything works.

Question from I. M. G. the eigenvalues of the stationary action have yet to be used.



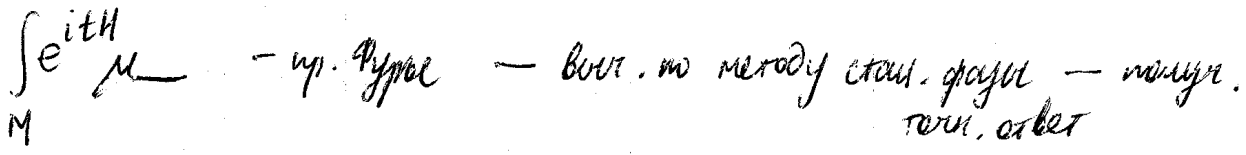
Original source details: GELFAND-P001109

Diagram and typed description. The question about the eigenvalues.

Idea of proof:

$$\int_M e^{itH}\mu$$

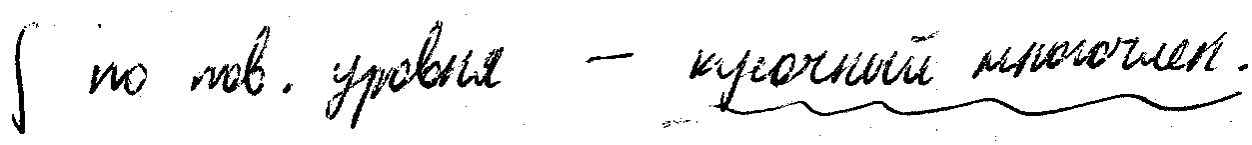
— Fourier transform — calculate by stationary phase — obtain [added below: exact answer].



Original source details: GELFAND-P001109

Diagram and typed description. The integral and the note about an exact answer.

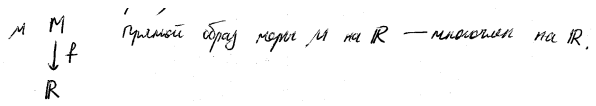
$\int$ over the level surface — piecewise polynomial.



Original source details: GELFAND-P001109

Diagram and typed description. The wavy underline below “piecewise polynomial.”

μ from the symplectic form $\Rightarrow$ everything is written in suitable local coordinates.
It suffices to prove that the sympl. form — is a polynomial (in fact a linear function)



Original source details: GELFAND-P001109

Diagram and typed description.

$$\begin{array}{c} \mu \quad M \\ \big\downarrow f \\ \mathbb{R} \end{array}$$

The pushforward of the measure μ onto $\mathbb{R}$ — is a polynomial on $\mathbb{R}$.

GELFAND-P001110

[source number: 58]

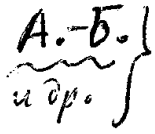


Diagram and typed description. A.–B. and others

Original source details: GELFAND-P001110

M — a compact manifold, G — a compact Lie group, acting on M.

$$H^*_G(M) \text{ — ?}$$

(cohomology)

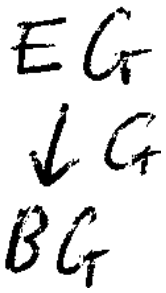
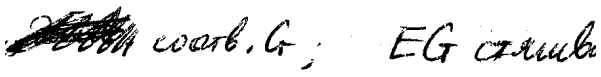


Diagram and typed description.

$$\begin{array}{c} EG \\ \big\downarrow G \\ BG \end{array}$$

Original source details: GELFAND-P001110

— universal bundle [crossed out] corresponding to G; EG is contractible.



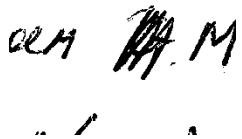
Uncertain reading. word after “bundle” illegible under the strokes

Original source details: GELFAND-P001110

Consider $EG \underset{G}{\times} M$

$$H^*_G(M) \stackrel{\text{def}}{=} H^*(EG \underset{G}{\times} M) \qquad \text{(Borel)}$$

$EG \underset{G}{\times} M$ — bundle over BG with fibre [crossed out] M.



Uncertain reading. the mark before M is heavily crossed out

Original source details: GELFAND-P001110

$H^*(EG \underset{G}{\times} M)$ — a module over $H^*(BG)$ (polynomials!)

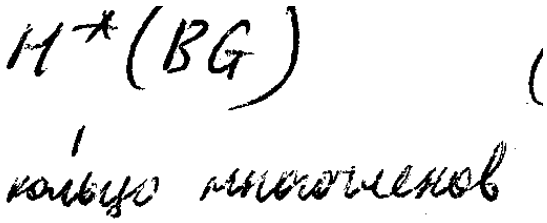


Diagram and typed description.

$H^*(BG)$

|

polynomial ring

Original source details: GELFAND-P001110

let $G = S^1$ and then $H^*(BG)$ — polynomial ring in 1 variable.

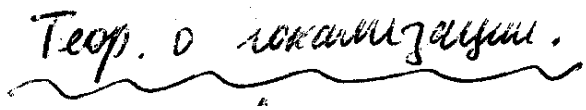


Diagram and typed description. Localization thm.

Original source details: GELFAND-P001110

M^G — set of fixed points of the action of G.

$$(1) \qquad H^*_G(M) \longrightarrow H^*_G(M^G)$$

On M^G the action is trivial $\Rightarrow$

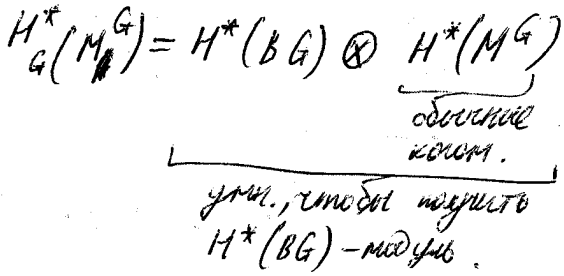


Diagram and typed description.

$H^*_G(M^G) = \underbrace{H^*(BG) \otimes \underbrace{H^*(M^G)}_{\text{ordinary cohom.}}}_{\text{mult. to obtain an } H^*(BG)\text{-module.}}$

Original source details: GELFAND-P001110



Uncertain reading. illegible crossed-out subscript at the first M

Original source details: GELFAND-P001110

Thm.

If we localize with respect to [crossed out ~~the whole~~ (?), with "the ring" inserted above[?]] of coefficients, we get an isomorphism



Uncertain reading. crossed out ~~the whole~~ (?), with "the ring" inserted above

Original source details: GELFAND-P001110

[crossed out]



Uncertain reading. illegible crossed-out mark after “isomorphism”

Original source details: GELFAND-P001110

$$(2) \qquad \mathrm{loc.}\big(H^*_G(M)\big) \xrightarrow{\sim} \mathrm{loc.}\big(H^*_G(M^G)\big)$$

(i.e. replace [inserted $H^*(BG)$] by the field [crossed out] of fractions).

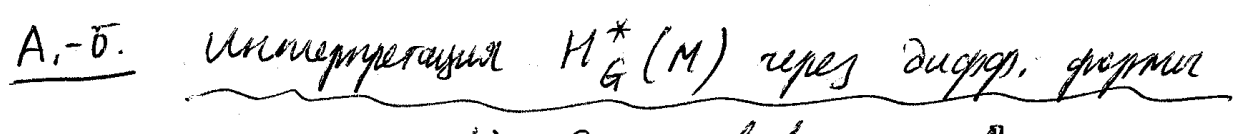


Uncertain reading. crossed-out writing between “field” and “of fractions”

Original source details: GELFAND-P001110

In K-theory — Atiyah, Segal.

In cohom. — Quillen.



Original source details: GELFAND-P001110

Diagram and typed description. A.–B. Interpretation of $H^*_G(M)$ via diff. forms



Original source details: GELFAND-P001110

Diagram and typed description. Localization thm. (2) is then equiv. to Witten’s thm. on a Killing field.

$G = S^1$

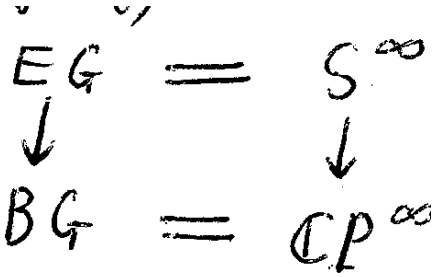


Diagram and typed description.

$$\begin{array}{ccc} EG & = & S^\infty \\ \downarrow & & \downarrow \\ BG & = & \mathbb{C}P^\infty \end{array}$$

Original source details: GELFAND-P001110

(inf.-dim. sphere)

$$H^*(BG) = \mathbb{Q}[u], \quad \deg u = 2.$$

GELFAND-P001111

ξ — the vector field determining the action of S^1.

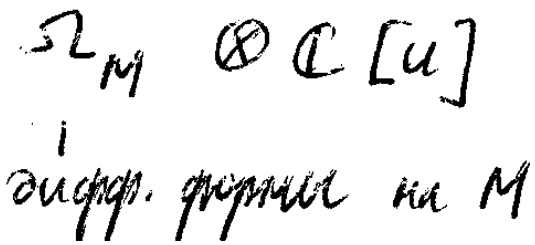
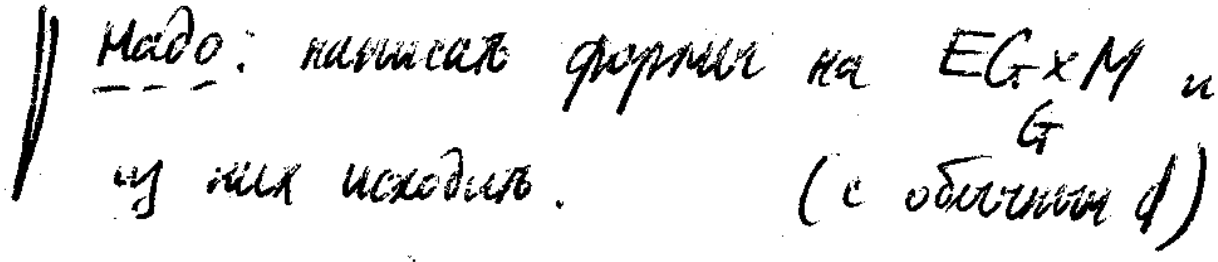


Diagram and typed description.
$\Omega_M \otimes \mathbb{C}[u]$
|
diff. forms on M

Original source details: GELFAND-P001111

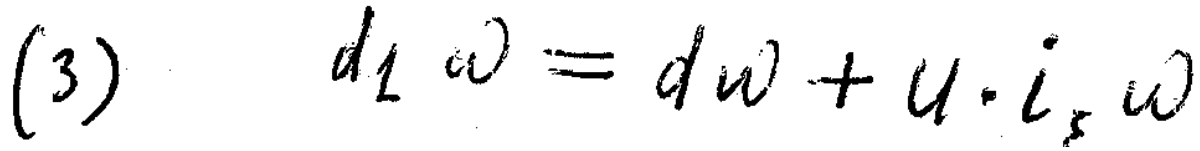
(to obtain a $\mathbb{C}[u]$-module)

Differential in $\Omega_M \otimes \mathbb{C}[u]$:



Original source details: GELFAND-P001111

Diagram and typed description. [at right] || Need to: write forms on $EG \underset{G}{\times} M$ and start from them. (with ordinary d)



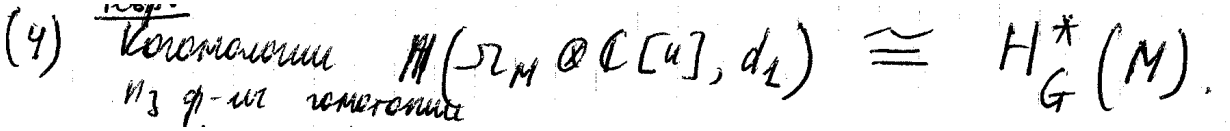
Original source details: GELFAND-P001111

Diagram and typed description. (3) $d_1\omega = d\omega + u \cdot i_\xi \omega$

(twist as in Witten)

The degrees agree!

Thm.



Original source details: GELFAND-P001111

Diagram and typed description. (4) Cohomology $H(\Omega_M \otimes \mathbb{C}[u], d_1) \cong H^*_G(M)$.

From the homotopy formula

$$L_\xi = d \circ i_\xi + i_\xi \circ d$$

it follows that



Diagram and typed description. [crossed out ~~from~~]

Original source details: GELFAND-P001111

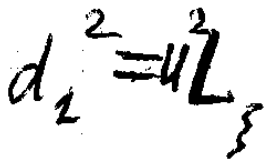


Diagram and typed description. $d_1^2 = u^2 L_\xi$

Original source details: GELFAND-P001111

Hence if $d_1\omega = 0$, then ω is invariant, i.e. $L_\xi\omega = 0$.

Twisted forms: [crossed out] ξ-invariant!! (from the very beginning).



Uncertain reading. heavily crossed-out mark before ξ

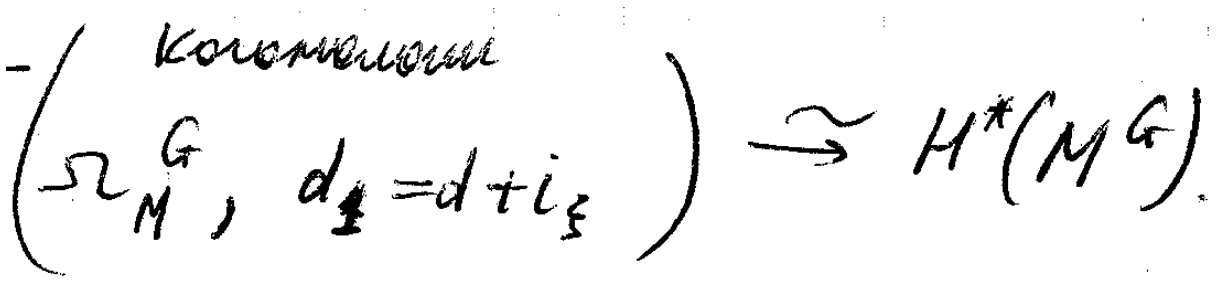
Original source details: GELFAND-P001111

$$\Omega^G = \Omega_M^G \otimes \mathbb{C}[u], \qquad \Omega_M^G \text{ — inv. forms on } M,$$

A. Goncharov: this is $\sim$ the Chern–Weil constr. of char. classes from connections.

Consequence of localization: u is erased.

(5) Thm.



Original source details: GELFAND-P001111

Diagram and typed description. $\left(\begin{matrix}\text{Cohomology} \\ \Omega_M^G, \quad d_1 = d + i_\xi\end{matrix}\right) \xrightarrow{\sim} H^*(M^G).$

I. M. G.'s problem. Formulate the same thing without G in the language of the double bundle.

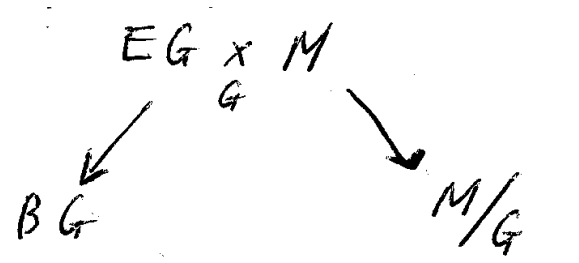


Diagram and typed description.

$$\begin{array}{ccc} & EG \underset{G}{\times} M & \\ \swarrow & & \searrow \\ BG & & M/G \end{array}$$

Original source details: GELFAND-P001111



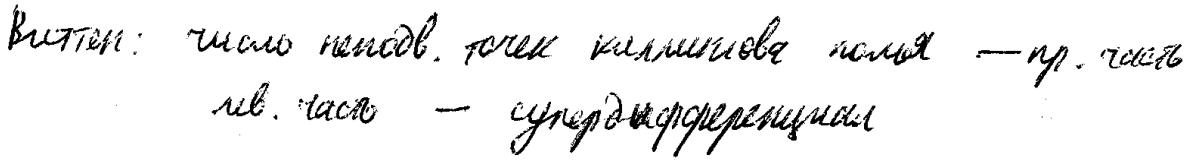
Diagram and typed description. — · —

Original source details: GELFAND-P001111

On the left in (5) there is no division into degrees — only parities.

We restrict to M^G: $i_\xi = 0$ there and we get the ordinary d!

This is precisely the map in (5)



Original source details: GELFAND-P001111

Diagram and typed description. Witten: the number of fixed points of a Killing field — right-hand side.
left-hand side — superdifferentials

GELFAND-P001112

[source number: 59]

Now the work of M. Vergne (its correct reading):

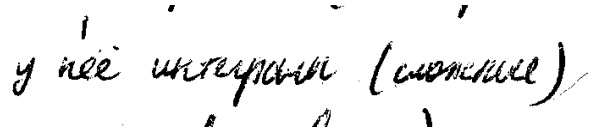


Diagram and typed description. [under the name] her integrals (complicated)

Original source details: GELFAND-P001112

Lemma (main). If there are no fixed points, the cohomology on the left is $= 0$.

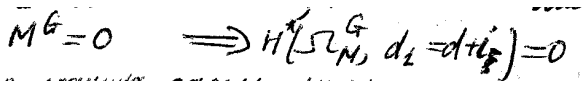


Diagram and typed description.
$M^G = 0 \implies H^*(\Omega_M^G, d_1 = d + i_\xi) = 0.$

Original source details: GELFAND-P001112

M. Vergne: wrote down explicitly what the cycle under consideration is a boundary of:

General argument: remove the fixed points and do the same thing.

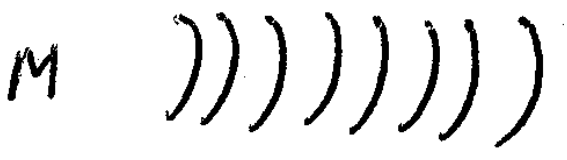
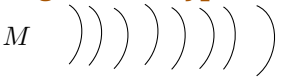


Diagram and typed description.



Original source details: GELFAND-P001112

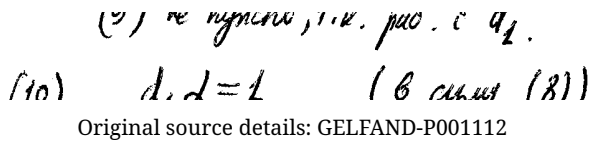
— circle S^1 acts without fixed points.

$$(6) \qquad \alpha = d\varphi \qquad (1\text{-form})$$

Properties (desired)

$$(7) \quad L_\xi \alpha = 0$$
$$(8) \quad i_\xi \alpha = 1$$
$$(9) \quad d\alpha = 0.$$

(*) Cond. (7), (8) can be satisfied $\alpha(\cdot) = \dfrac{\langle \xi, \cdot \rangle}{\langle \xi, \xi \rangle}$.



Original source details: GELFAND-P001112

Diagram and typed description. (9) is unnecessary, since we work with d_1. Let us rewrite (9):

$$(10) \qquad d_1 \alpha = 1 \qquad \text{(by (8))}$$

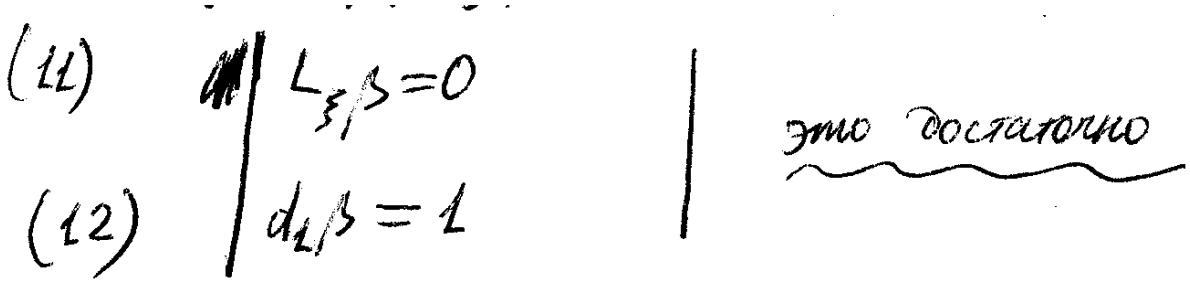
$d\alpha = 1$ is impossible.

Make a form β:

Uncertain reading. heavily crossed-out mark before the left rule at (11)



Original source details: GELFAND-P001112



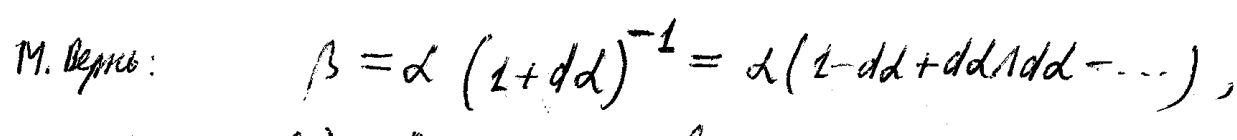
Original source details: GELFAND-P001112

Diagram and typed description.

$$\begin{vmatrix} (11) & L_\xi \beta = 0 \\ (12) & d_1 \beta = 1 \end{vmatrix}$$

this is sufficient

(13) M. Vergne:



Original source details: GELFAND-P001112

Diagram and typed description.

$$\beta = \alpha(1 + d\alpha)^{-1}$$
$$= \alpha(1 - d\alpha + d\alpha \wedge d\alpha - \ldots),$$

where α — is formula (*). Everything is easily checked.

Devise it: start with α and use perturbation theory.

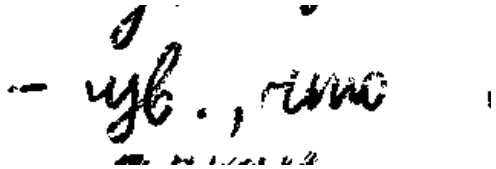
We have the identity:

$$\omega = d_1(\beta \wedge \omega) + \beta \wedge d_1\omega, \tag{14}$$

hence if $d_1\omega = 0$, then $\omega = d_1(\beta \wedge \omega)$.

Calculation of the integral over M (of a d_1-closed form): $\forall$ such integral reduces to integration over the set of fixed points. By Stokes' theorem! —

— "ув." (abbreviation unresolved), that $\omega = d_1(\beta \wedge \omega)$;



Uncertain reading. "ув." (abbreviation unresolved)

Original source details: GELFAND-P001112

the asymptotics must be calculated as we approach the fixed points.

GELFAND-P001113

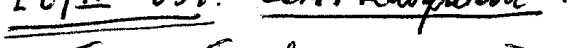
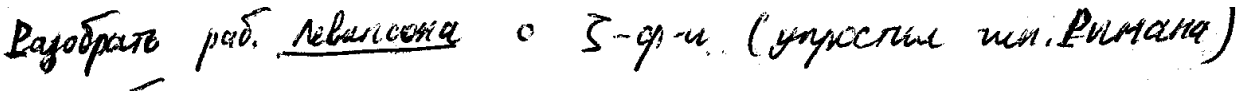


Diagram and typed description. 18/IV-83 Gelfand Sem..

Original source details: GELFAND-P001113



Original source details: GELFAND-P001113

Diagram and typed description. Work through the paper by Levinson on the ζ-function. (simplified the Riemann hyp.)

V. Ginzburg.

M — a symplectic compact manifold.

Let the torus $\mathbb{T}^n$ act on it.

Moment map $M \xrightarrow{\mu} t^*$

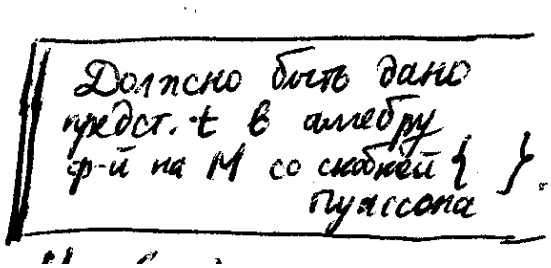


Diagram and typed description.

> A repr. must be given of t in the algebra of functions on M with the Poisson bracket $\{\}$.

Original source details: GELFAND-P001113

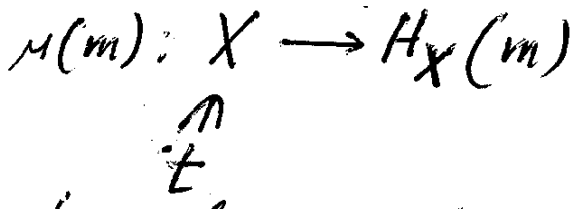


Diagram and typed description.

$$\mu(m): \quad X \longrightarrow H_X(m)$$
$$\uparrow$$
$$t$$

Original source details: GELFAND-P001113

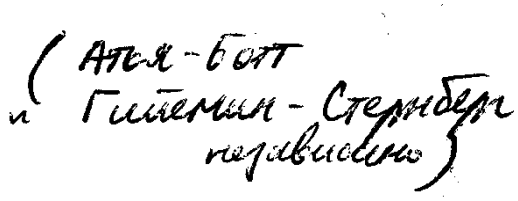
H_X — the Hamiltonian fn. corresponding to the elem. X (as a vector field on M)
(we are given H_X for each X).
[in the left margin !!]



Original source details: GELFAND-P001113

Uncertain reading. abbreviation below "!!": "к." (?)

(1) Claim. $\mu(M)$ — a convex polytope.



Original source details: GELFAND-P001113

Diagram and typed description. [at right] (Atiyah–Bott and Guillemin–Sternberg independently)

There is also the complex case.

(Duistermaat and [crossed out] Heckman — the fn. defined in the last talk — a piecewise polynomial on $\mu(M)$).



Original source details: GELFAND-P001113

Uncertain reading. illegible crossed-out writing before "Heckman"

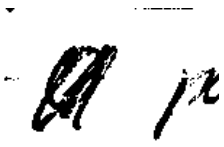
Faces (transition) — where $d\mu$ degenerates.

They are planes: apply to the degeneracy manifolds where the rank drops by several units.

A. Beilinson.

Values at integers of ζ-functions of varieties over number fields.

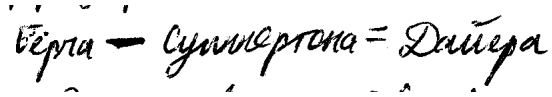
1. Statements up to multiplication by a [crossed out] rat. number $\neq 0$ (an element of $\mathbb{Q}^*$)



Original source details: GELFAND-P001113

Uncertain reading. crossed-out Q (?) before "rat."

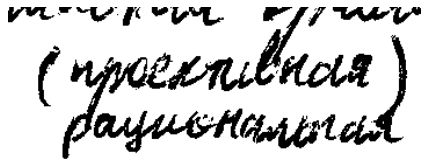
2. Precise statements.



Original source details: GELFAND-P001113

Diagram and typed description. Conjecture of Birch — Swinnerton = Dyer

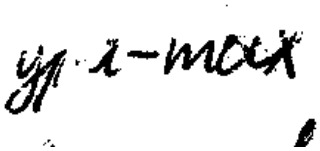
$X/\mathbb{Q}$ — a smooth curve; $\zeta(x,s)$ — its ζ-function.



Original source details: GELFAND-P001113

Diagram and typed description. [below "smooth curve"] (projective) rational

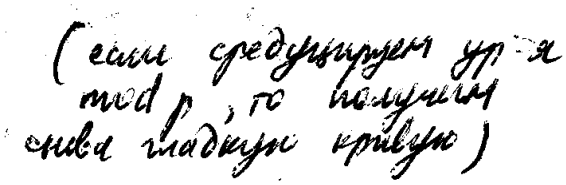
$X/\mathbb{Z}$ "уд. л-щая" (?), satisfying (?) nondeg. reduction for any p.



Original source details: GELFAND-P001113

Uncertain reading. "уд. л-щая" (?), satisfying (?)

(the eqs. are chosen sufficiently well).



Original source details: GELFAND-P001113

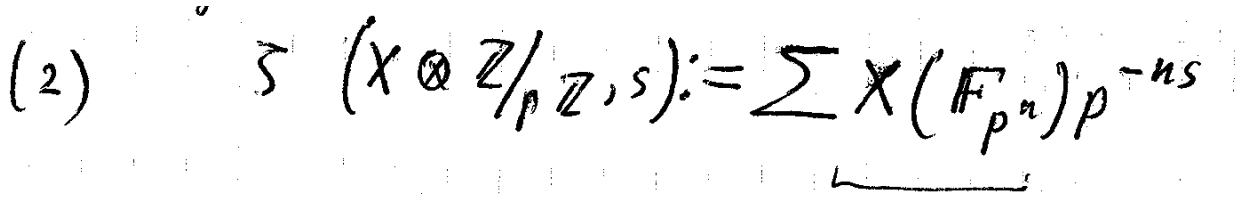
Diagram and typed description. [at right] (if we reduce the eqs. mod p, then we again obtain a smooth curve)

GELFAND-P001114

$$(1)\quad \zeta(X,s) := \prod_p \zeta(X \otimes \mathbb{Z}/p\mathbb{Z}, s).$$

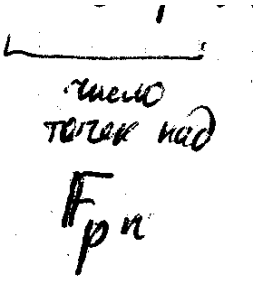
Here

$$(2)\quad \zeta(X \otimes \mathbb{Z}/p\mathbb{Z}, s) := \sum X(\mathbb{F}_{p^n}) p^{-ns}.$$



Original source details: GELFAND-P001114

Diagram and typed description. Formula (2) as written.



Original source details: GELFAND-P001114

Diagram and typed description. $\underbrace{X(\mathbb{F}_{p^n})}$ number of points over $\mathbb{F}_{p^n}$

Ordinary ζ-function

$$\prod \frac{1}{1-p^{-s}} = \prod (1 + p^{-s} + p^{-2s} + \ldots)$$

Here X — a point (i.e. $\mathbb{Q}$).



Diagram and typed description. — · —

Original source details: GELFAND-P001114

I. M. G.'s question.

$$\prod_p \sum a_n(p) p^{-ns}$$

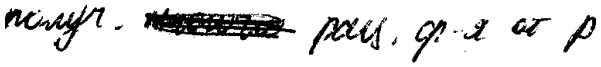
what are reasonable $a_n(p)$ (they are $\equiv 1$ for the ordinary ζ-function).

Weil conjecture (proved for a curve): one obtains [crossed out] a rat. function of p^{-s}



Uncertain reading. heavily crossed-out word between "obtains" and "rat."

Original source details: GELFAND-P001114

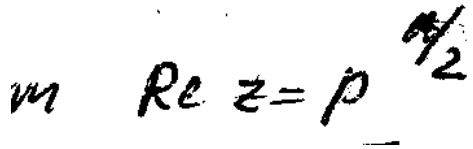


Undeciphered passage. crossed-out word illegible

Original source details: GELFAND-P001114

$$\frac{\prod(1-\alpha_i p^{-s})}{\prod(1-\beta_j p^{-s})}$$

Zeros at Re $z = p^{?/2}$.



Uncertain reading. overwritten numerator of the exponent: 1 or n (?)

Original source details: GELFAND-P001114

For curves even the fact of analytic continuation has not been proved for $\zeta(X, s)$.

(Over finite fields — full information [inserted below: is available]).



Diagram and typed description. inserted below: is available.

Original source details: GELFAND-P001114

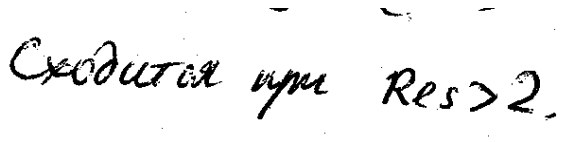
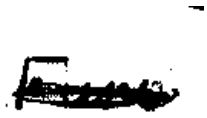


Diagram and typed description. [at right] Converges for Re $s > 2$.

Original source details: GELFAND-P001114

— question of compatibility for different p.

Known: ζ-functions of Langlands can be continued.



Uncertain reading. crossed-out word before “Conjecture”

Original source details: GELFAND-P001114



Undeciphered passage. initial stroke resembles “Б”, with the rest illegible

Original source details: GELFAND-P001114

Conjecture (strong): these 2 classes coincide, of ζ-functions.



Diagram and typed description. — · —

Original source details: GELFAND-P001114



Uncertain reading. crossed-out mark at left between the entries

Original source details: GELFAND-P001114

Wodzicki M. Which [crossed out ~~series~~ (?)] ζ-functions can be calculated?

Uncertain reading. crossed-out “series” (?)



Original source details: GELFAND-P001114



Undeciphered passage. the last letters cannot be read with confidence

Original source details: GELFAND-P001114

Shintani: Dedekind ζ-function [crossed out] reduces to ζ-functions of pseudodifferential operators.



Uncertain reading. crossed-out mark after “Dedekind”

Original source details: GELFAND-P001114

$$\zeta_{K/\mathbb{Q}}(s) = \sum N\mathfrak{a}^{-s} = \sum a_n n^{-s}.$$



Uncertain reading. heavily retraced sign after N: $\mathfrak{a}$ or a (?)

Original source details: GELFAND-P001114

Expr. via pseudodifferential operators on a torus (with constant coefficients).

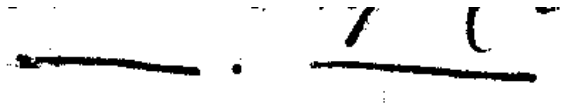


Diagram and typed description. — · —

Original source details: GELFAND-P001114

B. Birch — Swinn. = Dyer conjecture



Diagram and typed description. [in the left margin] X — a curve.

Original source details: GELFAND-P001114

1) Can be continued analytically.

2) functional eq.

$$(2\pi)^{-s}\Gamma(s)\zeta(X,s)$$

symmetric under the substitution $s \to 2 - s$

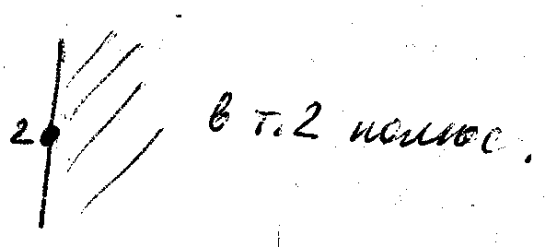
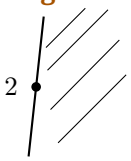


Diagram and typed description.



pole at pt. 2.

Original source details: GELFAND-P001114

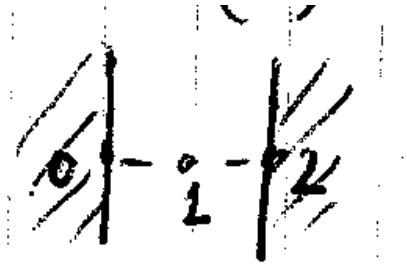
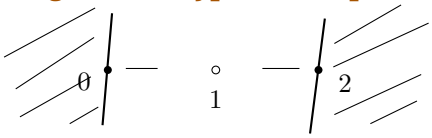


Diagram and typed description.



Original source details: GELFAND-P001114

$0 \leq \mathrm{Re}\, s \leq 2$ — critical strip.

GELFAND-P001115



Diagram and typed description. 5/IX-83 Sem. Gelfand.

Original source details: GELFAND-P001115



Diagram and typed description. Deligne.

Original source details: GELFAND-P001115

Finiteness theorems for the cohomology of families of algebraic varieties.

Work of FALTINGS and its variations. All over $\mathbb{C}$.

C — an alg. curve (compact Riem. surface)

S — a finite subset of C.

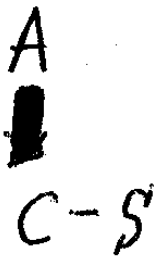


Diagram and typed description.

$$\begin{array}{c} A \\ | \\ C - S \end{array}$$

lower end of the connecting mark is blacked out.

Original source details: GELFAND-P001115

(family of alg. varieties parametrized by points of $C - S$)

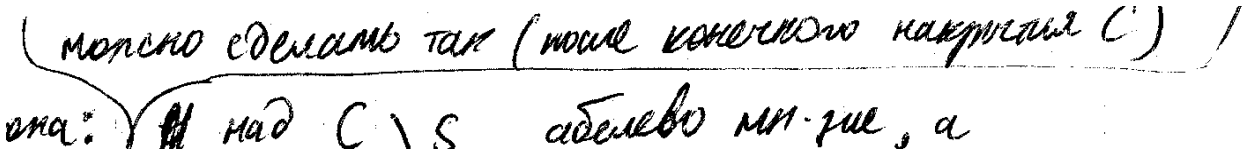


Diagram and typed description.

$$\begin{array}{c} \mathcal{A} \\ | \\ C \end{array}$$

Original source details: GELFAND-P001115

Néron model:



Original source details: GELFAND-P001115

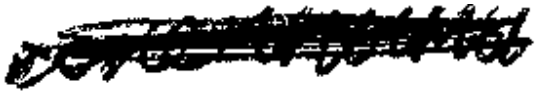
Diagram and typed description. [inserted above] this can be done (after a finite cover of C).

$\mathcal{A}$ (?) over $C \setminus S$ — an abelian variety, and at S — an extension of an ab. variety by [crossed out] mult. groups $\mathbb{C}^*$.



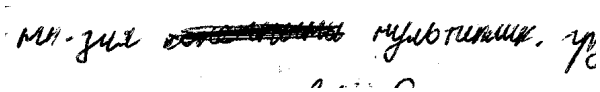
Original source details: GELFAND-P001115

Uncertain reading. inked-over letter before "over": $\mathcal{A}$ (?)



Original source details: GELFAND-P001115

Uncertain reading. long crossed-out word before "mult."



Original source details: GELFAND-P001115

Undeciphered passage. letters of the crossed-out word cannot be read with confidence

Problem — extend the bundle from $C - S$ to all of C.

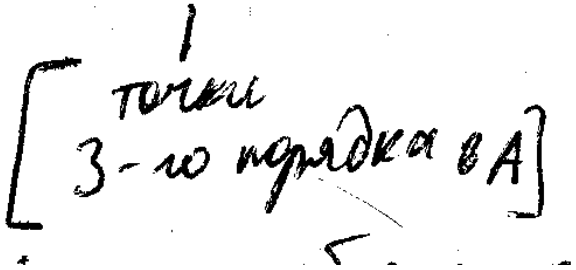
$$A = \text{Lie } A/\Gamma, \qquad \Gamma \text{ — a lattice, varying from fibre to fibre.}$$

$$A_3 = \tfrac{1}{3}\Gamma/\Gamma \simeq \underline{\underline{\text{locale...}}} \,.\, (\mathbb{Z}/3)^{2d}, \qquad d = \dim_{\mathbb{C}}.$$



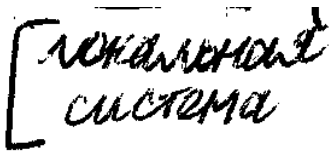
Original source details: GELFAND-P001115

Uncertain reading. "locale" is visible, with its ending heavily overwritten



Original source details: GELFAND-P001115

Diagram and typed description. [below A_3] [points of order 3 in A]



Original source details: GELFAND-P001115

Diagram and typed description. [below the Latin-script word] [local system

(any number is possible ≥ 3)

If the system is trivial, we can extend A to S.

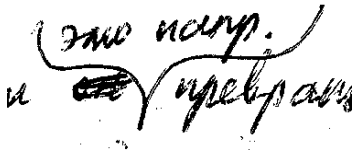
$$(\mathbb{C}^*)^e \longrightarrow \mathcal{A} \longrightarrow \underset{b}{B}; \qquad b + e = d.$$

One period of the torus increases and $\to \infty$ (and [crossed out] [inserted: this dir.] becomes $\mathbb{C}^*$) (Néron).



Original source details: GELFAND-P001115

Uncertain reading. crossed-out word after "and"

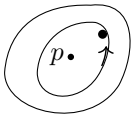


Original source details: GELFAND-P001115

Diagram and typed description. inserted above this dir..



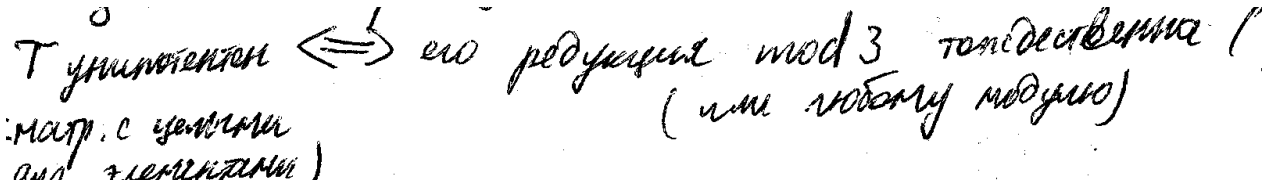
Original source details: GELFAND-P001115

Diagram and typed description. 

$T : \Gamma \to \Gamma$ (monodromy) around the point $p \in S$.

Eigenvalues of T — roots of 1.

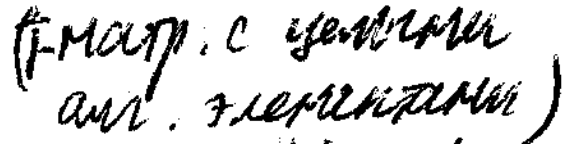
T is unipotent $\Longleftrightarrow$ its reduction mod3 is the identity (unit) (or any modulus).



Original source details: GELFAND-P001115

Diagram and typed description. The two-way arrow and addition in the manuscript.

(Γ (?)-matrix with alg. integer entries)



Original source details: GELFAND-P001115

Uncertain reading. the initial looks like Γ; T is possible

$$\left(\bigwedge^d \underline{\text{Lie}}\right)^\vee = \omega = \text{line bundle}/C.$$

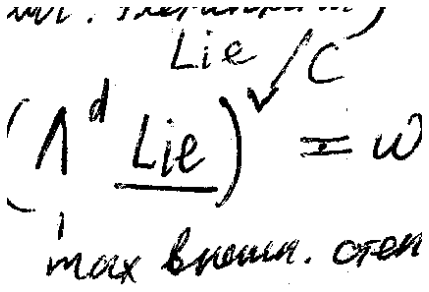
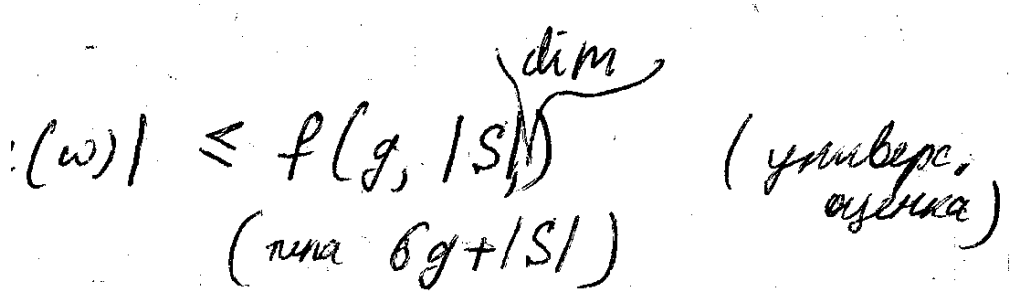


Diagram and typed description. [above] Lie $/C$.
[below $\bigwedge^d$] max exterior power.

Original source details: GELFAND-P001115

Thm. (Faltings)

$$|\deg(\omega)| \leq f(g, |S|, \overset{\text{ins.}}{\dim})$$



Original source details: GELFAND-P001115

Diagram and typed description. (universal bound)
(of the type $6g + |S|$)

Example.

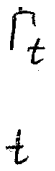


Diagram and typed description. [at left] Γ_t / t

Original source details: GELFAND-P001115

$$\pi_1(C - S, t) \longrightarrow GL(\Gamma_t)$$

This is one of the few ways to construct interesting discrete groups.

GELFAND-P001116

[source number: 76]

A way to construct a family of ab. varieties.
$C \subset$ [crossed out] $(\mathbb{P}^1)^5$



Uncertain reading. heavily crossed-out mark before $(\mathbb{P}^1)^5$

Original source details: GELFAND-P001116

$$y^N = \prod (x - x_i)^{m_i}$$

(curve depending on 5 [inserted above: distinct] points x_i)

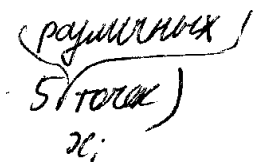


Diagram and typed description. inserted above: distinct.

Original source details: GELFAND-P001116

Take the Jacobian $\longrightarrow$ we obtain a family of abelian varieties.

(C — a curve in the set of quintuples of points)

S — the intersection of C with the set where some x_i coincide.

This is how, in $SU(2,1)$ and $SU(3,1)$ nonarithmetic subgroups were constructed...

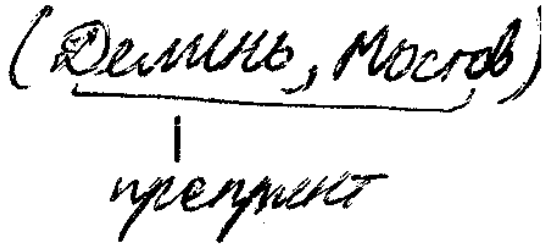


Diagram and typed description.

(Deligne, Mostow)

|

preprint

Original source details: GELFAND-P001116

This is related to hyperg. (?) functions



Uncertain reading. the abbreviation before “functions” reads “гип.” (?)

Original source details: GELFAND-P001116

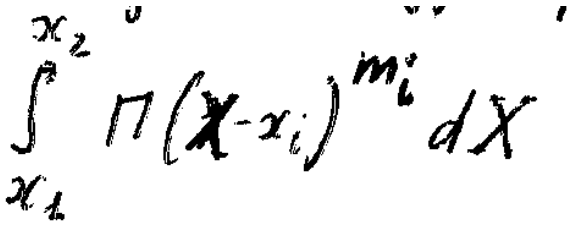


Diagram and typed description.

$$\int_{x_1}^{x_2} \prod (X - x_i)^{m_i}\, dX$$

Original source details: GELFAND-P001116

Terada (recently)

Lauricella (last century)

2 consequences of Faltings' thm.

a) Fix C, S and consider families with a principal polarization (e.g. Jacobians) $\longrightarrow$ only finitely many representations arise in this way (up to isom.)

b) vary the curve but fix the genus and the number of points. We get a repr. of another group. Still a "finite number": the curves and S are connected by a homotopy...

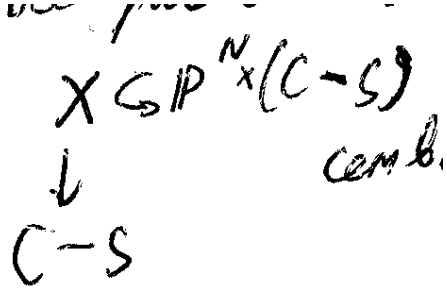


Diagram and typed description.

$$\begin{array}{c} X \subset \mathbb{P}^N \times (C - S) \\ \downarrow \\ C - S \end{array}$$

Original source details: GELFAND-P001116

a family of proj. nonsing. varieties (now nonabelian, generally speaking).

$$\pi_1(C - S, t) \longrightarrow GL\big(H_i(X_t, \mathbb{Z})/\text{Torsion}\big).$$



Diagram and typed description. inserted above: GL.

Original source details: GELFAND-P001116

c) dim H_i is bounded $\Longrightarrow$ we obtain a finite number of reprs..

Diagram and typed description. $\longrightarrow$ Hodge structure on homology.



Original source details: GELFAND-P001116

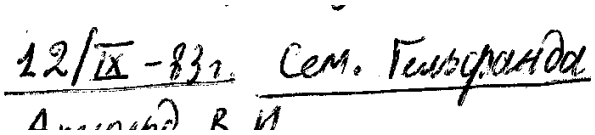


Diagram and typed description. 12/IX-83 Gelfand Sem.

Original source details: GELFAND-P001116

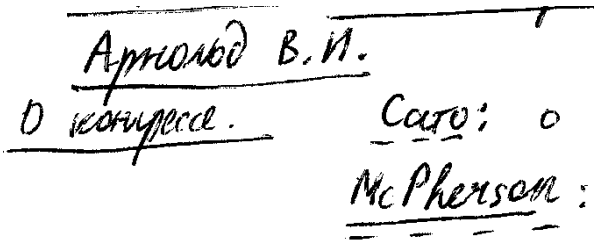


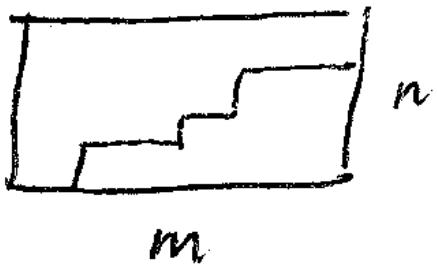
Diagram and typed description. Arnold V. I. On the congress. Sato: on correlation functions and the like
McPherson: global questions in the theory of singular spaces

Original source details: GELFAND-P001116

Stanley's talk. Combinatorial consequences of the hard Lefschetz thm.

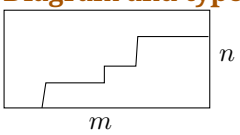
(on the hyperplane section)

$m, n \in \mathbb{Z}, \quad m, n > 0.$



Original source details: GELFAND-P001116

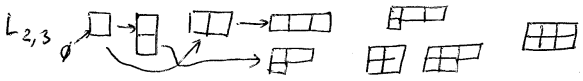
Diagram and typed description.



Consider all Young diagrams that fit inside.

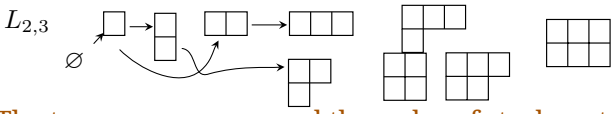
Order all the diagrams (partially, by inclusion).

$L_{m,n}$ — the resulting partially ordered set.



Original source details: GELFAND-P001116

Diagram and typed description.



The two curves cross, and the order of strokes at the crossing is unclear.

GELFAND-P001117

Width of the lattice = max. number of pairwise incomparable diagrams.

Thm. Width of the lattice $L_{m,n}$ = the number of diagrams of middle size (i.e. of size $[\frac{1}{2}mn]$).

See above; the width of $L_{2,3}$ equals 2.

(There is no simple combinatorial proof)

Proof: consider the Grassmannian $G_{m,m+n}$. Its homology is computed by means of Schubert cells (constructed by means of a frame)

(Flag; rows are dimensions of intersections with different spaces of the flag)

Cells = homology (since all the dimensions are even)

Area of the diagram = dimension of the cell.

Inclusion of diagr. = inclusion of cells.

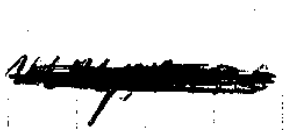
The variety is Kähler (and even projective — the Plücker map, a special case of the Veronese map for toric varieties)

$$H^i \xrightarrow{\wedge\omega^{N-i}} H^{2N-i}$$ This is an isomorphism (Lefschetz thm.)

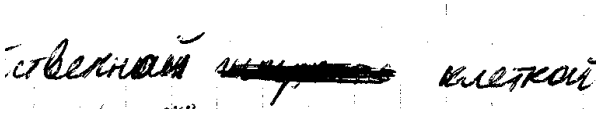
(exterior mult.)

In homology: intersection with the dual [crossed out: ~~Schubert~~ (?)] cell.

Uncertain reading. crossed-out "Schubert" (?)



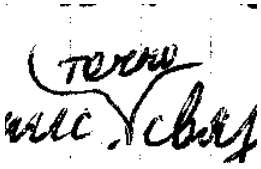
Original source details: GELFAND-P001117



Original source details: GELFAND-P001117

Undeciphered passage. middle of the crossed-out word is illegible

I. M. G. Horrocks (comparatively recently) — computed [inserted above: exactly] the relation of the Chern classes (basis) to the basis obtained from Schubert cells.



Original source details: GELFAND-P001117

Diagram and typed description. inserted above: exactly.

A. 2nd part.

$$\alpha_0 - \alpha_1 + \alpha_2 = 2 \qquad \text{(Euler thm. for polyhedra)}$$

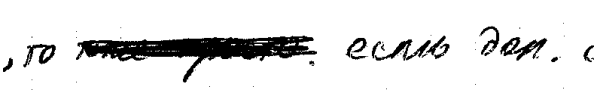
$$\alpha_0 - \alpha_1 + \ldots + (-1)^{d-1}\alpha_{d-1} = 1 + (-1)^{d-1}.$$

If all faces are simplicial, then [crossed out:] there are further relations.



Original source details: GELFAND-P001117

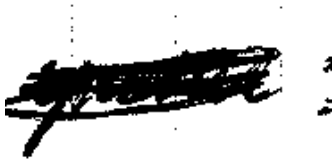
Uncertain reading. crossed-out words after “then”



Original source details: GELFAND-P001117

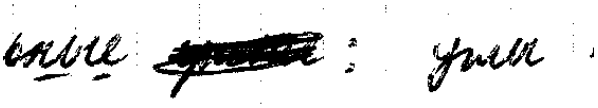
Undeciphered passage. words under the strokes cannot be read with confidence

Cosimplicial [crossed out:]: all angles are simplicial (dual to simplicial).



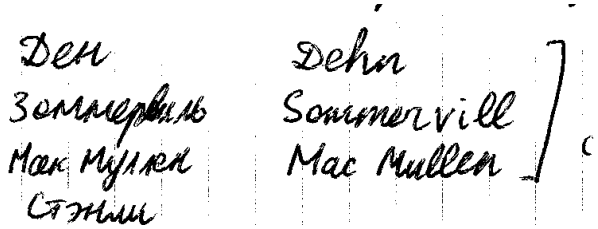
Original source details: GELFAND-P001117

Uncertain reading. crossed-out word before the colon



Original source details: GELFAND-P001117

Undeciphered passage. word beneath the strokes is illegible



Original source details: GELFAND-P001117

Diagram and typed description.

Dehn	Dehn
Sommerville	Sommervill
McMullen	Mac Mullen

Stanley

[to the right of the bracket] see Arnold's 1978 article from FA.

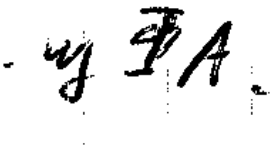


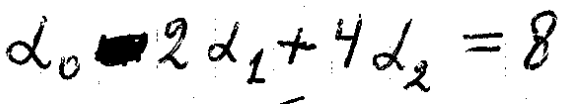
Diagram and typed description. Before "ФА" the word reads "из" (from).

Original source details: GELFAND-P001117

see references in Stanley's talk at the 1983 congress.

Mac Mullen — there is a book «Convex polyhedra»

$$\alpha_0 \,?\, 2\alpha_1 + 4\alpha_2 = 8$$



Uncertain reading. the sign before $2\alpha_1$ is covered by a solid stroke

Original source details: GELFAND-P001117

(see the cube, for example)

Def.

$$p(t) = \alpha_0 + (t-1)\alpha_1 + (t-1)^2\alpha_2 + \ldots + (t-1)^{d-1}\alpha_{d-1} + (t-1)^d$$

(Poincaré polynomial of the polyhedron)



Diagram and typed description. [crossed out: ~~Thm.~~]

Original source details: GELFAND-P001117

Dehn's thm.: 1) the polynomial $p(t)$ is reciprocal ($\Rightarrow$ identity).

2) coeffs.



Diagram and typed description.



Original source details: GELFAND-P001117

(unimodal — a hump) (Stanley)

Problem: determine whether a set of numbers can serve as the set $\alpha_0, \alpha_1, \alpha_2, \ldots, \alpha_d$.

Thm. (Billera, Lee; another proof — Stanley) Some nec. and suff. condition.

GELFAND-P001118

[source number: 77]

h_j — coefficients.

$$h_0 = 1 = h_d, \qquad h_{d-i} = h_i \qquad \text{(reciprocity)}$$

Nec. and suff. that there exist a graded algebra (assoc., commut.)

$$R = R_0 \oplus R_1 \oplus \ldots \oplus , \qquad \text{generated by } R_1 \text{ over } R_0 = \mathbb{C},$$

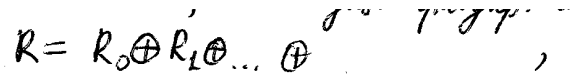


Diagram and typed description. There is a blank after the final direct sum sign.

Original source details: GELFAND-P001118

$$R_i R_j \subseteq R_{i+j}, \qquad R_0 = \mathbb{C}$$

and also

$$\dim_{\mathbb{C}} R_i = h_i - h_{i-1}, \qquad 0 \le i \le [d/2]. \quad \text{(from Lefschetz's thm.)}$$

($\Rightarrow$ unimodality, since $h_i \ge 0$)



Diagram and typed description. The manuscript has $h_i \ge 0$.

Original source details: GELFAND-P001118

Proof: from Lefschetz's thm..

From the polyhedron we construct a variety (algebraic).



Original source details: GELFAND-P001118

Diagram and typed description. [inserted between the lines] toric $\longrightarrow$ see Danilov in Uspekhi Mat. Nauk.

The dual polyhedron — simplicial.

Make it integral — approximation + dilation.

$\Rightarrow$ the variety can be constructed.

Algebra:

$$R = H^*(X)/(\omega), \qquad h_i = \dim H^{2i}(X) \qquad \text{(see Danilov)}$$

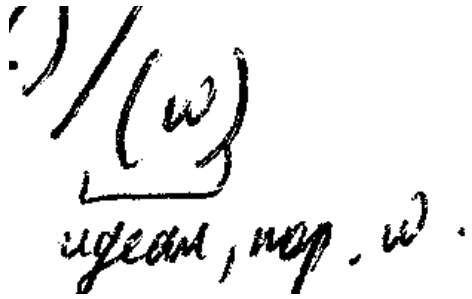


Diagram and typed description.

$\underbrace{(\omega)}$

ideal gen. by ω

Original source details: GELFAND-P001118

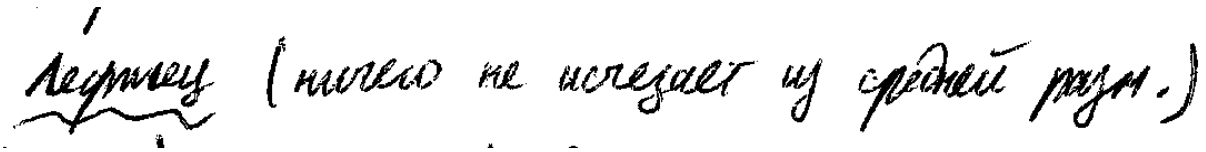
It is important that simplicial $\Rightarrow$ we get a Kähler variety.



Uncertain reading. mark at left resembling “7” or “4”

Original source details: GELFAND-P001118

$$\dim R_i = \dim H^{2i} - \dim(\omega H^{2i-2}) = \dim H^{2i} - \dim H^{2i-2} =$$



Original source details: GELFAND-P001118

Diagram and typed description. [below] Lefschetz (nothing disappears from the middle dim.).

There was also:

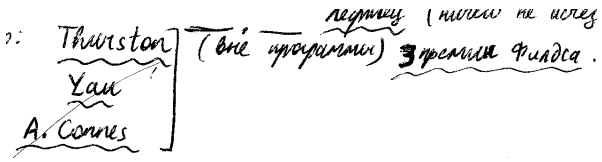


Diagram and typed description. Thurston
Yau
A. Connes
(outside the programme) 3 Fields Medals.

Original source details: GELFAND-P001118



Uncertain reading. at left, two indistinct marks before a closing parenthesis

Original source details: GELFAND-P001118

Yau — there is a survey and a list of unsolved problems (Princeton, geometry conf.).



Diagram and typed description. [below] 1–2 years ago.

Original source details: GELFAND-P001118

From the talk by Vinberg (after Mostow) it is clear these relations are the basis of the theorem of Vinberg: in Lobachevsky space of dim. ≥ 30 there are no groups gen. by reflections.

Thurston. $z \mapsto z^2 + i$

Iterations (rat. functions in general). There are works by Milnor, Douady, Sullivan. (dyn. system and so on)

Look at the image of the point 0 (crit.).

$$0 \mapsto i \mapsto -1+i \mapsto -i \mapsto -1+i \mapsto \ .$$

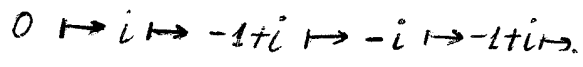


Diagram and typed description. The line in the manuscript ends with an arrow.

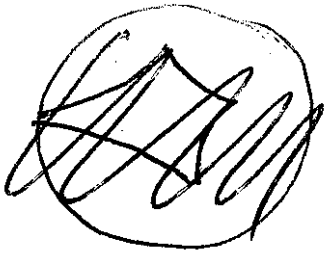
Original source details: GELFAND-P001118

The crit. point lands in a 2-cycle and stays there.

Conjecture: maps with this property are dense: the image of the crit. point becomes periodic from a certain point on.

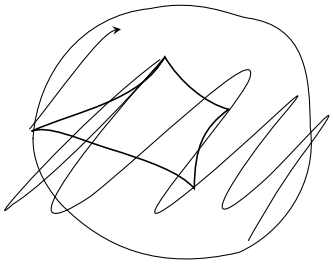
Main results — with the help of machines.

Thm. Consider a fund. domain in the Lobachevsky plane.



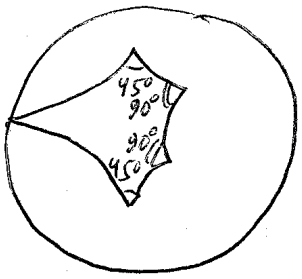
Original source details: GELFAND-P001118

Diagram and typed description.



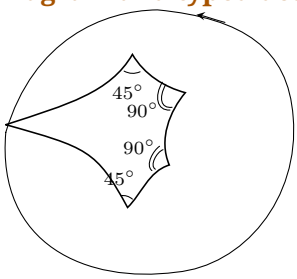
The outlines and the visible arrowhead at upper left are redrawn. The function of the overlapping diagonal strokes is unclear.

GELFAND-P001119



Original source details: GELFAND-P001119

Diagram and typed description.



Map: there is a model with negative curvature,

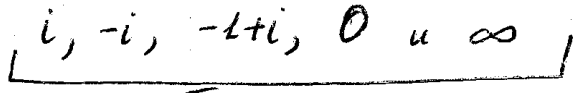
Introduce a Riem. (?) metric of neg. curvature on the plane



Original source details: GELFAND-P001119

Uncertain reading. abbreviation before “metric” reads “рим.” or “рив.” (?)

without the points



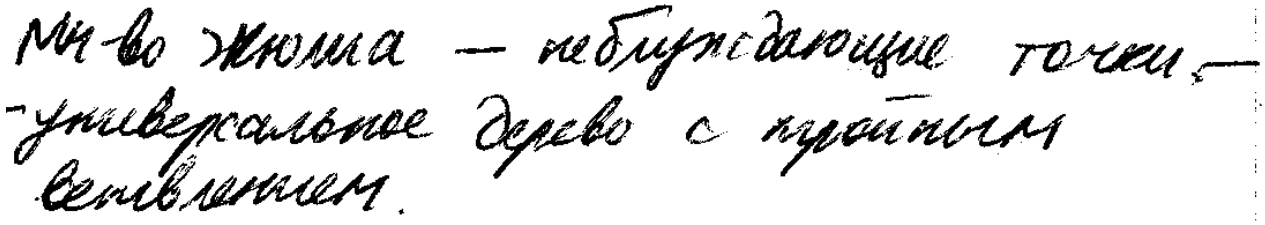
Original source details: GELFAND-P001119

Diagram and typed description.

$i,\ -i,\ -1+i,\ 0$ and ∞

(the crit. point and its images)

Take the universal cover and there a metric of const. neg. curvature.
Inverse map — expansion in this metric (uniform).
Useful — 2-sheeted cover.
On the covering surface the singular points disappear, and so on.
A tower of Riem. surfaces, and so on.
Final result:



Original source details: GELFAND-P001119

Diagram and typed description. [at right] Julia set — nonwandering points, — a universal tree with triple branching.

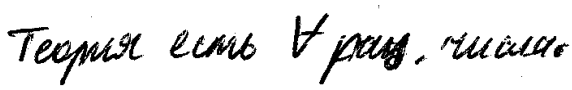


Diagram and typed description. There is a theory for ∀ rat. numbers.

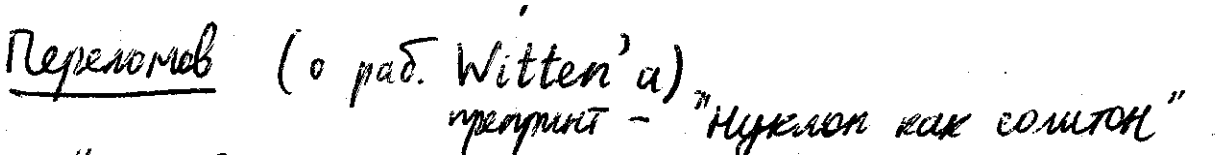
Original source details: GELFAND-P001119

GELFAND-P001120



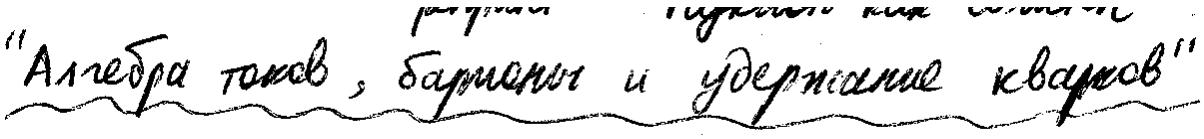
Diagram and typed description. 26/IX-83 Gelfand Sem.

Original source details: GELFAND-P001120



Original source details: GELFAND-P001120

Diagram and typed description. Perelomov (on Witten's work) preprint — «Nucleon as a soliton»



Original source details: GELFAND-P001120

Diagram and typed description. «Current algebra, baryons, and quark confinement»

«A baryon is a particle with nonzero baryon charge»

Conservation of charges (from analysis of transformations):

Q — el. charge
B — bar. charge
ζ — strangeness
L — lepton charge

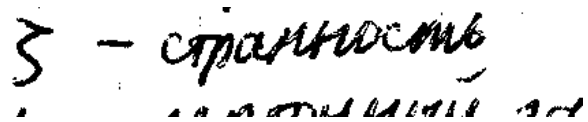
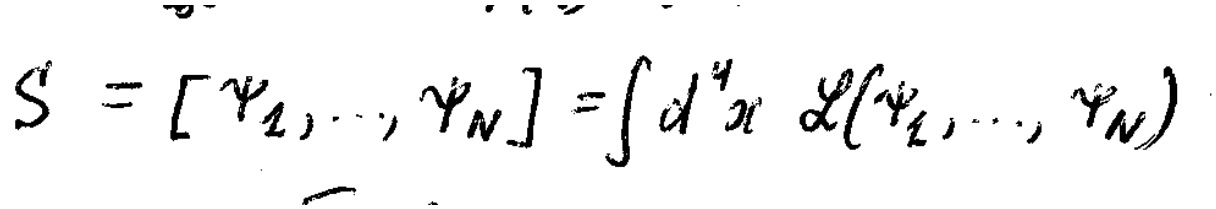


Diagram and typed description. ζ — strangeness.

Original source details: GELFAND-P001120

Before 1974 For a particle — a field $\Psi(x)$ and an action

$$S = [\Psi_1, \ldots, \Psi_N] = \int d^4x \, \mathcal{L}(\Psi_1, \ldots, \Psi_N).$$



Original source details: GELFAND-P001120

Diagram and typed description. The action line has an equals sign after S, before the brackets.

It is written so that all charges are conserved.

Particles enter in pairs $\Psi(x), \overline{\Psi}(x)$ (particle and antiparticle).

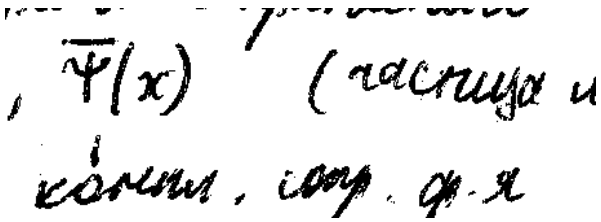


Diagram and typed description.

$\overline{\Psi}(x)$

complex conj. fn.

Original source details: GELFAND-P001120

$B = 1 \Rightarrow$ cannot write $\int \Psi^2(x) d^4x$; can write $\int \overline{\Psi}\Psi \, d^4x$.

(there would be a matrix elem. corresp. to a transition with creation of a baryon)

$\forall$ elem. particle $\rightsquigarrow$ field. The Lagrangian is written from general principles:

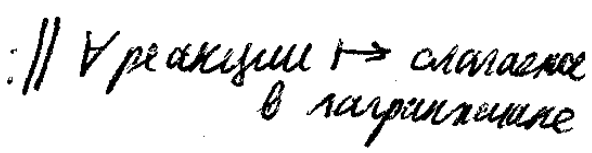
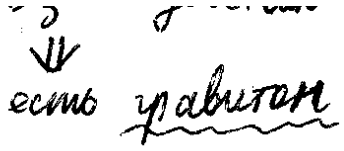


Diagram and typed description. $\|\, \forall$ reaction $\mapsto$ a summand in the Lagrangian

Original source details: GELFAND-P001120

∀ interaction also corresponds to a particle.

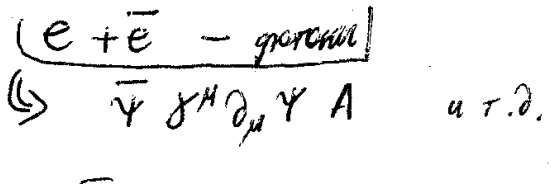


Original source details: GELFAND-P001120

Diagram and typed description.

⇓

there is a graviton



Original source details: GELFAND-P001120

Diagram and typed description.

$e + \bar{e}$ — photon

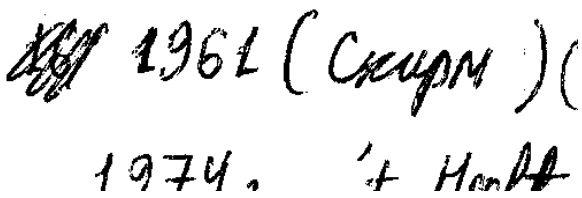
$\overline{\Psi}\gamma^{\mu}\partial_{\mu}\Psi A$ and so on

[crossed out] 1961 (Skyrme) (no attention was paid to it) — there was a geom. [crossed out] structure.



Original source details: GELFAND-P001120

Uncertain reading. crossed-out digits before 1961



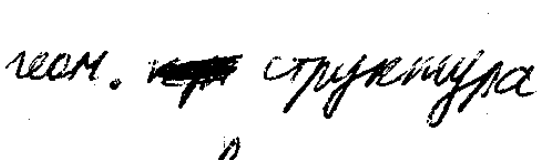
Original source details: GELFAND-P001120

Undeciphered passage. the full crossed-out date is illegible



Original source details: GELFAND-P001120

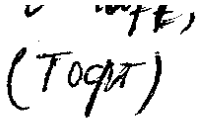
Uncertain reading. crossed-out writing before “structure”



Original source details: GELFAND-P001120

Undeciphered passage. a descending stroke is visible, but the crossed-out writing is illegible

1974 ’t Hooft, Polyakov ⇒ the geom. direction began to develop after them.



Original source details: GELFAND-P001120

Diagram and typed description. [below ’t Hooft] (Тофт).

Witten also belongs to the geom. direction.

GELFAND-P001121

[source number: 80]

The models are called chiral models or σ-models.

M^4 — Minkowski space,

$$\varphi\colon M^4 = \mathbb{R}^3 \times \mathbb{R}^1 \longrightarrow (\Phi, g).$$

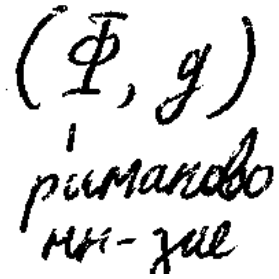


Diagram and typed description.

(Φ, g)
|
Riemannian manifold

Original source details: GELFAND-P001121

$\varphi(x)$ — field with values in a Riem. manifold ||

Action:

$$(1)\quad S = \int (\partial_\mu \varphi, \partial_\mu \varphi) d^4x$$
$$= \int (h_{\alpha\beta} \partial^\alpha \varphi, \partial^\beta \varphi) d^4x; \qquad \partial_\mu \varphi = \varphi_* \frac{\partial}{\partial x_\mu}.$$

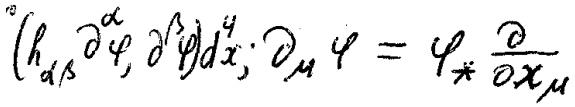


Diagram and typed description. The second integral has $h_{\alpha\beta}$, with the upper indices on ∂.

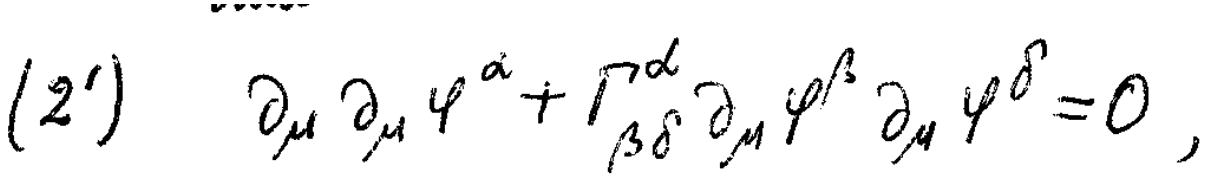
Original source details: GELFAND-P001121

Equation:

$$(2)\qquad \delta S = 0$$

or

$$(2')\quad \partial_\mu \partial_\mu \varphi^\alpha + \Gamma^\alpha_{\beta\delta} \partial_\mu \varphi^\beta \partial_\mu \varphi^\delta = 0,$$



Original source details: GELFAND-P001121

Diagram and typed description. hence this generalizes the geodesic eq..

Requirement:

$$(3) \quad \pi_3(\Phi) \neq 0, \qquad \varphi(x) = \varphi(\bar{x}, t)$$

For fixed t there is a map $\mathbb{R}^3 \to \Phi$.

$$(4) \quad \text{We require:} \quad \exists \lim_{|\bar{x}| \to \infty} \varphi(\bar{x}, t) = \varphi_0(t)$$

$$\mathbb{R}^3 \cup \{\infty\} = S^3.$$

The fields split into homot. classes (according to the map $S^3 \to \Phi$)

Let us require:

$$(5) \qquad S \geq \varepsilon \quad \text{on a homot. class}$$

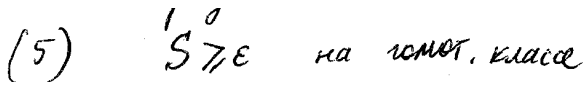


Diagram and typed description. The lower bound in (5).

Original source details: GELFAND-P001121

$\Rightarrow$ in $\forall$ homot. class there is a solution.

Examples.

(6) Φ — a simple [inserted above: compact] Lie group $\Phi = G$ (principal chiral field)

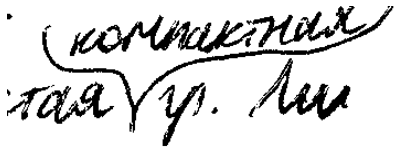


Diagram and typed description. inserted above: compact.

Original source details: GELFAND-P001121

or

(7) Φ — orbit of the adj. repr. of a compact Lie group $\mathfrak{g}$ (?)



Uncertain reading. mark after “Lie” resembles $\mathfrak{g}$ and may be crossed out

Original source details: GELFAND-P001121

(8) $\Phi = S^2$ — $\bar{n}$-field

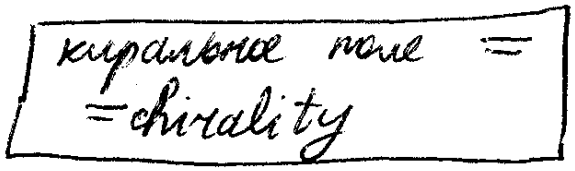
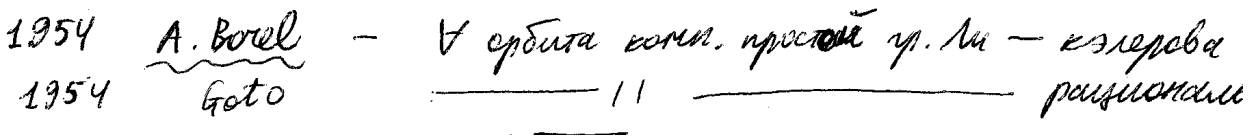


Diagram and typed description.

chiral field =
= chirality

Original source details: GELFAND-P001121



Original source details: GELFAND-P001121

Diagram and typed description. 1954 A. Borel — $\forall$ orbit of a comp. simple Lie group — is Kähler.

1954 Goto —— || —— is rational



Diagram and typed description. — · —

Original source details: GELFAND-P001121

2-dimensional case:

$$S^2 \to \Phi, \quad \Phi = G/H \qquad (G \text{ — simple compact})$$

$$\pi_2(\Phi) = \pi_1(H) \neq 0$$

Pass to a Kähler metric. The eqs. become the Cauchy–Riemann eqs.:

GELFAND-P001122

$$(9) \qquad \partial_{\bar{z}}\varphi = 0 \qquad (\Leftrightarrow \ \varphi \text{ is holomorphic})$$

(i.e. the solns. are holom. maps)

(Complex structure: $G/H \sim$ the flag space; in several ways ($|W|$ ways))

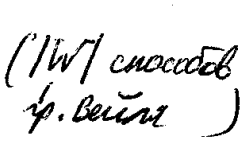


Diagram and typed description. [below W] Weyl group.

Original source details: GELFAND-P001122

Matr. = (triang.) × (unitary)

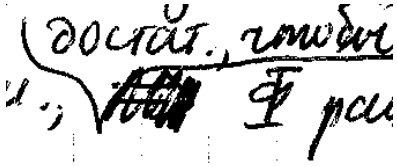
Φ is compact [crossed out]; it is unknown whether there are nonconstant solutions.



Uncertain reading. the crossed-out mark after "compact" is illegible

Original source details: GELFAND-P001122

To have a large number of solns.; [crossed out] [inserted above: (it suffices that)] Φ be rational (i.e. the field of fns. on Φ is isomorphic to the field of rat. fns. of some number of variables).



Uncertain reading. the crossed-out writing below the insertion is illegible

Original source details: GELFAND-P001122

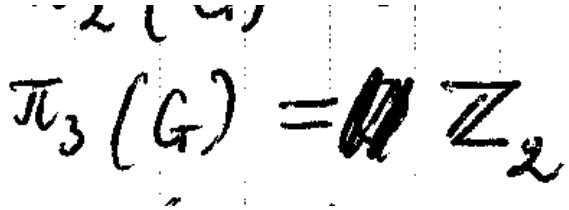
$$\mathbb{C}P^1 \longrightarrow \Phi \quad \text{(a Kähler manifold)}$$

$\Rightarrow$ there are many instanton solutions

$\Phi = S^2$ — Belavin, Polyakov. (from stereogr. projection)

$\Phi = G$ — a simple compact Lie group.

$$\pi_1(G) = 0$$
$$\pi_2(G) = 0$$
$$\pi_3(G) = \text{[crossed out] } \mathbb{Z}_2$$



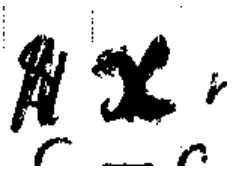
Uncertain reading. a mark resembling 0 is crossed out before $\mathbb{Z}_2$

Original source details: GELFAND-P001122

$$\pi_4(SU(3)) = 0, \quad \pi_4(SU(2)) = \mathbb{Z}_2, \quad \pi_5(SU(2)) = \mathbb{Z}_2.$$

There is no lower bound:

on replacing [crossed out] **x** by λ**x**: S by $\lambda^2 S \Rightarrow S$ can be made arbitrarily small.



Uncertain reading. the mark before **x** is crossed out, possibly φ

Original source details: GELFAND-P001122

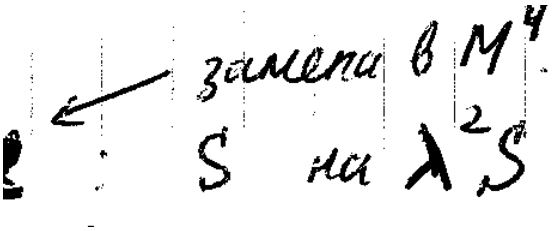


Diagram and typed description.
substitution in M^4

$\swarrow$ λ**x**

Original source details: GELFAND-P001122

(10) In fact $S = S_0 + S_1 + S_2$

$$(11) \quad S_0 = \int \operatorname{tr}(g^{-1}\partial_\mu g \cdot g^{-1}\partial_\mu g)\, d^4x$$

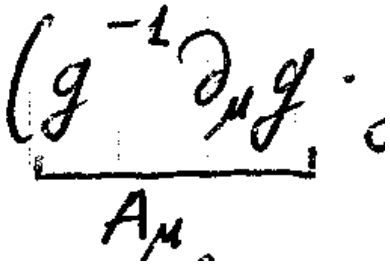


Diagram and typed description. $\underbrace{g^{-1}\partial_\mu g}_{A_\mu}$

Original source details: GELFAND-P001122

for unit. matrices ($SU(3)$ or $SU(2)$)



Diagram and typed description. [at right] A_μ — elem. of the Lie algebra.

Original source details: GELFAND-P001122

$$(12) \quad S_2 = \int d^4x \ \mathrm{tr}\big([A_\mu, A_\nu] : [A_\mu, A_\nu]\big)$$

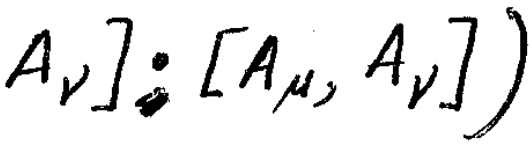
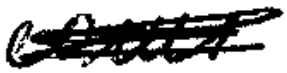


Diagram and typed description. Two dots are visible between the commutators in (12).

Original source details: GELFAND-P001122

Derivs. in t enter quadratically $\to$ already invariant under homothety in M^4

$S_0 + S_2$ already has [crossed out ~~soli...~~ (?)] instanton solutions.



Uncertain reading. the crossed-out beginning may read "soli..."

Original source details: GELFAND-P001122



Undeciphered passage. the remainder of the crossed-out word is illegible

Original source details: GELFAND-P001122

S_1 is added so as to be able to distinguish bosons from fermions

Wess, Zumino (1970) added S_1, but the formula is complicated

Witten greatly simplified this formula for S_1

Compactify time:

$$\begin{array}{c} M^4 = S^3 \times S^1 \longrightarrow G = SU(3) \\ \| \\ \partial Q, \quad Q = S^3 \times D^2 \end{array}$$

(13) Claim. The map extends to a continuous map of Q

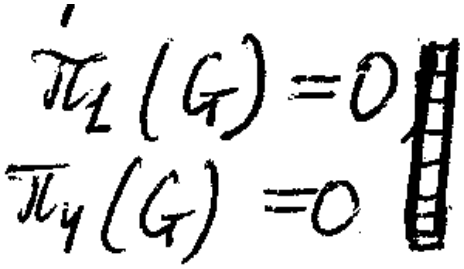


Diagram and typed description. (from the fact that $\pi_1(G) = 0$ $\pi_4(G) = 0$)

Original source details: GELFAND-P001122

[at right] Rewrite as a map

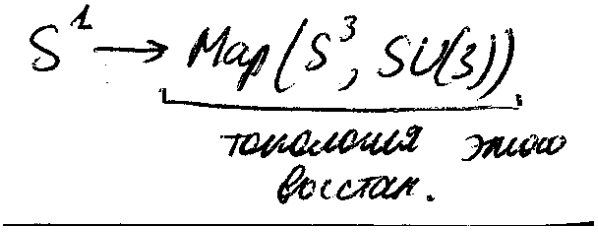


Diagram and typed description.
$S^1 \longrightarrow \underbrace{\mathrm{Map}(S^3, SU(3))}_{\text{the topology of this reconstr.}}$

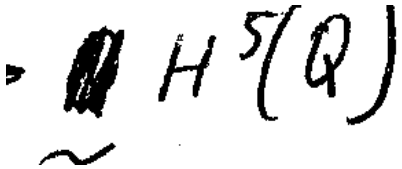
Original source details: GELFAND-P001122

GELFAND-P001123

[source number: 81]

We obtain:

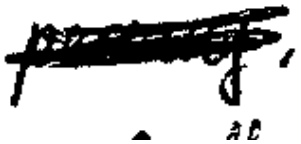
$$\varphi\colon Q \longrightarrow SU(3)$$
$$\varphi^*\colon H^5(SU(3)) \longrightarrow \text{[crossed out] } H^5(Q)$$
$$\omega \longmapsto \widetilde{\omega}$$



Uncertain reading. a mark resembling 0 is crossed out before $H^5(Q)$

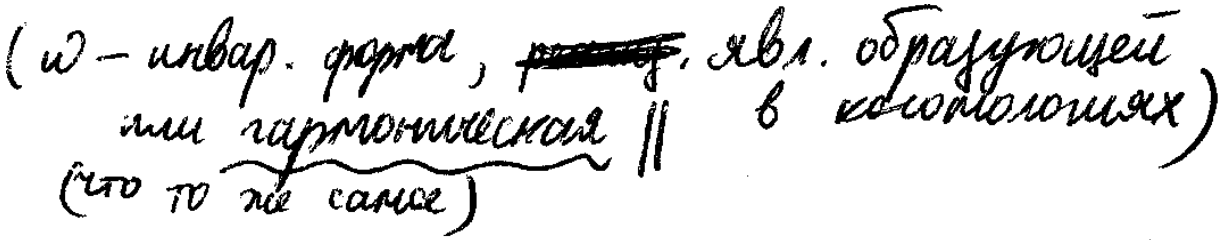
Original source details: GELFAND-P001123

[at right] (ω — an inv. form, [crossed out], generating the cohomology)



Uncertain reading. crossed-out writing after "inv. form"

Original source details: GELFAND-P001123



Original source details: GELFAND-P001123

Undeciphered passage. lower loops are visible, but the crossed-out word is illegible

or harmonic ‖ (which is the same thing)

$$(14) \qquad S_1 = N \int_Q \widetilde{\omega}$$

N is chosen to get an integer.

The choice of extension [inserted: φ] may change [crossed out] $\int$ by an [crossed out 2π (?); inserted: (integer)] multiple of 2π.



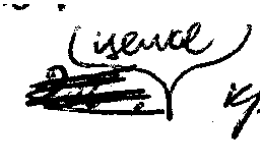
Diagram and typed description. inserted: φ.

Original source details: GELFAND-P001123



Uncertain reading. the crossed-out mark before the integral is illegible

Original source details: GELFAND-P001123



Uncertain reading. the crossed-out mark resembles 2π, with “integer” inserted above

Original source details: GELFAND-P001123

Cohomology of $SU(3)$: $(1+t^3)(1+t^5)$

Under deformations of [inserted: (the extn. of φ to)] Q the integral is unchanged

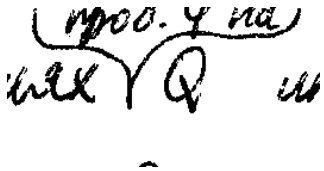


Diagram and typed description. inserted: (the extn. of φ to Q).

Original source details: GELFAND-P001123

$$(15) \qquad \int_Q \widetilde{\omega} = \int_{\varphi(Q)} \omega$$

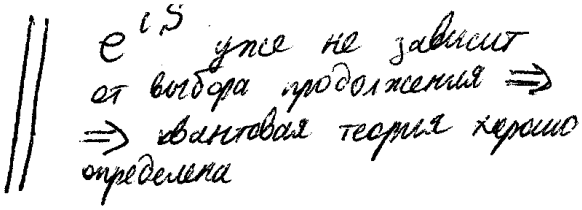


Diagram and typed description. [at right] ‖ e^{iS} is already independent of the choice of extension ⇒
⇒ the quantum theory is well defined

Original source details: GELFAND-P001123

Zumino — d-dimensional case already (preprint) and secondary char. classes of Chern–Simons appear there.

1) Baryon charge.

$$\pi_3(\Phi) = \mathbb{Z} \quad \text{and by Hurewicz's thm.} \quad H^3(\Phi) = \mathbb{Z}.$$

Expr. for baryon charge:

$$(16) \quad B = \int d^3x\, B_0 = \frac{1}{24\pi^2} \int d^3x\, \varepsilon^{ijk}\, \mathrm{tr}(A_i A_j A_k)$$

(Witten proposes calling this the baryon charge)

(invariant 3-form, the unique generator in $H^3(\Phi)$)

2) Fermion or boson?

$$\int e^{iS[\varphi]} d\varphi \qquad \text{— transition amplitude}$$

Suppose it is stationary $\Rightarrow$ the integral equals e^{-iMT}, M — mass, T — time

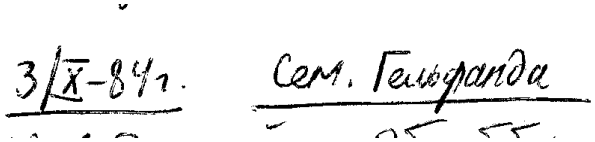


Diagram and typed description. 3/X-84 Gelfand Sem.

Original source details: GELFAND-P001123

Yu. L. Daletskii On a generalization of the Vlasov eq.

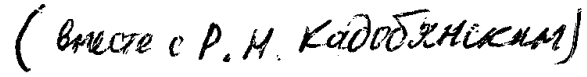
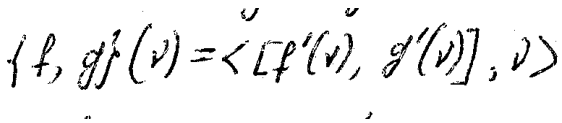


Diagram and typed description. (with R. N. Kadobianskii)

Original source details: GELFAND-P001123

$\mathfrak{g}$ — Lie alg., $\mathfrak{g}'$ — dual space, $\langle\,\rangle$ — duality.

$$\{f, g\}(\nu) = \langle [f'(\nu), g'(\nu)], \nu \rangle \qquad (f, g \text{ — fns. on } \mathfrak{g}')$$



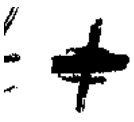
Uncertain reading. the argument letter resembles ν or v

Original source details: GELFAND-P001123

$H(\nu)$ — a fn. on $\mathfrak{g}'$

$$\frac{\partial u}{\partial t} = \{u(\nu), H(\nu)\}$$

$$(*) \quad \frac{d\nu}{dt} \pm [\mathrm{ad}^*\, H'(\nu)]\nu = 0 \qquad \text{(adjoint eq.)}$$

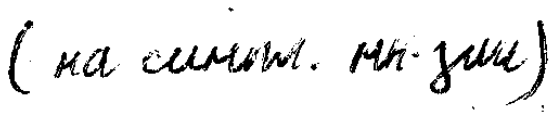


Uncertain reading. the sign is retraced, with a separate lower stroke beneath the crossbar; it may be $\pm$ or a corrected $+$

Original source details: GELFAND-P001123

GELFAND-P001124

$\mathfrak{g}$ is the Lie alg. of functions $\varphi(x)$, $x \in X$ (on a symplectic manifold).



Uncertain reading. manifold

Original source details: GELFAND-P001124

The dual space consists of measures $\nu(dx)$.

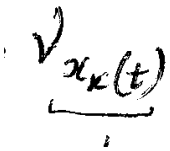
$h_1(x)$ is the particle energy.

$h_2(x_1, x_2)$ — // — of interaction.

$$H(\nu) = \int_X h_1(x)\nu(dx) + \frac{1}{2}\int_X \int_X h_2(x, x')\nu(dx)\nu(dx').$$

$$X^n \qquad y = (x_1, \ldots, x_n)$$

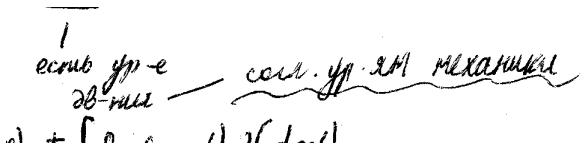
Claim. $\nu = \sum_{k=1}^{n} \nu_{x_k(t)}$ is a solution of the eq. (Klimontovich).



Uncertain reading. $\nu_{x_k(t)}$

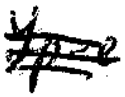
Original source details: GELFAND-P001124

[Under $x_k(t)$ a brace and an annotation:]



Uncertain reading. is the eq. of motion — according to the eqs. of mechanics.

Original source details: GELFAND-P001124



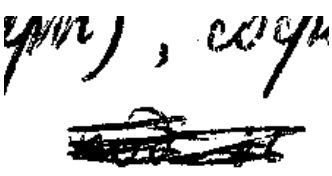
Uncertain reading. [Crossed out on the left: ~~Eq.?~~.]

Original source details: GELFAND-P001124

$$H'(\nu) = h_1(x) + \int h_2(x, x')\nu(dx').$$

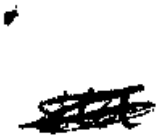
One gets a flow on the space of generalized functions.

Generalized functions (measures) supported on a finite number of points. It turns out that [crossed out:] functions supported on n points [crossed out:] are invariant under the flow. This is equivalent to the Vlasov eq.



Uncertain reading. crossed out: illegible

Original source details: GELFAND-P001124



Uncertain reading. crossed out: illegible

Original source details: GELFAND-P001124

$$\nu(dx) = C(q,p)\,dp\,dq$$

$$\frac{\partial C}{\partial t} + \frac{p}{m}\frac{\partial C}{\partial q} = \frac{\partial C}{\partial p}\left(V'(q) + \int \frac{\partial \Phi(q,q')}{\partial q} C(q',p')\,dq'\,dp'\right).$$

(Vlasov eq. — collisionless plasma).

$$h_1(x) = \frac{p^2}{2m} + V(q); \qquad h_2(x,x') = \Phi(q,q').$$

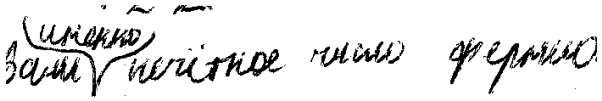
All this holds for Lie superalgebras (see the Berezin–Marinov eqs.).

A. Beilinson. On a paper by Witten.

There is no gauge $SU(2)$ theory with an odd number of fermions

Reason: $\pi_4(SU(2)) = \mathbb{Z}/2\mathbb{Z}$.

(Previously physicists considered [inserted above: specifically] an odd number of fermions)

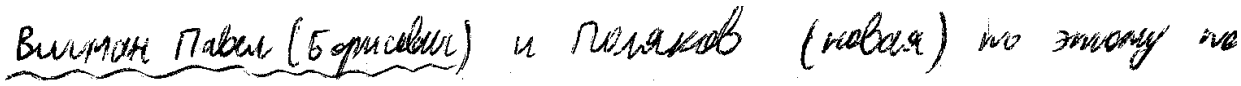


Uncertain reading. [inserted above: specifically] an odd number of fermions

Original source details: GELFAND-P001124

$$L = \int d^4x \left(\mathrm{Tr}\, F_{\mu\nu}^2 + \overline{\Psi} i D \Psi\right)$$

There is a paper by Wiegmann, Pavel (Borisovich), and Polyakov (a new one) on this subject.



Original source details: GELFAND-P001124

Uncertain reading. Wiegmann, Pavel (Borisovich),

$$\int d\Psi\, d\overline{\Psi}\, dA\, \exp(-L)$$



Uncertain reading. [Crossed out between exp and $(-L)$: illegible mark.]

Original source details: GELFAND-P001124

$$\pi_4(SU(2)) = \pi_0\left(SU(2)^{S^4}\right).$$

GELFAND-P001125

[source number: 82]

Fuchs:

Suspension: $\pi_3(S^2) \longrightarrow \pi_4(S^3)$ — an epimorphism with kernel $2\mathbb{Z}$.

I.M.G. Donaldson: on $\mathbb{R}^4$ there are smooth structures that depend on a parameter and are different for different values of the parameter!!

Fuchs:

$$0 \longrightarrow \mathbb{Z} \longrightarrow \pi_{4n-1}(S^{2n}) \xrightarrow{\text{epi}} \pi_{4n}(S^{2n+1}) \longrightarrow 0$$

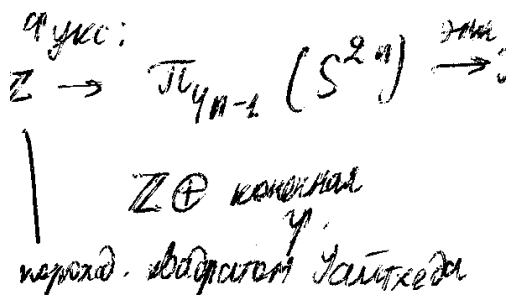


Diagram and typed description. [Under $\pi_{4n-1}(S^{2n})$] $\mathbb{Z} \oplus$ a finite group.
[Leader from $\mathbb{Z}$] generated by the Whitehead square.

Original source details: GELFAND-P001125

B. Theorem (Witten).

$\int d\Psi$ is interpreted as $(\det D)^{-1/2} \exp(\dots)$.

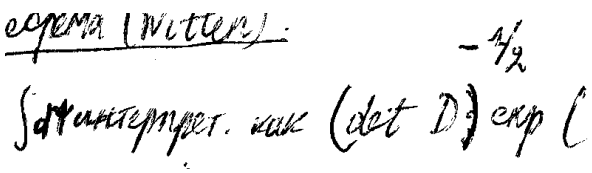


Diagram and typed description. Exponent in the manuscript $-1/2$.

Original source details: GELFAND-P001125

(A fixed)

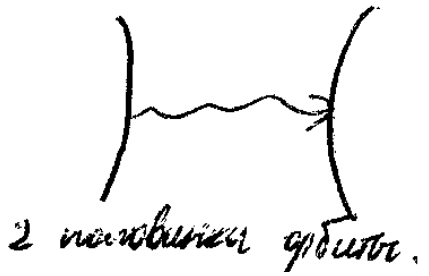
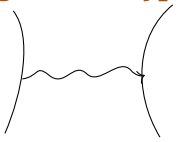


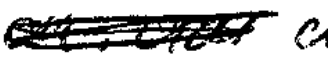
Diagram and typed description.



2 halves of the orbit.

Original source details: GELFAND-P001125

$\sqrt{\det D}$ will change sign, since some [crossed out] eigenvalues will change sign on passing through $0 \rightsquigarrow$



Uncertain reading. The crossed-out word is illegible.

Original source details: GELFAND-P001125

$\rightsquigarrow$ we have to count the zero modes of the Dirac operator on $S^4 \times S^1$.

The eigenvalues split into pairs [under the word "pairs": $(\lambda, -\lambda)$]; $\sqrt{\det}$ will change sign on passing through a zero mode.

Then one uses the Atiyah–Singer theorem. It gives the number of zero modes $\bmod 2$.

I.M.G. A general problem for mathematicians:

Use operators to do what is done with Feynman integrals.

Example. σ-model, simple (see Perelomov's earlier talk) $\longrightarrow$ write down an operator.



Diagram and typed description. simple.

Original source details: GELFAND-P001125

GELFAND-P001126



Diagram and typed description. 17 October 1983. Gelfand Seminar.

Original source details: GELFAND-P001126

A. Beilinson. On Witten's work

$$(1) \qquad S = \int (F, F) + i(\Psi, \mathcal{D}\Psi), \qquad F \text{ — curvature of the connection } A$$



Uncertain reading. [Before S an illegible mark is crossed out.]
[From S a leader points down to "action".]

Original source details: GELFAND-P001126

$$\mathcal{D} : \Gamma(E \otimes S) \longrightarrow \Gamma(E \otimes \bar{S}) \quad \text{— Dirac operator}$$

S — canonical spinor bundle on a 4-dimensional manifold.



Diagram and typed description. (2-dimensional)

Original source details: GELFAND-P001126

Ψ — 2-component spinor.

e^{iS} — a function on the space of pairs (A, Ψ), gauge invariant

$$(2) \qquad \int e^{iS}\, d(A, \Psi) \text{ — ?}$$

Claim. $(2) = 0$, if the number of components of Ψ is odd.

Consider.

$$(3) \qquad \int \left(\int e^{-S}\, d\Psi \right) dA \qquad \text{(integral over the set of orbits)}$$

Each orbit splits into 2 pieces. [Crossed out.] With any natural definition the function takes values of opposite sign on these components! $\Rightarrow$ hence $(2) = 0$.



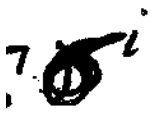
Uncertain reading. The crossed-out writing is illegible.

Original source details: GELFAND-P001126

$$(4) \qquad \int e^{-S}\, d\Psi = \sqrt{\det \mathcal{D}} \quad \text{— a function of } A_i$$

$$(5) \qquad \mathcal{D} = \sum \sigma^i \left(\frac{\partial}{\partial x_i} + A_i \right),$$

σ^i — matrices of order 2 (Pauli)



Uncertain reading. Overwritten symbol read as σ^i.

Original source details: GELFAND-P001126

$$[\sigma_1, \sigma_2] = i\sigma_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

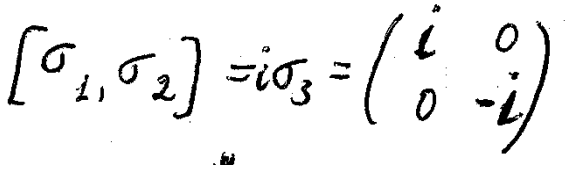


Diagram and typed description. $[\sigma_1, \sigma_2] = i\sigma_3$.

Original source details: GELFAND-P001126

$$(6) \qquad \pi_0 \left(SU(2)^{S^4} \right) = \pi_4(SU(2)) = \pi_4(S^3) = \mathbb{Z}_2$$

GELFAND-P001127

[source number: 84]

Consider $S^4 \times S^1$ and on it a bundle E with a connection.

We deform the connection [written underneath: A] into another [written underneath: A'] so that A and A' are related by a transformation from the 2nd component of the gauge group.

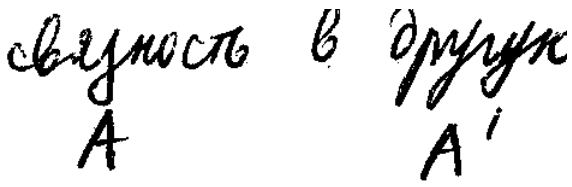


Diagram and typed description. [written underneath: A, A'.]

Original source details: GELFAND-P001127

(Connections form an affine space)

Then $\sqrt{\det \mathcal{D}}$ changes sign!

$\sqrt{\det \mathcal{D}}$ takes into account half the eigenvalues. [Unclear mark at right.] Eigenvalues occur in pairs $(\lambda, -\lambda)$.



Uncertain reading. The mark before "Eigenvalues" looks like a cancelled letter or an asterisk.

Original source details: GELFAND-P001127

An odd number of eigenvalues go from positive to negative and vice versa.

— · —

Dirac eq.

$$\sum \gamma^i \frac{\partial}{\partial x_i} + A_i(x_0, \ldots, x_3, t) \tag{7}$$

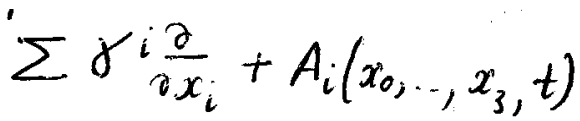


Diagram and typed description. Equation (7) as written, without additional parentheses.

Original source details: GELFAND-P001127

(Along S^4 it is the same as before)

t — the parameter for deforming the connection; let the start be the same as the end (gauge equiv.)

The bundle on $S^4 \times S^1$ is trivial on $S^4 \times [0, 1)$ and then glue using the chosen element of the gauge group $\Rightarrow$ it becomes nontrivial.

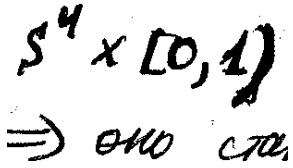


Diagram and typed description. $S^4 \times [0, 1)$.

Original source details: GELFAND-P001127

$$(8) \qquad \left(\sum \gamma^i \frac{\partial}{\partial x_i} + A_i(x_0, \ldots, x_3, t)\right) \Psi = 0$$

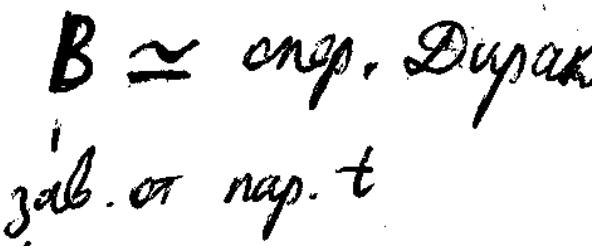
regard this as an evolution equation

$$(9) \qquad \frac{\partial}{\partial t}\Psi = B\Psi, \quad \text{where } B \simeq \text{ Dirac operator on a 4-dimensional manifold.}$$



Original source details: GELFAND-P001127

Uncertain reading. At the beginning of eq. (9) illegible marks are crossed out.



Original source details: GELFAND-P001127

Diagram and typed description. [Under B: depends on the parameter t.]

Let B be independent of t, and replace the interval by the line.

Choose as Ψ an eigenfunction with eigenvalue $\lambda > 0$.

$$\begin{aligned} t \longrightarrow -\infty \quad &\Rightarrow \quad \text{function} \longrightarrow 0. \\ t \longrightarrow +\infty \quad &\Rightarrow \quad \text{the function grows} \end{aligned}$$

It would be good if it decayed as $t \longrightarrow \pm\infty$.

We need: the eigenvalue to change sign.

A solution with these boundary conditions $\approx$ a function on the circle.

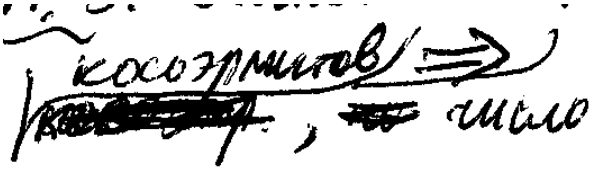
$\Rightarrow$ from an eigenvalue that changes sign one constructs a solution of the Dirac eq..

Thus the number of solutions to (8) on $S^4 \times S^1$ is related to changes in sign of the eigenvalues of operator (7).

Witten says «we will vary it adiabatically» (so that we have an eigenfunction throughout)

The A–S theorem counts mod2 the solutions of eq. (8) on $S^4 \times S^1$.

Operator [crossed out; inserted above: skew-Hermitian $\Rightarrow$], [crossed out] the number of solutions mod2 — a topological invariant.



Original source details: GELFAND-P001127

Uncertain reading. [crossed out; inserted above: skew-Hermitian $\Rightarrow$.]



Uncertain reading. The crossed-out writing before “the number” is illegible.

Original source details: GELFAND-P001127

I.M.G. Problem. Devise a general mathematical framework in which this is a special case.

GELFAND-P001128

A. Zelevinsky.

$$\Lambda_n = \mathbb{Z}[x_1, \ldots, x_n]^{S_n} \quad \text{— symmetric polynomials}$$

Λ_n^k — homogeneous [inserted above: symm.] polynomials of degree k.

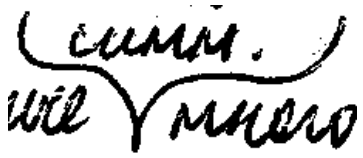


Diagram and typed description. [inserted above: symm.]

Original source details: GELFAND-P001128

$n \longrightarrow \infty \quad \Rightarrow \quad \Lambda_n^k$ ceases to depend on n.

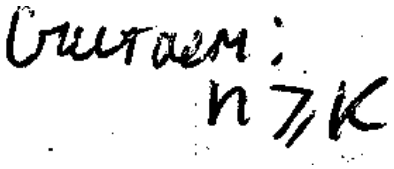


Diagram and typed description. [On the left:] Assume $n \geq k$.

Original source details: GELFAND-P001128

Λ — symm. polynomials in an “inf.” number of variables (formalized by a projective limit)

$$\text{1st degree} \quad x_1 + x_2 + \ldots = m_{(1)}$$

$$\text{2nd degree} \quad \begin{bmatrix} x_1^2 + x_2^2 + \ldots = m_{(2)} \\ x_1x_2 + x_1x_3 + x_2x_3 + \ldots = m_{(1^2)} \end{bmatrix}$$

$$(1) \qquad \lambda = (\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n \geq \ldots \geq 0)$$

(an inf. sequence that breaks off somewhere)

$$(2) \qquad m_\lambda = x_1^{\lambda_1} x_2^{\lambda_2} \ldots + \text{ monomials obtained by permuting the letters}$$

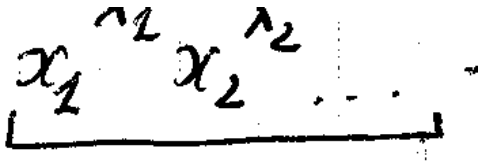


Diagram and typed description. Straight bracket below $x_1^{\lambda_1} x_2^{\lambda_2} \ldots$.

Original source details: GELFAND-P001128

Each is an integral linear combination, so, m_λ — a basis.
(monomial basis)

$$m_{(3\,1^2)} = x_1^3 x_2 x_3 + \dots$$

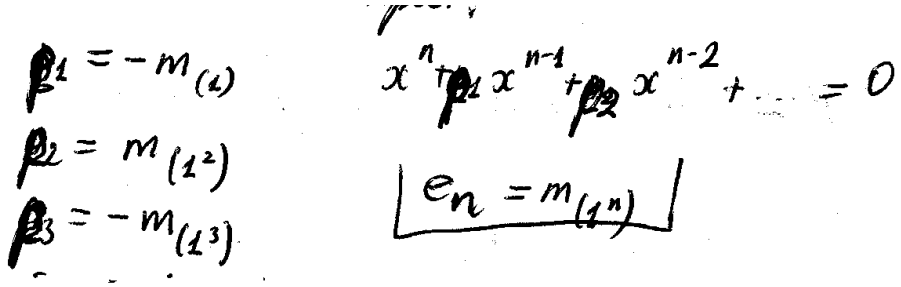


Diagram and typed description. Straight bracket below ... and label: perm.

Original source details: GELFAND-P001128

$$\beta_1 = -m_{(1)}$$
$$\beta_2 = m_{(1^2)}$$
$$\beta_3 = -m_{(1^3)}$$
$$\dots$$

$$x^n + \beta_1 x^{n-1} + \beta_2 x^{n-2} + \dots = 0$$



Original source details: GELFAND-P001128

Uncertain reading. The overwritten coefficient letters are tentatively read as β_i.

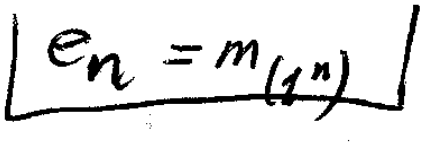


Diagram and typed description. [Boxed on three sides] $e_n = m_{(1^n)}$.

Original source details: GELFAND-P001128

$$(3) \qquad E(t) = \sum_{n\geq 0} e_n t^n = \prod_i (1 + t x_i) \qquad \text{(generating function)}, \quad e_0 = 1,$$

$$(4) \qquad h_n = \sum_{|\lambda| = n} m_\lambda,$$
$$h_2 = m_{(2)} + m_{(1^2)} = (x_1^2 + x_2^2 + \dots) + (x_1 x_2 + x_1 x_3 + \dots)$$

(sum of all monomials)

$$(5) \qquad H(t) = \sum_{n\geq 0} h_n t^n = \prod_i (1 - t x_i)^{-1} \qquad \text{(Exercise!)}, \quad h_0 = 1$$

$$(6) \qquad p_n = m_{(n)} \quad \text{— power sum}$$

$$(7) \qquad P(t) = \sum_{n\geq 1} p_n t^n = \sum_i \frac{t x_i}{1 - t x_i} = t H'(t)/H(t)$$

Bases: from $\lambda = (\lambda_1, \lambda_2, \dots)$ construct:

GELFAND-P001129

[source number: 85]

$$(8)\qquad \begin{cases} e_\lambda = \prod e_{\lambda_i} & (e_\lambda) \text{ — } \mathbb{Z}\text{-basis in } \Lambda \\ h_\lambda = \prod h_{\lambda_i} & (h_\lambda) \text{ — } \mathbb{Z}\text{-basis in } \Lambda \\ p_\lambda = \prod p_{\lambda_i} & (p_\lambda) \text{ — } \mathbb{Q}\text{-basis in } \Lambda \end{cases}$$



Diagram and typed description. [In the left margin by the second line] and.

Original source details: GELFAND-P001129

$$\Lambda = \mathbb{Z}[e_1, e_2, \ldots] = \mathbb{Z}[h_1, h_2, \ldots]$$

(fundamental theorem on symm. polynomials)

$$\Lambda_{\mathbb{Q}} = \mathbb{Q}[p_1, p_2, \ldots]$$

$$p_{(2,1^2)} = (x_1^2 + x_2^2 + \ldots)(x_1 + x_2 + \ldots)^2$$

5th basis. S-functions of Schur (appeared in Jacobi)

$$\alpha = (\alpha_1, \ldots, \alpha_n)$$

$$a_\alpha(x_1, \ldots, x_n) = \det(x_i^{\alpha_j}) = \begin{vmatrix} x_1^{\alpha_1} & x_1^{\alpha_2} & \ldots & x_1^{\alpha_n} \\ x_2^{\alpha_1} & x_2^{\alpha_2} & \ldots & x_2^{\alpha_n} \\ \ldots & \ldots & \ldots & \ldots \end{vmatrix}$$

[Below the determinant:] skew-symmetric function of $x_1, \ldots, x_n$.

$$a_\delta = \prod_{i<j}(x_i - x_j)$$

$$(\delta = (n-1, n-2, \ldots, 1, 0))$$

$\frac{a_\alpha}{a_\delta}$ — symm. polynomial.

Assume $\alpha_1 > \alpha_2 > \ldots > \alpha_n$ (one can always achieve this up to sign)



Diagram and typed description. Put differently:

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$$\alpha = \lambda + \delta, \qquad \lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$$

$$(9) \qquad s_\lambda \overset{\text{def}}{=} a_{\lambda+\delta}/a_\delta \quad \text{— another basis.}$$

[crossed out] (over $\mathbb{Z}$)



Uncertain reading. The start of the cancelled writing reads “в” (in). The rest is illegible.

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For $n \geq |\lambda|$ the ratio does not depend on n.

Theory: find relations between them and the structure constants (for multiplication)

Example — Theorem.



Uncertain reading. The crossed-out signs before the formula are illegible.

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$$s_\lambda = \sum_\mu K_{\lambda\mu} m_\mu$$

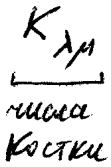
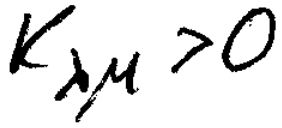


Diagram and typed description. Under the bracket below $K_{\lambda\mu}$: Kostka numbers.

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$$K_{\lambda\mu} \in \mathbb{Z}, \qquad K_{\lambda\mu} > 0$$

Diagram and typed description. $K_{\lambda\mu} > 0$.



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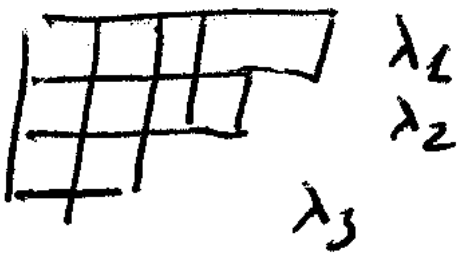
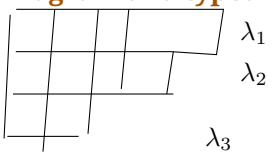


Diagram and typed description.

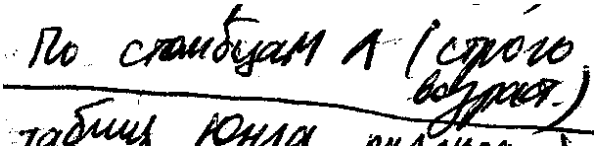


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λ defines a Young diagram.

μ — μ_1 ones, μ_2 twos, . . . ||

Fill in this diagram. Along the rows $\leq$ (nondecreasing). Along the columns $\nearrow$ (strictly increasing).



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Uncertain reading. The slanted mark is tentatively read as $\nearrow$.

Call this a tableau of Young.



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Diagram and typed description.

1	1	2
3	4	
5		

The number of Young tableaux of size λ and weight μ equals $K_{\lambda\mu}$.

GELFAND-P001130

Easy exercise: find the change-of-basis matrices between the others.

6th basis f_λ (from the word forgotten)



Original source details: GELFAND-P001130

Diagram and typed description. 24 October 1983. Gelfand Seminar.

I.M. Gelfand with work by Flashka.



Original source details: GELFAND-P001130

Diagram and typed description. Flashka.

$$L_n = \sum_{s=0}^{n} u_s \frac{\partial^s}{\partial x_1^s}, \qquad u_n = 1. \tag{1}$$

$$L_n \widetilde{\Psi} = k^n \widetilde{\Psi}. \tag{2}$$

Consider an asymptotic solution as $k \to \infty$

$$\widetilde{\Psi}^n = e^{kx} \sum_{0}^{\infty} c_j k^{-j}, \qquad c_0 = 1. \tag{3}$$

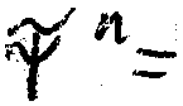


Diagram and typed description. $\widetilde{\Psi}^n$.

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Ex.: write recurrence formulas to determine c_j.

$$\chi = \widetilde{\Psi}_{x_i} \widetilde{\Psi}^{-1} = k + \sum_{1}^{\infty} \chi_j k^{-j}. \tag{4}$$



Diagram and typed description. $\widetilde{\Psi}_{x_i}$.

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Here χ_j does not depend on the choice of $\widetilde{\Psi}$

$\forall$ integer m, $\exists$ an operator L_m of order m ([crossed out] with leading coeff. 1)



Uncertain reading. The crossed-out writing before “with leading” is illegible.

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$$(L_m\widetilde{\Psi})\widetilde{\Psi}^{-1} = k^m + \varepsilon_m k^{-1} + \dots \tag{5}$$

Here:

$$L_m = [(L_n)^{m/n}] \quad \text{(Gelfand–Dikii)} \quad = [(L_n)^{m/n}]_+$$

(5) determines the operator L_m uniquely

Introduce variables $x_2, x_3, \dots$ such that

$$\frac{\partial L_n}{\partial x_m} = [L_n, L_m] \tag{6}$$

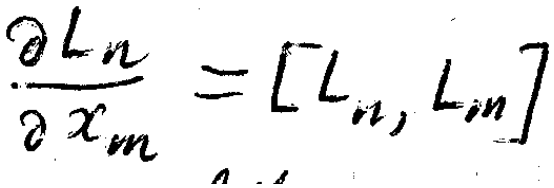


Diagram and typed description. $[L_n, L_m]$.

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or equivalently



Uncertain reading. The overwritten word is tentatively read as equivalently.

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$$\frac{\partial \Psi}{\partial x_m} = L_m \Psi \tag{7}$$

where

$$\Psi = e^{\sum_2^\infty x_j k^j}\widetilde{\Psi}. \tag{8}$$

Assume that u_j — functions of the variables $x_2, x_3, \dots$

$$\text{Put} \quad \varepsilon(k) \equiv -1 + \sum_{m=1}^{\infty} \varepsilon_m k^{-m-1} \tag{9}$$

$$\Lambda(k) = \sum_0^\infty k^{-m-1} L_m \tag{10}$$

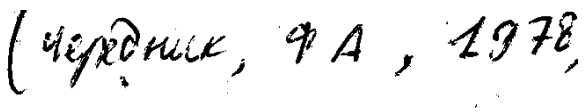


Diagram and typed description. [At right:] (Cherednik, FA, 1978)

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GELFAND-P001131

[source number: 86]

Cherednik's Thm. 3.1 states that

$$\left(\frac{\partial}{\partial x_1} - \chi(k)\right)\Lambda(k) = \varepsilon(k) \tag{11}$$

Cherednik:

$$\sum_{m=0}^{\infty} k^{-m-1}\left(\frac{\partial}{\partial x_1}(L_m\Psi(k))\Psi^{-1}(k)\right) = \frac{\partial\chi}{\partial k} + \mathcal{E}(k). \tag{12}$$

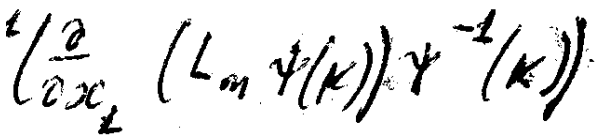
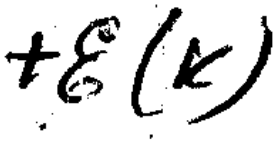


Diagram and typed description. Parentheses in eq. (12).

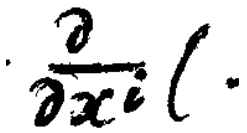
Original source details: GELFAND-P001131



Uncertain reading. The last letter is tentatively read as $\mathcal{E}(k)$.

Original source details: GELFAND-P001131

$$-\ell\chi_\ell + \varepsilon_\ell = \frac{\partial}{\partial x^i}(\dots) \tag{13}$$



Uncertain reading. The denominator is tentatively read with a superscript x^i.

Original source details: GELFAND-P001131

$$\begin{aligned}\frac{\partial}{\partial x_1}(L_m\Psi(k)\cdot\Psi^{-1}(k)) &= \frac{\partial}{\partial x_1}\left(\frac{\partial\Psi}{\partial x_m}\Psi^{-1}\right)\\ &= \frac{\partial}{\partial x_m}\left(\frac{\partial\Psi}{\partial x_1}\Psi^{-1}\right) = \frac{\partial\chi}{\partial x_m} \quad !!!\end{aligned} \tag{14}$$



Uncertain reading. Crossed-out number in the left margin (15?).

Original source details: GELFAND-P001131

$$-\ell\chi_\ell + \varepsilon_\ell = \frac{\partial\chi_{\ell-1}}{\partial x_1} + \ldots + \frac{\partial\chi_1}{\partial x_{\ell-1}}; \tag{15}$$

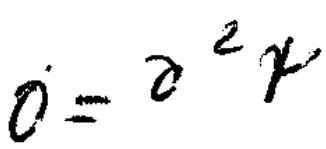


Diagram and typed description. [At right] $0 = \partial^2\Psi$.

Original source details: GELFAND-P001131

$$\varepsilon_j = \frac{\partial\tau}{\partial x_j} \qquad — \qquad \tau\text{-function} \tag{16}$$

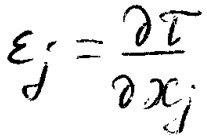


Diagram and typed description. $\varepsilon_j = \frac{\partial\tau}{\partial x_j}$.

Original source details: GELFAND-P001131

$$\Psi(x,k) = \frac{e^{\sum_{j=1}^{\infty} x_j k^j} e^{-\sum \frac{1}{\ell}\frac{\partial}{\partial x_\ell}}\tau(x)}{\tau(x)} \tag{17}$$

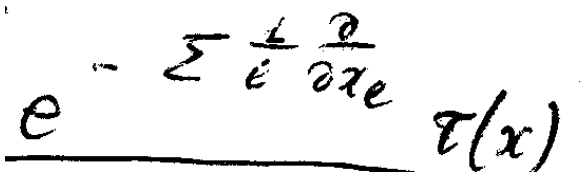


Diagram and typed description. Exponent of the second exponential $-\sum \frac{1}{\ell}\frac{\partial}{\partial x_\ell}$.

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(⇒ the structure of the recurrence formulas)
G. Olshanski.

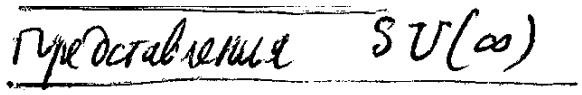


Diagram and typed description. Representations of $SU(\infty)$

Original source details: GELFAND-P001131

W — the Weil repr. of $Sp(\infty,\mathbb{R})^\sim$ on $\mathcal{H}(z_1, z_2, \ldots)$.

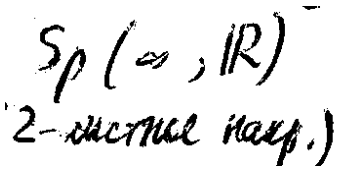


Diagram and typed description. [Under $Sp(\infty,\mathbb{R})^\sim$] (2-sheeted cover)

Original source details: GELFAND-P001131

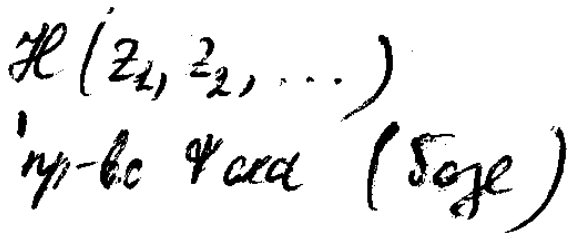
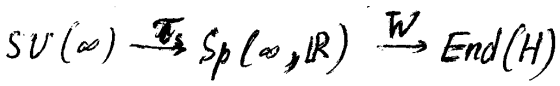


Diagram and typed description. [Under $\mathcal{H}(z_1, z_2, \ldots)$] Fock space (Bose)

Original source details: GELFAND-P001131



Original source details: GELFAND-P001131

Diagram and typed description.

$$SU(\infty) \xrightarrow{\tau_s} Sp(\infty, \mathbb{R}) \xrightarrow{W} \mathrm{End}(H)$$

[At right, past ||]

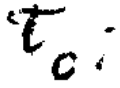
$$Sp(\infty, \mathbb{R}) = U(\infty, \infty) \cap Sp(\infty, \mathbb{C})$$

(complex realization)

$$T_s^+ = W \circ \tau_s \qquad \text{(fund. repr.)}$$

[At right] $s \neq 0 \Rightarrow$ irreducible

Embedding τ_0:



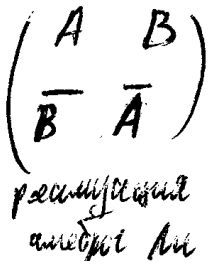
Original source details: GELFAND-P001131

Uncertain reading. Tentative reading τ_0 (the subscript is open in shape).

$$\begin{pmatrix} \bar{g} & 0 \\ 0 & g \end{pmatrix} \subset Sp(\infty, \mathbb{R})^{\sim}; \qquad g \in SU(\infty).$$

$$\tau_s(g) = \begin{pmatrix} \mathrm{ch}\, s & \mathrm{sh}\, s \\ \mathrm{sh}\, s & \mathrm{ch}\, s \end{pmatrix} \begin{pmatrix} \bar{g} & 0 \\ 0 & g \end{pmatrix} \begin{pmatrix} \mathrm{ch}\, s & -\mathrm{sh}\, s \\ -\mathrm{sh}\, s & \mathrm{ch}\, s \end{pmatrix}$$

$$\begin{pmatrix} A & B \\ \bar{B} & \bar{A} \end{pmatrix} \qquad \begin{cases} A = -A^* \\ B = B' \end{cases}$$



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Diagram and typed description. [Below the matrix] realization ↓ of the Lie algebra

$$T_t^- = S \circ \tau_t^-, \qquad S \text{ — spinor repr. of the group } \mathrm{Spin}(\infty).$$

GELFAND-P001132

Weil repr.

$$W : \begin{cases} \begin{pmatrix} -E_{ji} & 0 \\ 0 & E_{ij} \end{pmatrix} \longmapsto z_i \partial_j + \dfrac{\delta_{ij}}{2} \\ \begin{pmatrix} 0 & E_{ij} + E_{ji} \\ 0 & 0 \end{pmatrix} \longmapsto \partial_i \partial_j \\ \begin{pmatrix} 0 & 0 \\ E_{ij} + E_{ji} & 0 \end{pmatrix} \longmapsto -z_i z_j \end{cases}$$



Uncertain reading. [At lower left] 26/...83... [tentative reading of the visible parts of the marks].

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Undeciphered passage. The lower parts of the date are cut off at the horizontal edge of the written area. The month and continuation cannot be read.

Original source details: GELFAND-P001132